

BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-00613	PCBA, MLB, BETTER, HY-4GB, X430	MLB_CMNPTS, CPU:1.6GHZ, DDR3:HYNIX_4GB, ALTERNATE
639-00614	PCBA, MLB, BETTER, HY-8GB, X430	MLB_CMNPTS, CPU:1.6GHZ, DDR3:HYNIX_8GB, ALTERNATE
639-00616	PCBA, MLB, BETTER, SM-4GB, X430	MLB_CMNPTS, CPU:1.6GHZ, DDR3:SAMSUNG_4GB, ALTERNATE
639-00617	PCBA, MLB, BETTER, SM-8GB, X430	MLB_CMNPTS, CPU:1.6GHZ, DDR3:SAMSUNG_8GB, ALTERNATE
639-00621	PCBA, MLB, BETTER, EL-4GB, X430	MLB_CMNPTS, CPU:1.6GHZ, DDR3:ELPIDA_4GB
639-00622	PCBA, MLB, BETTER, EL-8GB, X430	MLB_CMNPTS, CPU:1.6GHZ, DDR3:ELPIDA_8GB
639-00695	PCBA, MLB, BETTER, EL-16GB, X430	MLB_CMNPTS, CPU:1.6GHZ, DDR3:ELPIDA_16GB
685-00043	CMN PTS, PCBA, MLB, X430	MLB_COMMON, J110_MLB
685-00044	VCORE FET, REN, X430	VCORE_FET:REN
685-00045	VCORE FET, VSHY, X430	VCORE_FET:VSHY

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00044	685-00045		ALL	Replaces alt to Vishay

333S0704	333S0700		ALL	Elpida CAN DRAM all to Hynix
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BOM Groups

BOM GROUP	BOM OPTIONS
MLB_PROGPARTS	BOOTROM:PROG, SMC:PROG, TBTROM:PROG

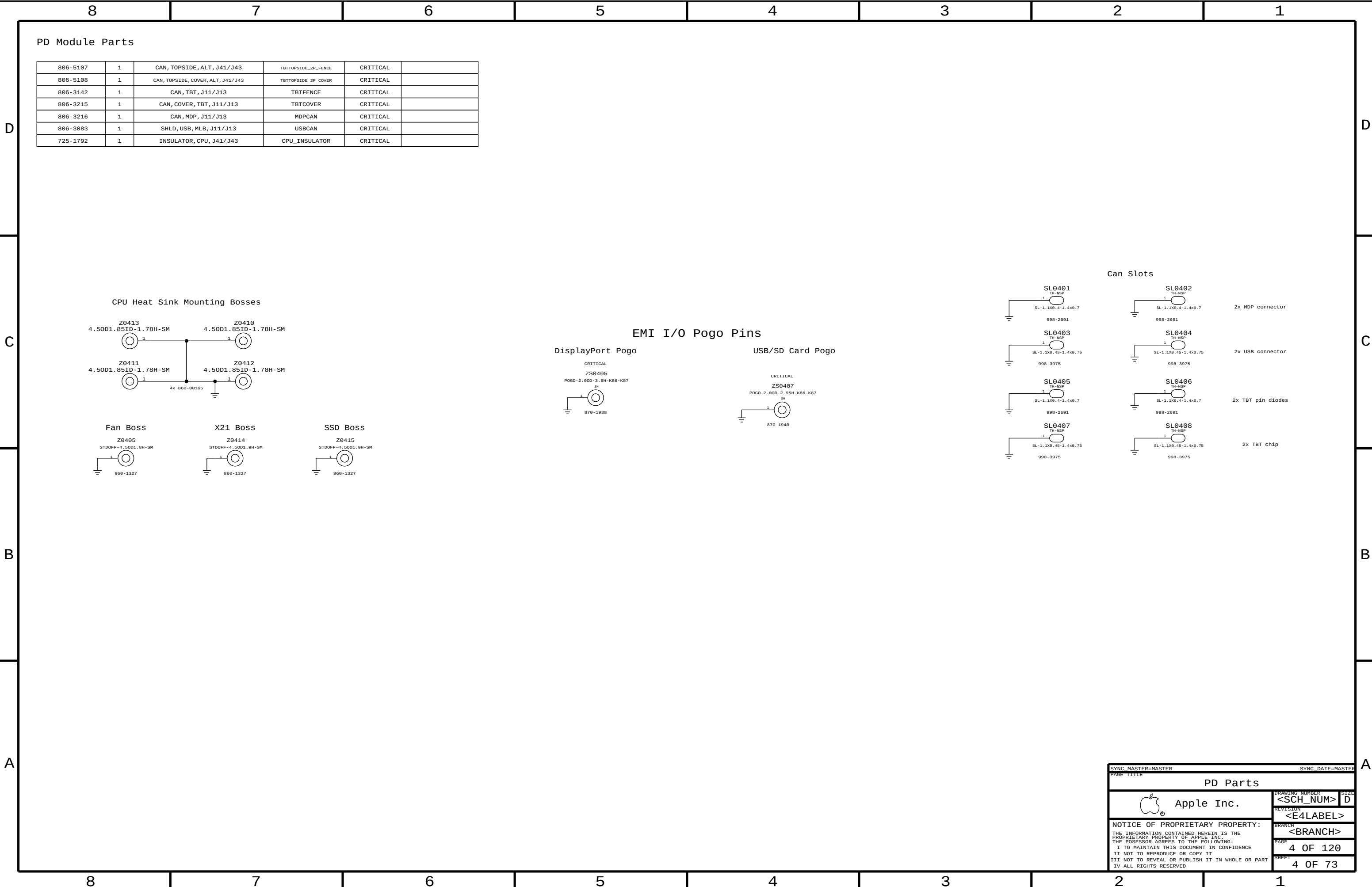
Programmable Parts

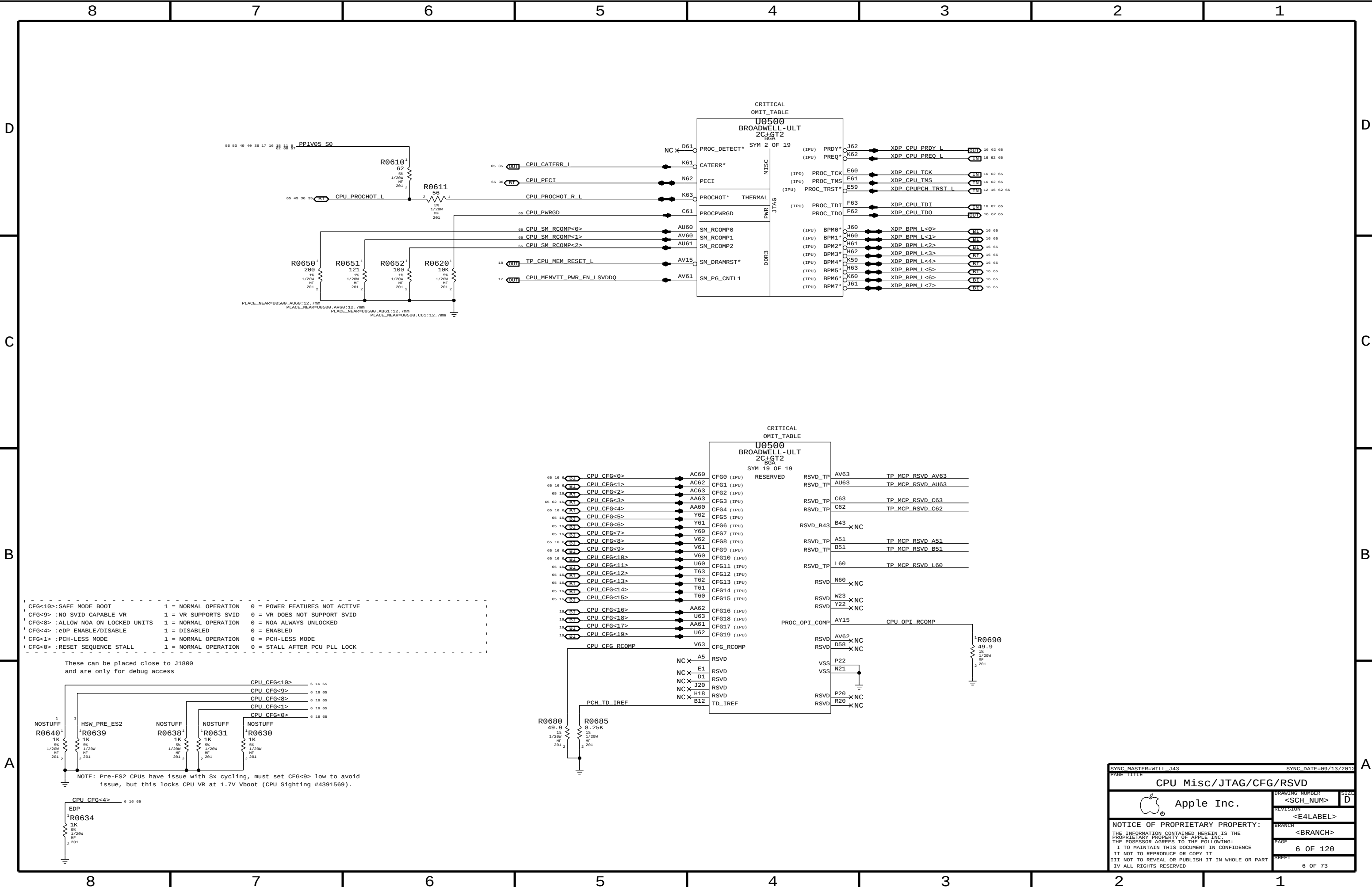
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341S00147	1	IC, SMC-A3,EXT, Vxxx, PROTO 0, 3110	U5000	CRITICAL	SMC:PROG

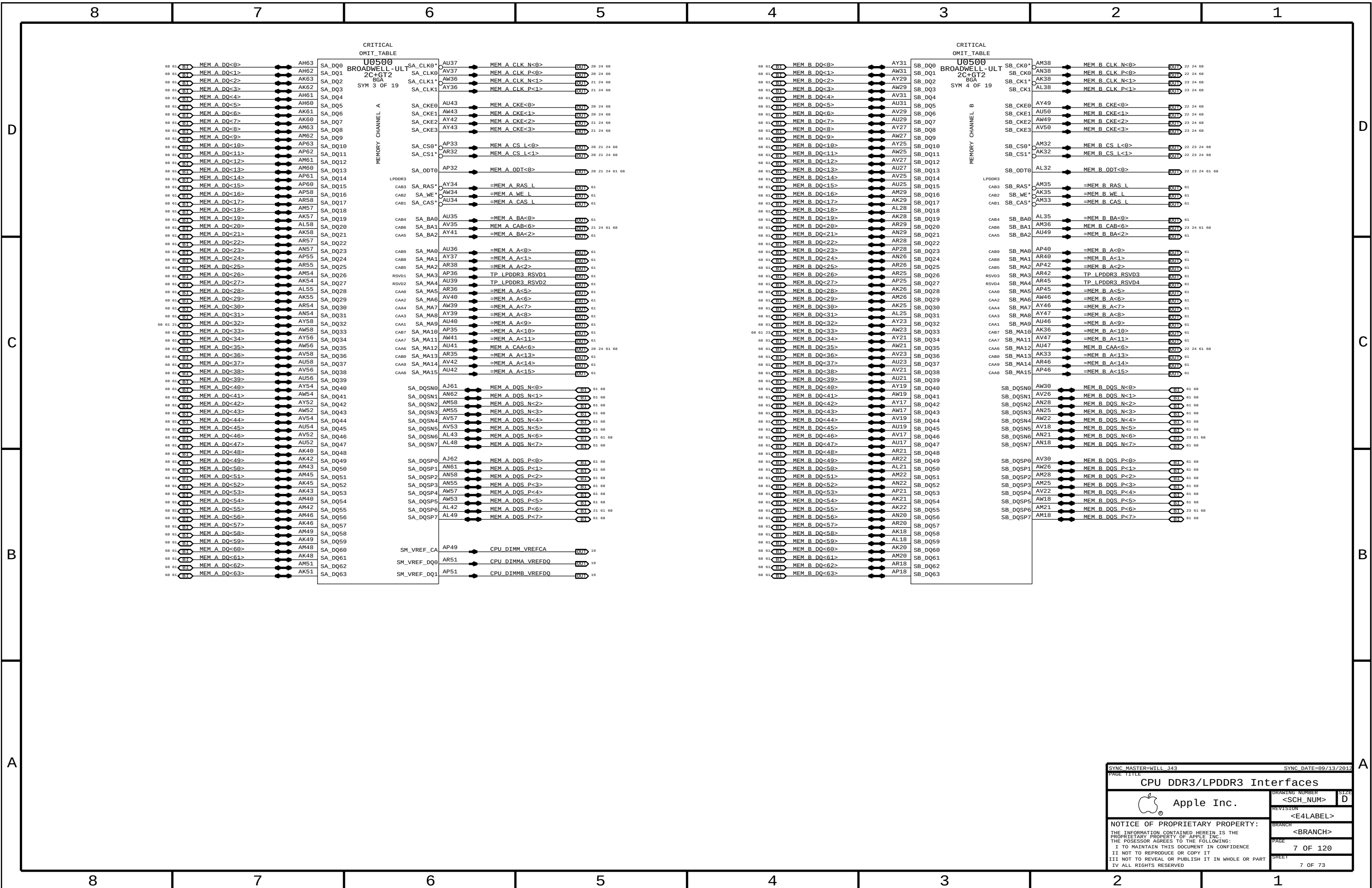
Sub - BOMs

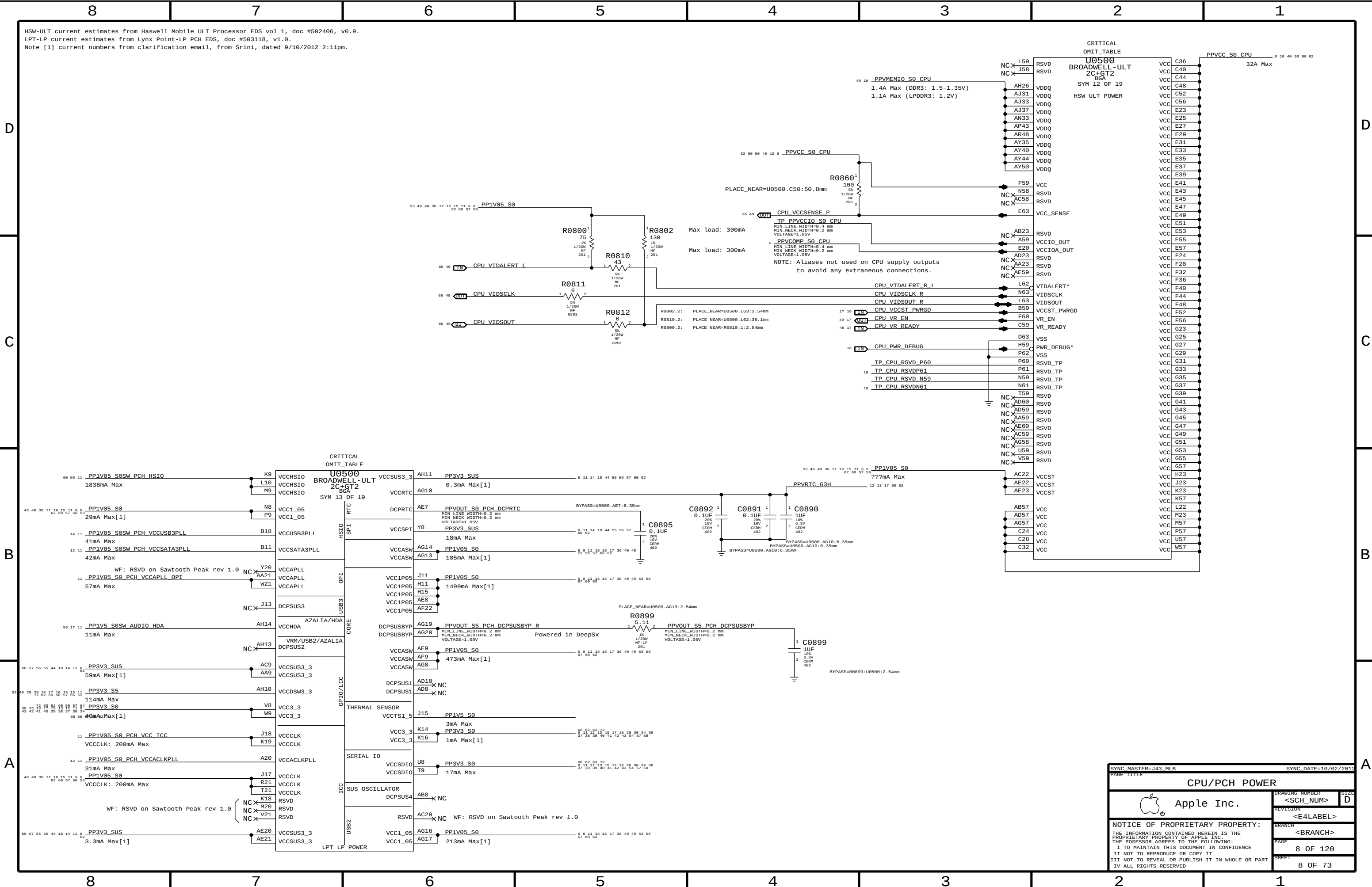
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685-00045	1	VCORE FET,VSHY,J110	VCOREFETS	CRITICAL	VCORE_FETS

SYNC_MASTER=MASTER		SYNC_DATE=MASTER	
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BOM Variants			
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		PAGE	3 OF 120
		SHEET	3 OF 73









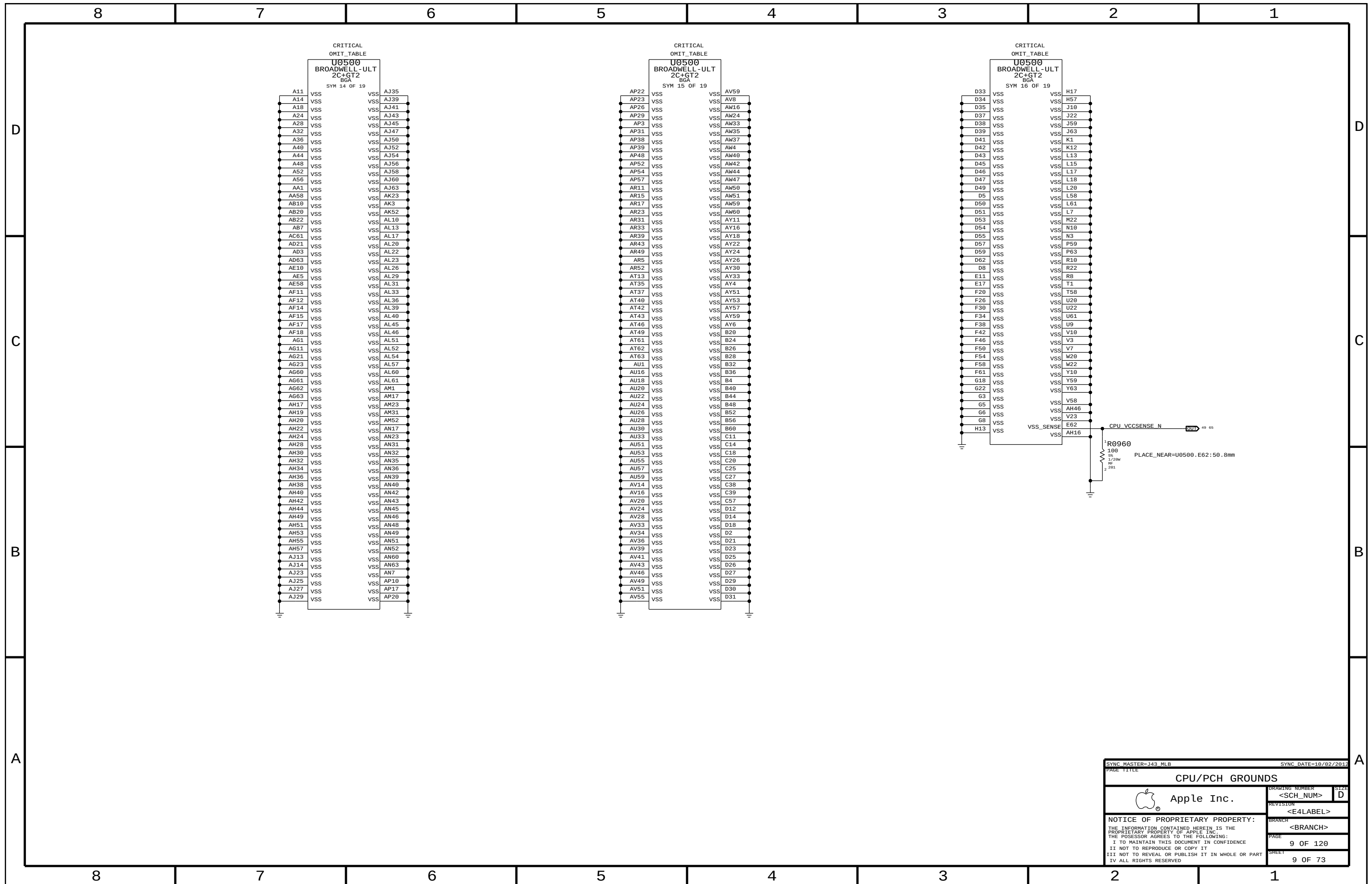
HSW-ULT current estimates from Haswell Mobile ULT Processor EDS vol 1, doc #502406, v0.9.
LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0.
Note [1] current numbers from clarification email, from Srini, dated 9/10/2012 2:11pm.

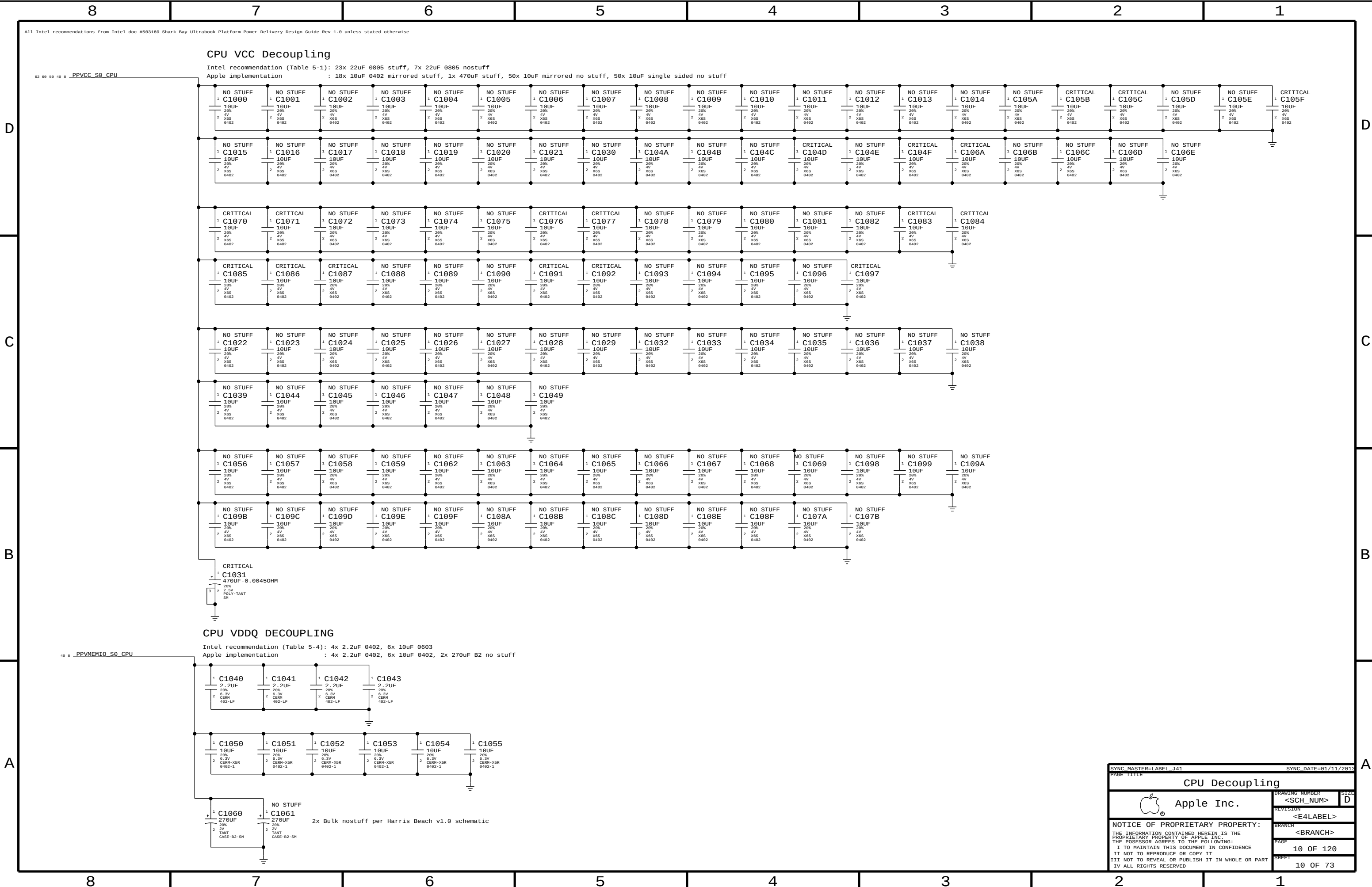
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U0500
BROADWELL-ULT
2C+GT2
BGA
SYM 12 OF 19
HSW ULT POWER

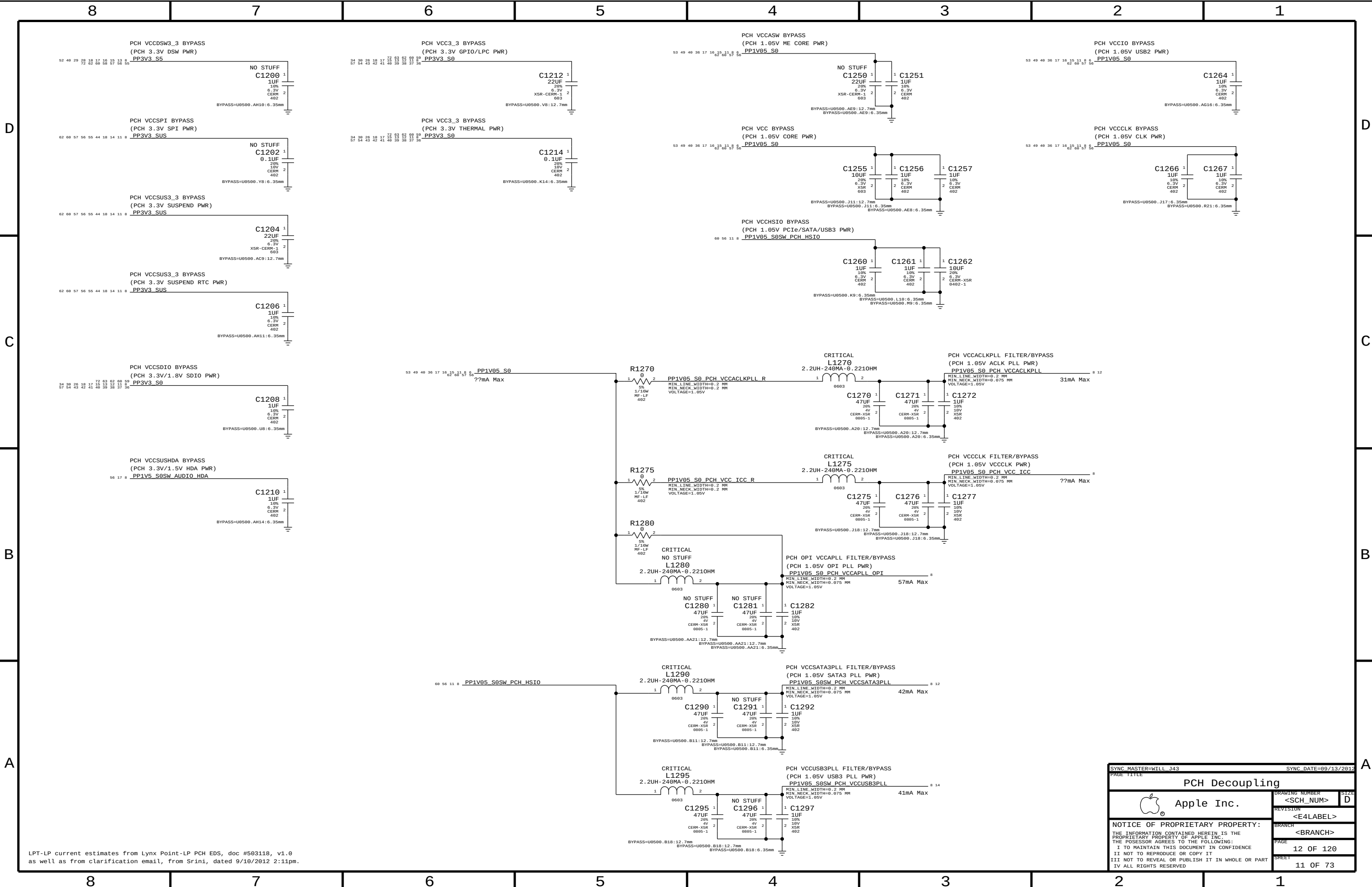
PPVCC_S0_CPU
32A Max
8 10 40 50 60 62

CRITICAL
OMIT_TABLE
U0500
BROADWELL-ULT
2C+GT2
BGA
SYM 13 OF 19

SYNC MASTER=143 MLB		SYNC DATE=10/02/2012	
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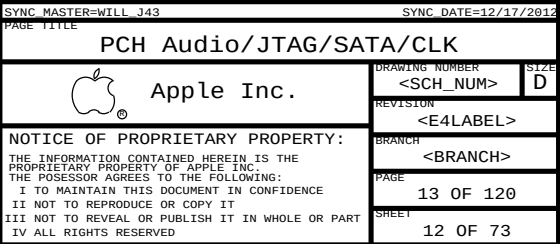






LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0
as well as from clarification email, from Srini, dated 9/10/2012 2:11pm.

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		SHEET	11 OF 73



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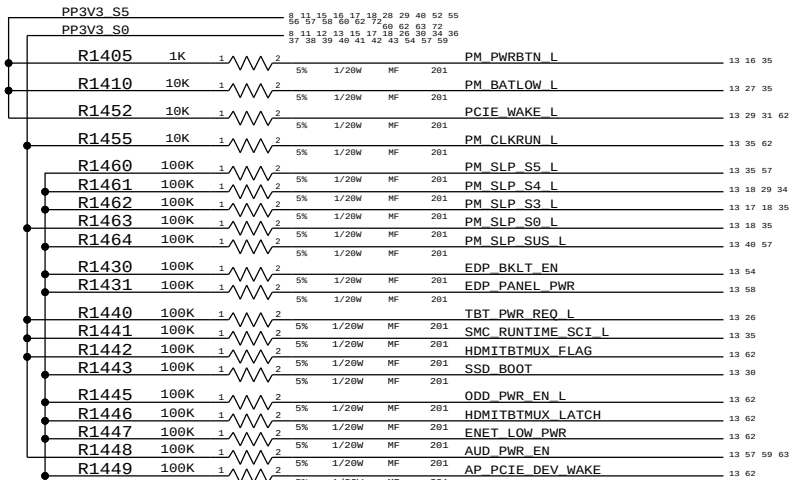
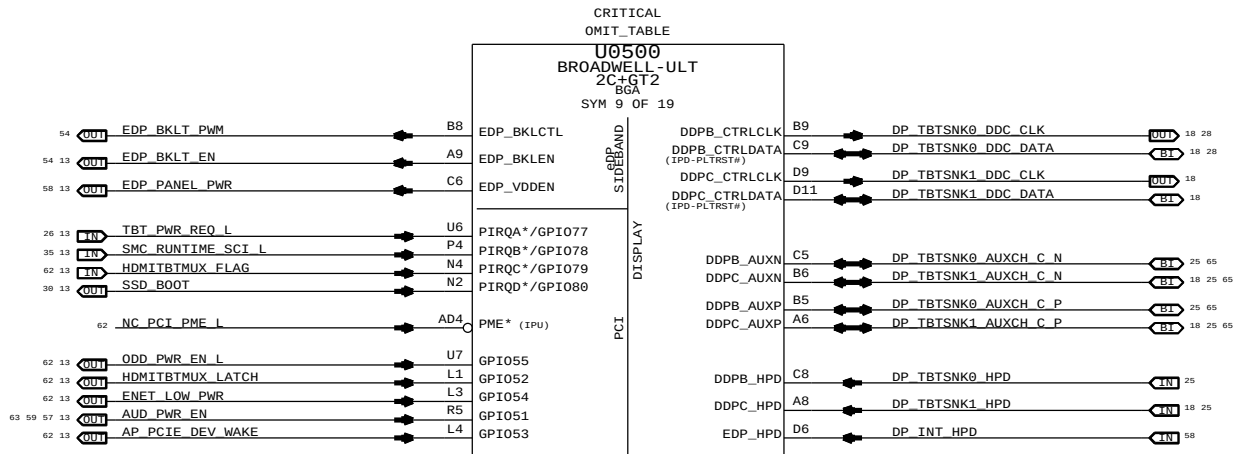
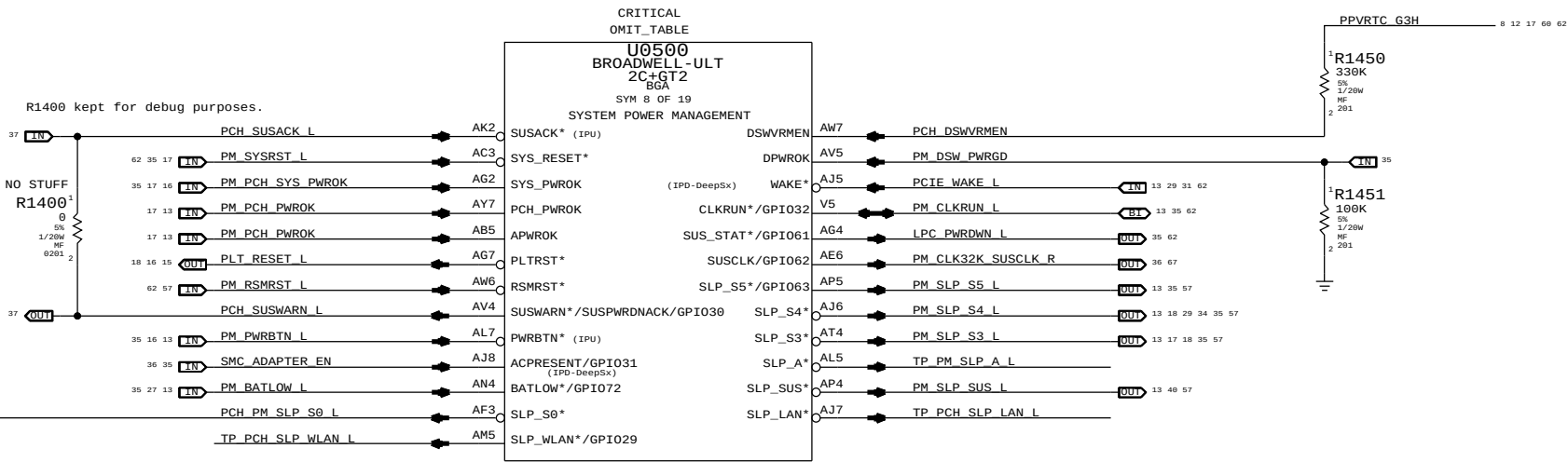
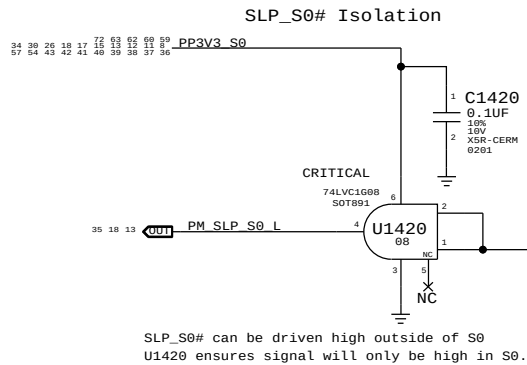
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SYNC_MASTER=143_MLB SYNC_DATE=02/20/2013

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PCH PM/PCI/GFX

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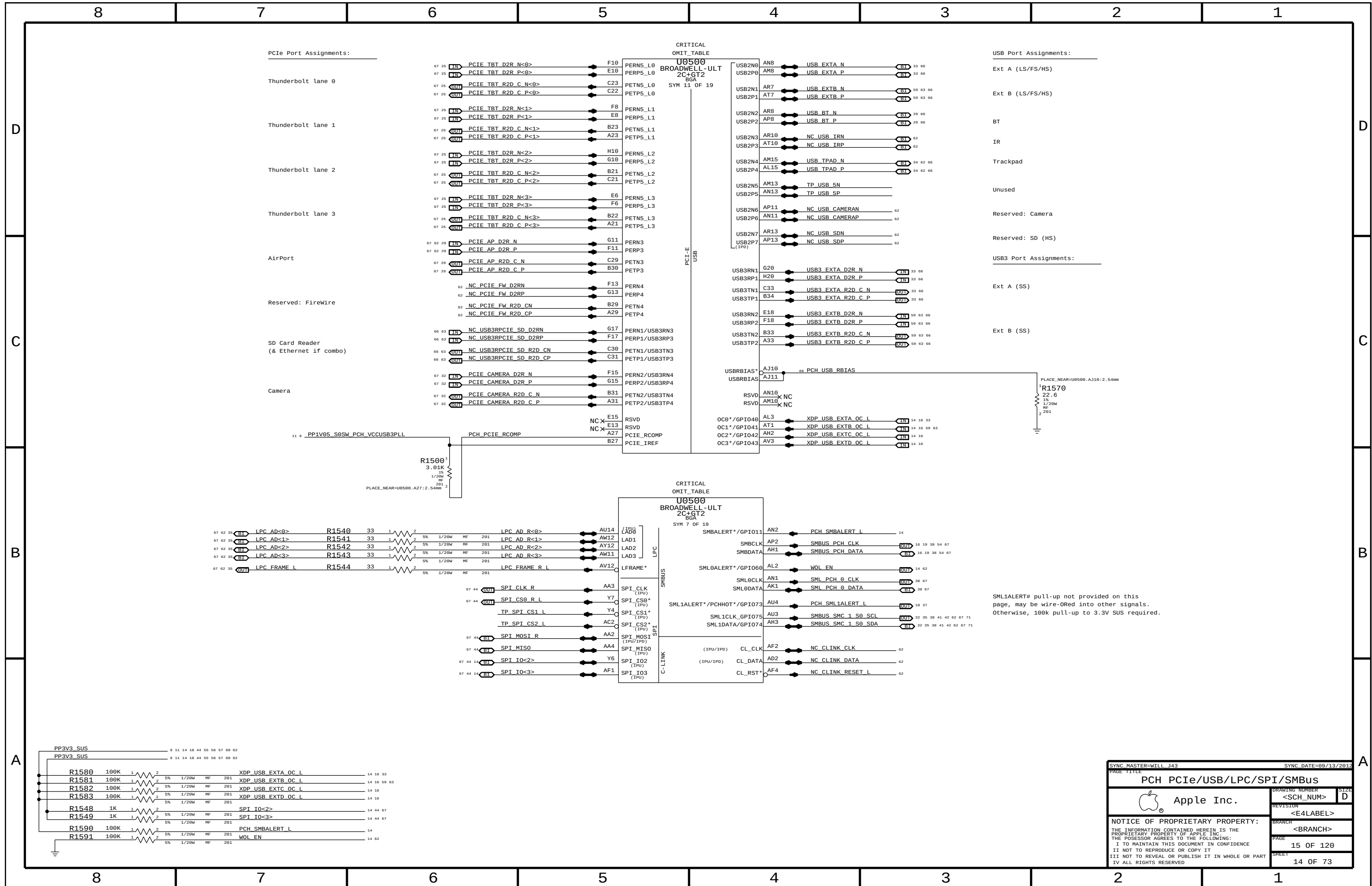
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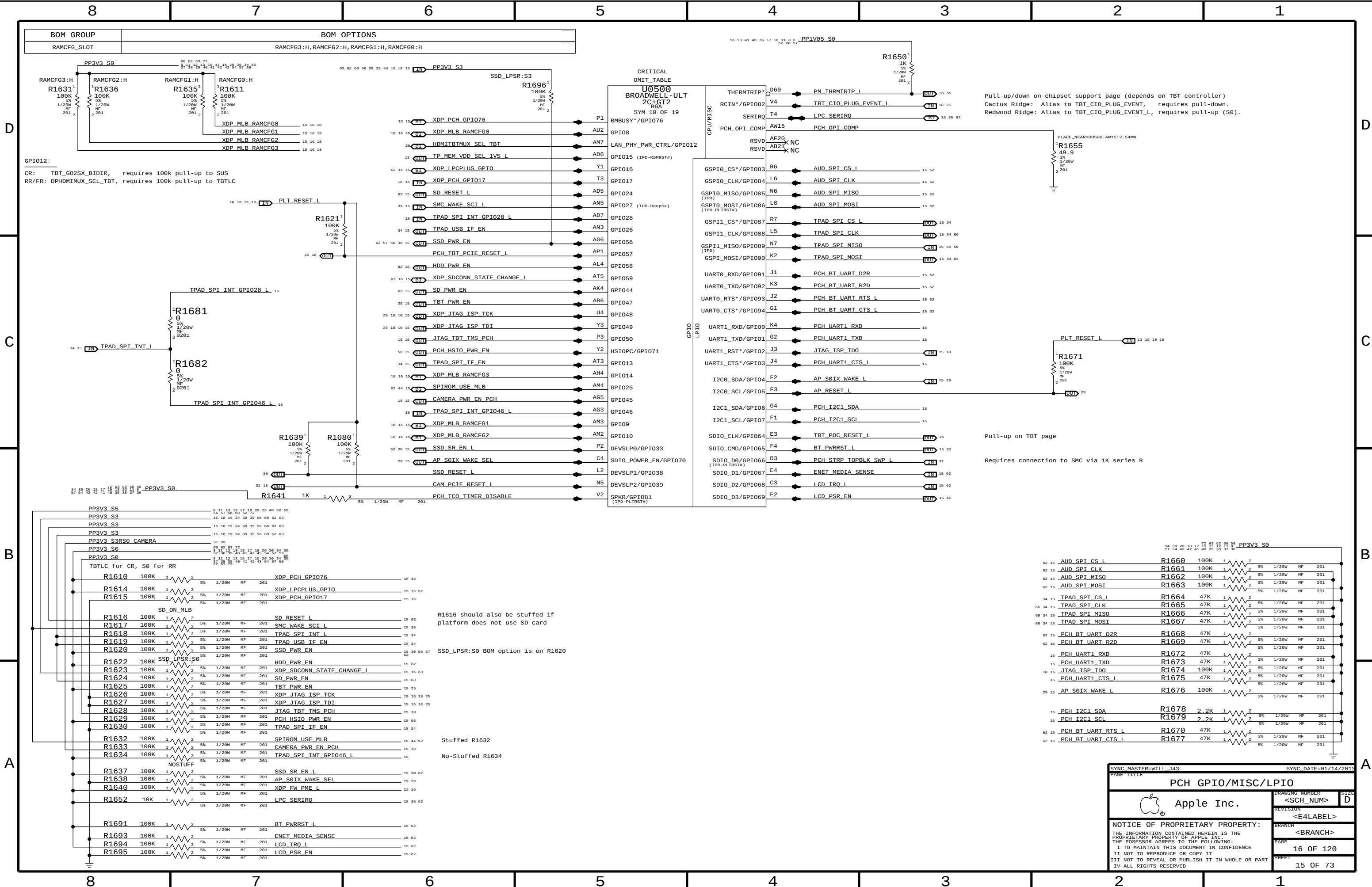
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SHEET
13 OF 73

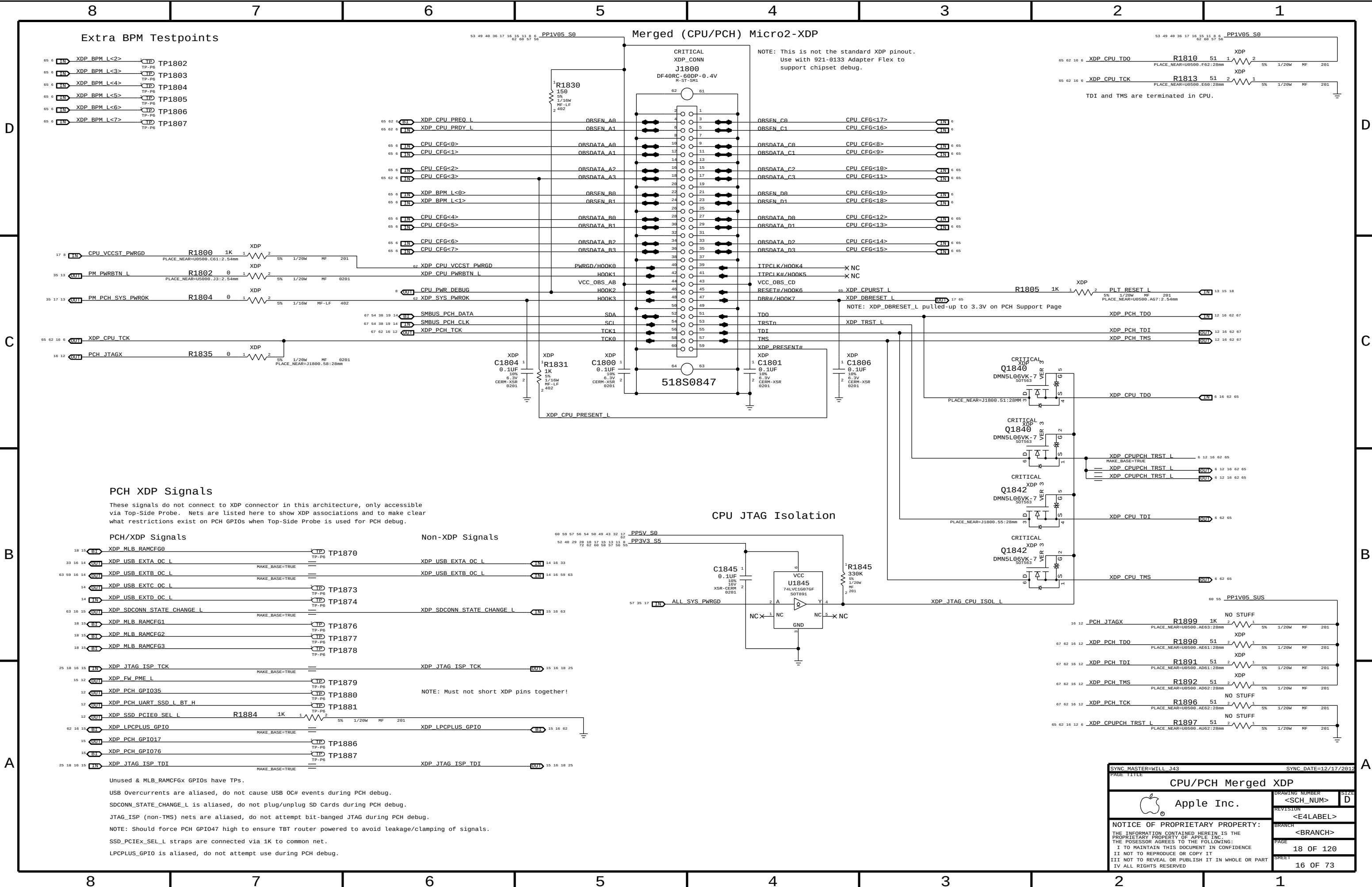
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Pull-up/down on chipset support page (depends on TBT controller)
Cactus Ridge: Alias to TBT_CIO_PLUG_EVENT, requires pull-down.
Redwood Ridge: Alias to TBT_CIO_PLUG_EVENT_L, requires pull-up (S0).

Pull-up on TBT page
Requires connection to SMC via 1K series R



PCH XDP Signals

These signals do not connect to XDP connector in this architecture, only accessible via Top-Side Probe. Nets are listed here to show XDP associations and to make clear what restrictions exist on PCH GPIOs when Top-Side Probe is used for PCH debug.

PCH/XDP Signals

18 15	BT	XDP_MLB_RAMCFG0	TP-P6	TP1870
33 16 14	OUT	XDP_USB_EXTB_OC_L	MAKE_BASE=TRUE	
63 59 16 14	OUT	XDP_USB_EXTC_OC_L	MAKE_BASE=TRUE	TP-P6
14	OUT	XDP_USB_EXTD_OC_L	TP-P6	TP1873
14	IN	XDP_USB_EXTD_OC_L	TP-P6	TP1874
63 16 15	OUT	XDP_SDCONN_STATE_CHANGE_L	MAKE_BASE=TRUE	
18 15	BT	XDP_MLB_RAMCFG1	TP-P6	TP1876
18 15	BT	XDP_MLB_RAMCFG2	TP-P6	TP1877
18 15	BT	XDP_MLB_RAMCFG3	TP-P6	TP1878
25 18 16 15	IN	XDP_JTAG_ISP_TCK	MAKE_BASE=TRUE	
15 12	OUT	XDP_FW_PME_L	TP-P6	TP1879
12	OUT	XDP_PCH_GPIO35	TP-P6	TP1880
12	OUT	XDP_PCH_UART_SSD_L_BT_H	TP-P6	TP1881
12	OUT	XDP_SSD_PCIE0_SEL_L	R1884 1K 1 2	
62 16 15	BT	XDP_LPCPLUS_GPIO	MAKE_BASE=TRUE	
10	OUT	XDP_PCH_GPIO17	TP-P6	TP1886
15	BT	XDP_PCH_GPIO76	TP-P6	TP1887
25 18 16 15	IN	XDP_JTAG_ISP_TDI	MAKE_BASE=TRUE	

Unused & MLB_RAMCFGx GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

SDCONN_STATE_CHANGE_L is aliased, do not plug/unplug SD cards during PCH debug.

JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.

NOTE: Should force PCH GPIO1047 high to ensure TBT router powered to avoid leakage/clamping of signals.

SSD_PCIE_SEL_L straps are connected via 1K to common net.

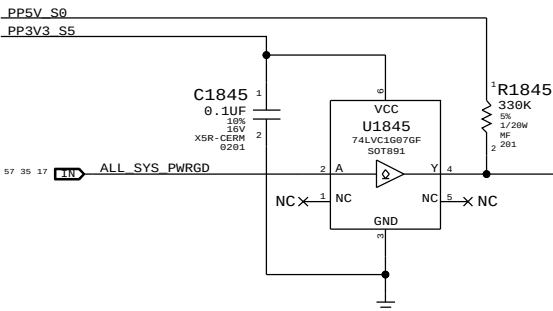
LPCPLUS_GPIO is aliased, do not attempt use during PCH debug.

Non-XDP Signals

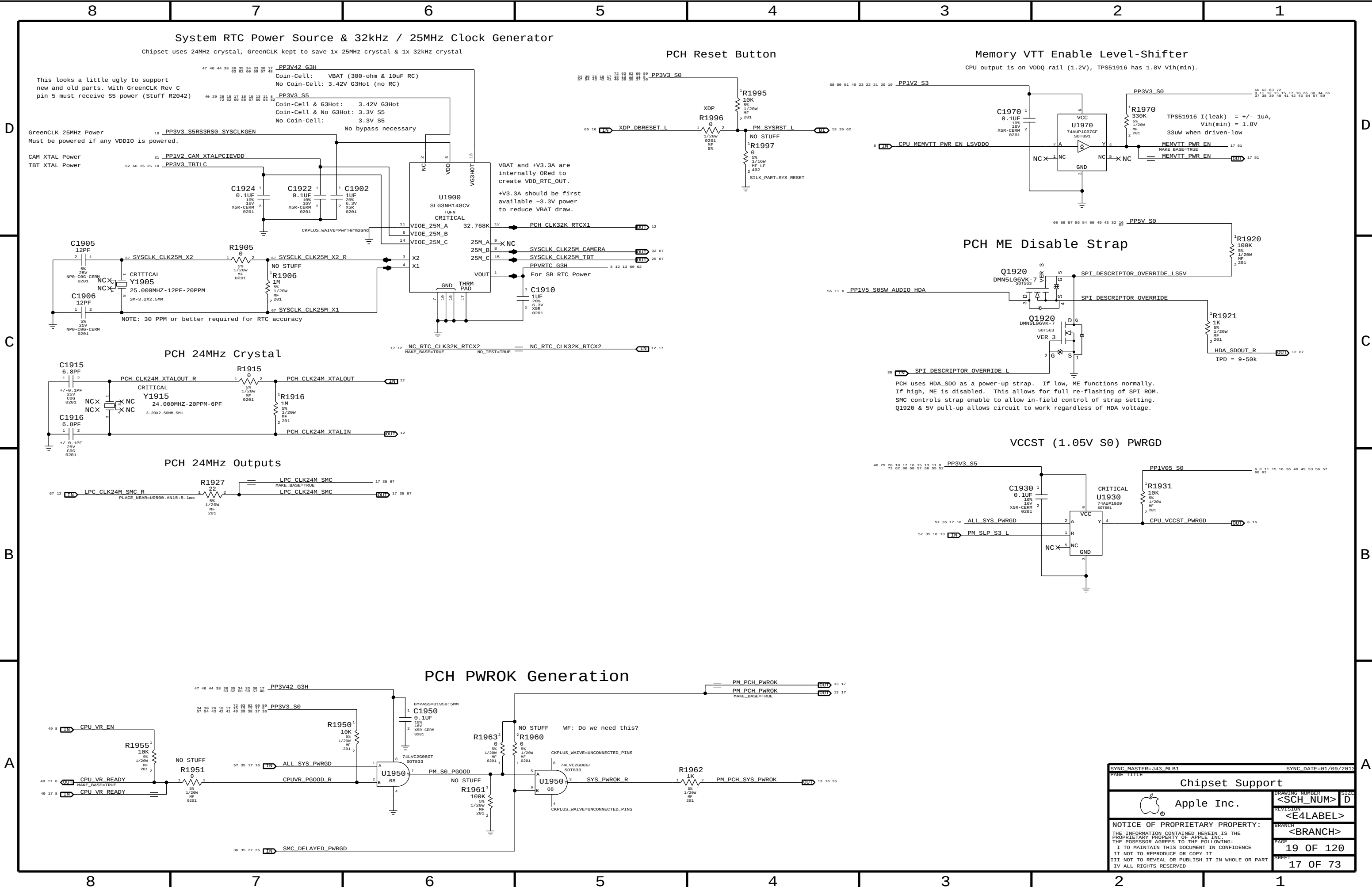
14 16 33	IN	XDP_USB_EXTB_OC_L	
14 16 59 63	IN	XDP_USB_EXTC_OC_L	
15 16 63	IN	XDP_SDCONN_STATE_CHANGE_L	
15 16 25	OUT	XDP_JTAG_ISP_TCK	
15 16 18 25	OUT	XDP_JTAG_ISP_TDI	

NOTE: Must not short XDP pins together!

CPU JTAG Isolation



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		<BRANCH>	
		PAGE	18 OF 120
		SHEET	16 OF 73



System RTC Power Source & 32kHz / 25MHz Clock Generator

Chipset uses 24MHz crystal, GreenCLK kept to save 1x 25MHz crystal & 1x 32kHz crystal

This looks a little ugly to support new and old parts. With GreenCLK Rev C pin 5 must receive S5 power (Stuff R2042)

GreenCLK 25MHz Power
Must be powered if any VDDIO is powered.

CAM XTAL Power

TBT XTAL Power

PCH Reset Button

Memory VTT Enable Level-Shifter

CPU output is on VDDQ rail (1.2V), TPS51916 has 1.8V Vih(min).

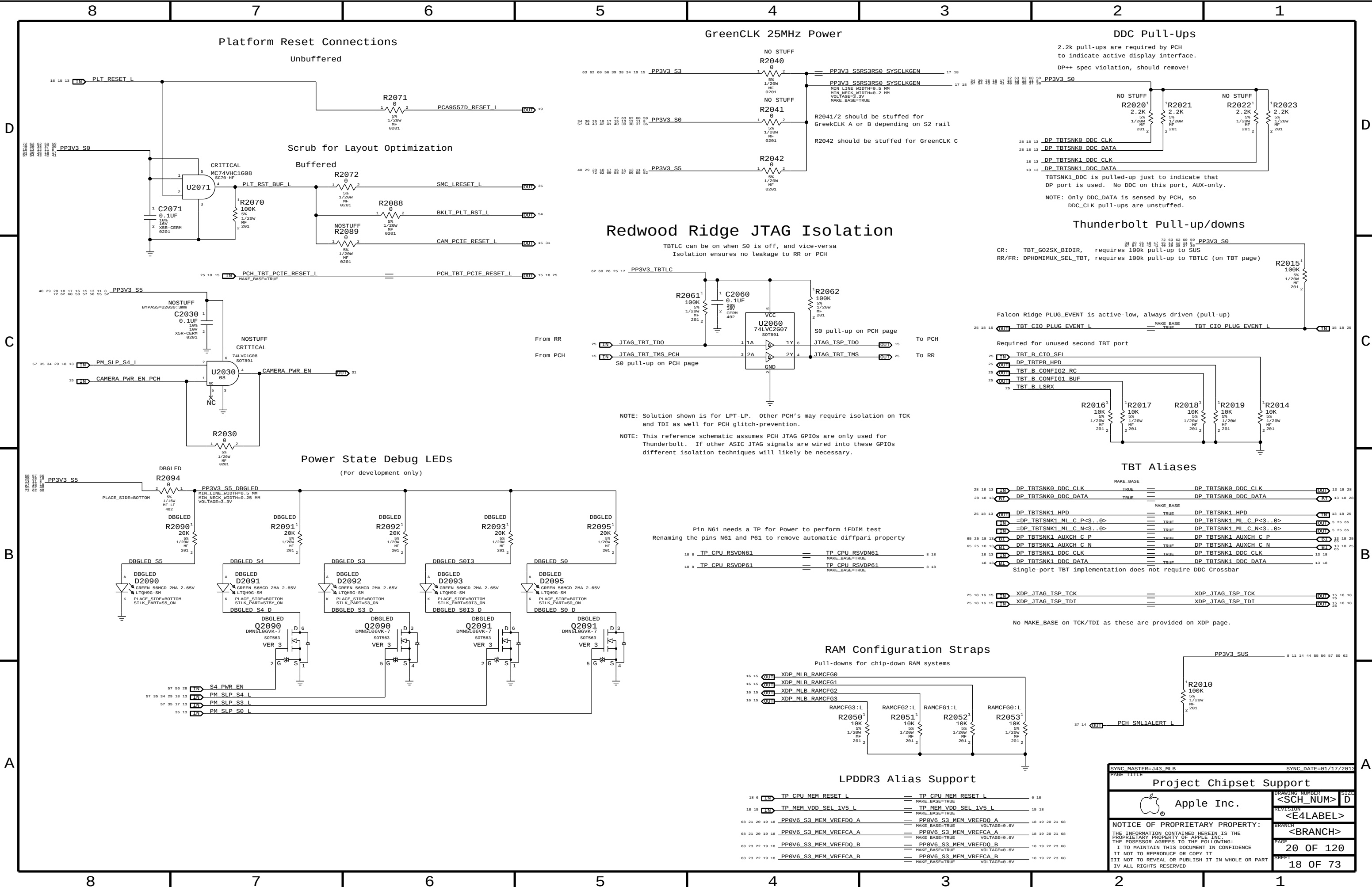
PCH ME Disable Strap

PCH uses HDA_SDO as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. Q1920 & 5V pull-up allows circuit to work regardless of HDA voltage.

VCCST (1.05V S0) PWRGD

PCH PWROK Generation

SYNC MASTER=143 MLB1		SYNC DATE=01/09/2013	
PAGE TITLE			
Chipset Support			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	SIZE D
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		BRANCH <BRANCH>	
		PAGE 19 OF 120	
		SHEET 17 OF 73	



SYNC MASTER=143_MLB

SYNC DATE=01/17/2013

PAGE TITLE

Project Chipset Support

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<SCH_NUM>

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PAGE

20 OF 120

SHEET

18 OF 73

Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

- DDRVREF_DAC - Stuffs DAC margining circuit.

CPU-Based Margining

FETs for CPU isolation during DAC margining

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step
LPDDR3 (1.2V) 7.77mV per step

NOTE: CPU has single output for VREFCA. Split into two signals for independent DAC margining support. When DAC margining VREFCA ensure VREFMRGN_CPU_EN is low to remove short due to CPU.

DAC-Based Margining

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.

OMIT

PP3V3_S3

R2218 SHORT

PP3V3_S3_VREFMRGN_DAC

DDRREF_DAC C2200 2.2UF

DDRREF_DAC C2201 0.1UF

SMBUS_PCH_CLK

SMBUS_PCH_DATA

Addr=0x98(WR)/0x99(RD)

VDD

VOUTA

VOUTB

VOUTC

VOUTD

VREFMRGN_DQ A

VREFMRGN_DQ B

VREFMRGN_CA AB

VREFMRGN_MEMVREG

NOTE: MEMVREG and SPARE share a DAC output, cannot enable both at the same time!

DDRREF_DAC C2202 0.1UF

CRITICAL DDRREF_DAC

U2201 PCA9557

Addr=0x30(WR)/0x31(RD)

SMBUS_PCH_CLK

SMBUS_PCH_DATA

RST* on 'platform reset' so that system watchdog will disable margining.

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

PCA9557D_RESET_L

VRef Dividers

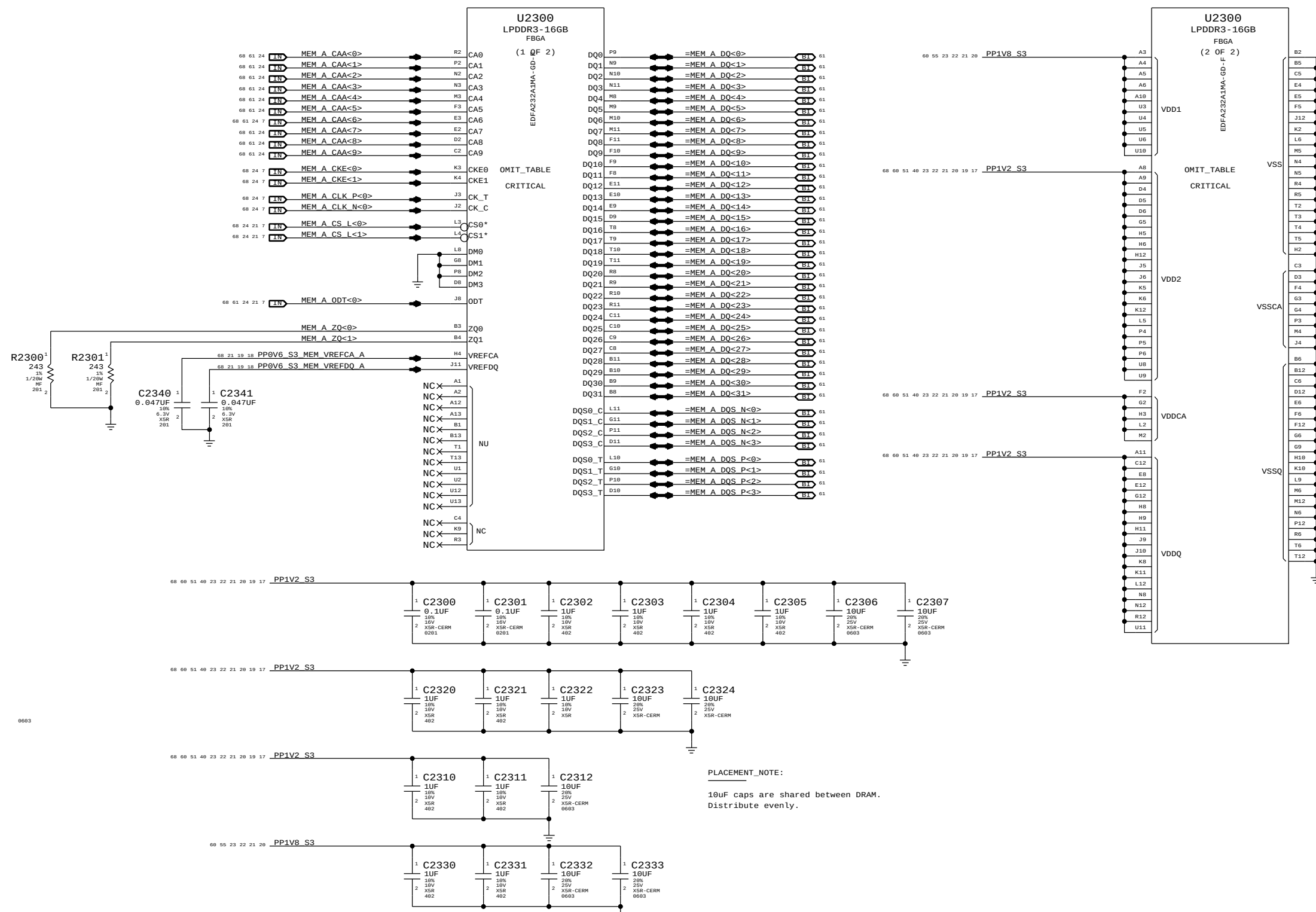
Always used, regardless of margining option.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
Nominal value	LPDDR3 (1.2V) 0.600V (DAC: 0x2E.5)		DDR3L (1.35V) 0.675V (DAC: 0x34)		LPDDR3 (1.2V) 1.200V (DAC: 0x5D)
Margin target:	0.300V - 0.900V (+/- 300mV)		0.337V - 1.013V (+/- 337.5mV)		0.800V - 1.600V (+/- 400mV)
DAC range:	0.000V - 1.199V (0x00 - 0x5D)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 2.397V (0x00 - 0xBA)
VRef current:	+73uA - -73uA (- = sourced)		+82uA - -82uA (- = sourced)		+25uA - -25uA (- = sourced)
DAC step size:	6.36mV / step @ output		6.36mV / step @ output		4.28mV / step @ output

NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

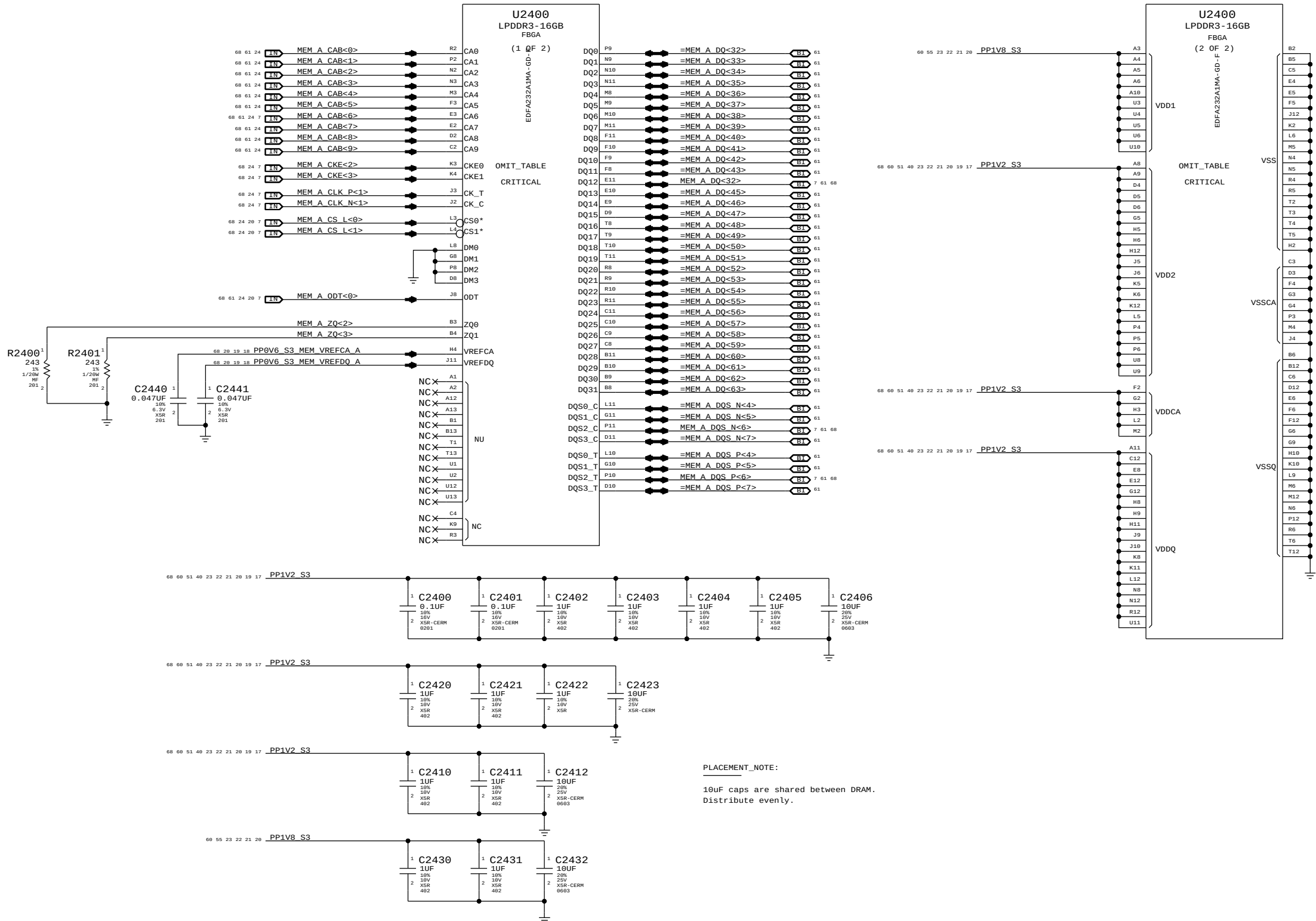
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PAGE		22 OF 120	
SHEET		19 OF 73	
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LPDDR3 CHANNEL A (0-31)

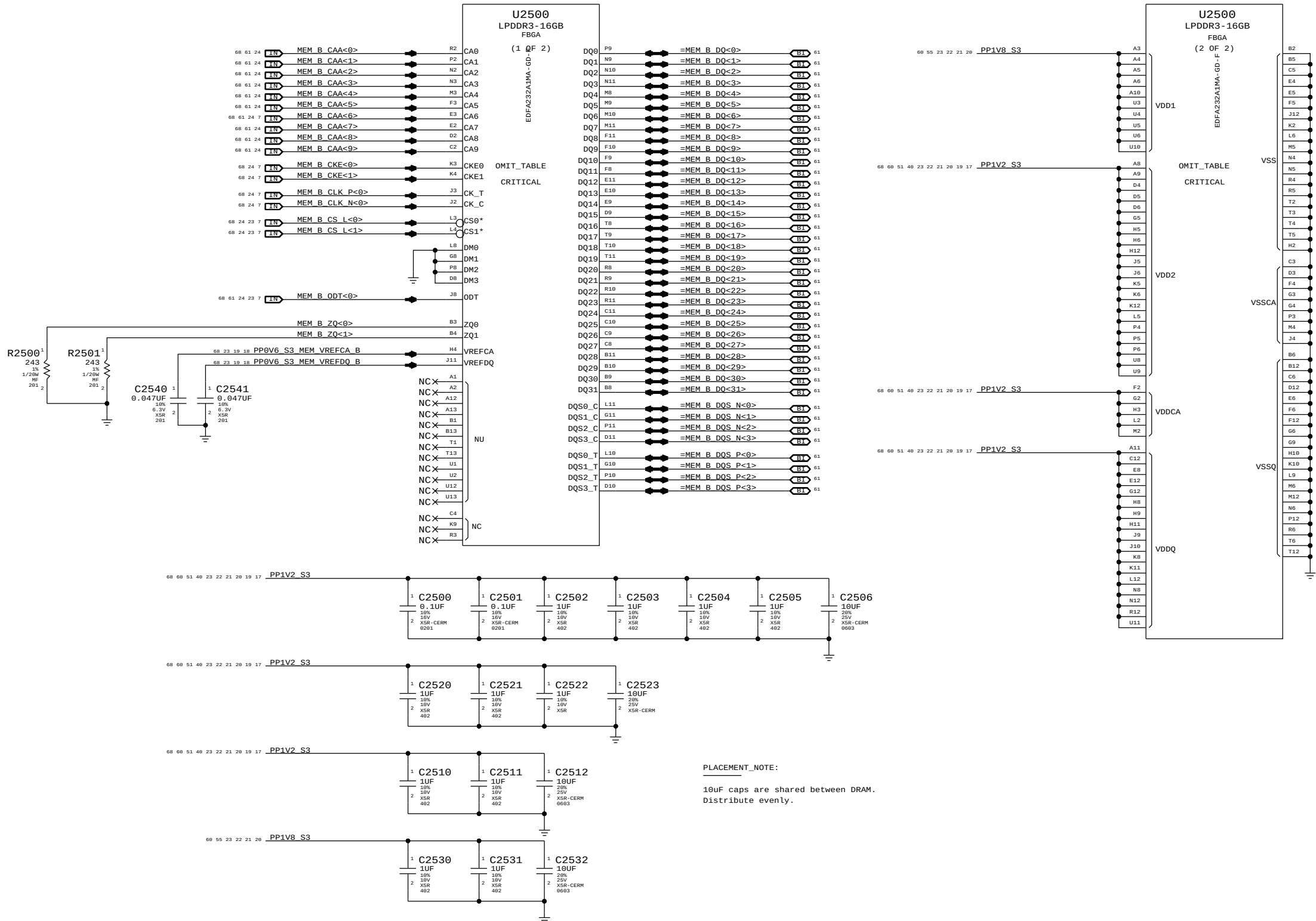


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		PAGE 23 OF 120	
		SHEET 20 OF 73	

LPDDR3 CHANNEL A (32-63)



LPDDR3 CHANNEL B (0-31)



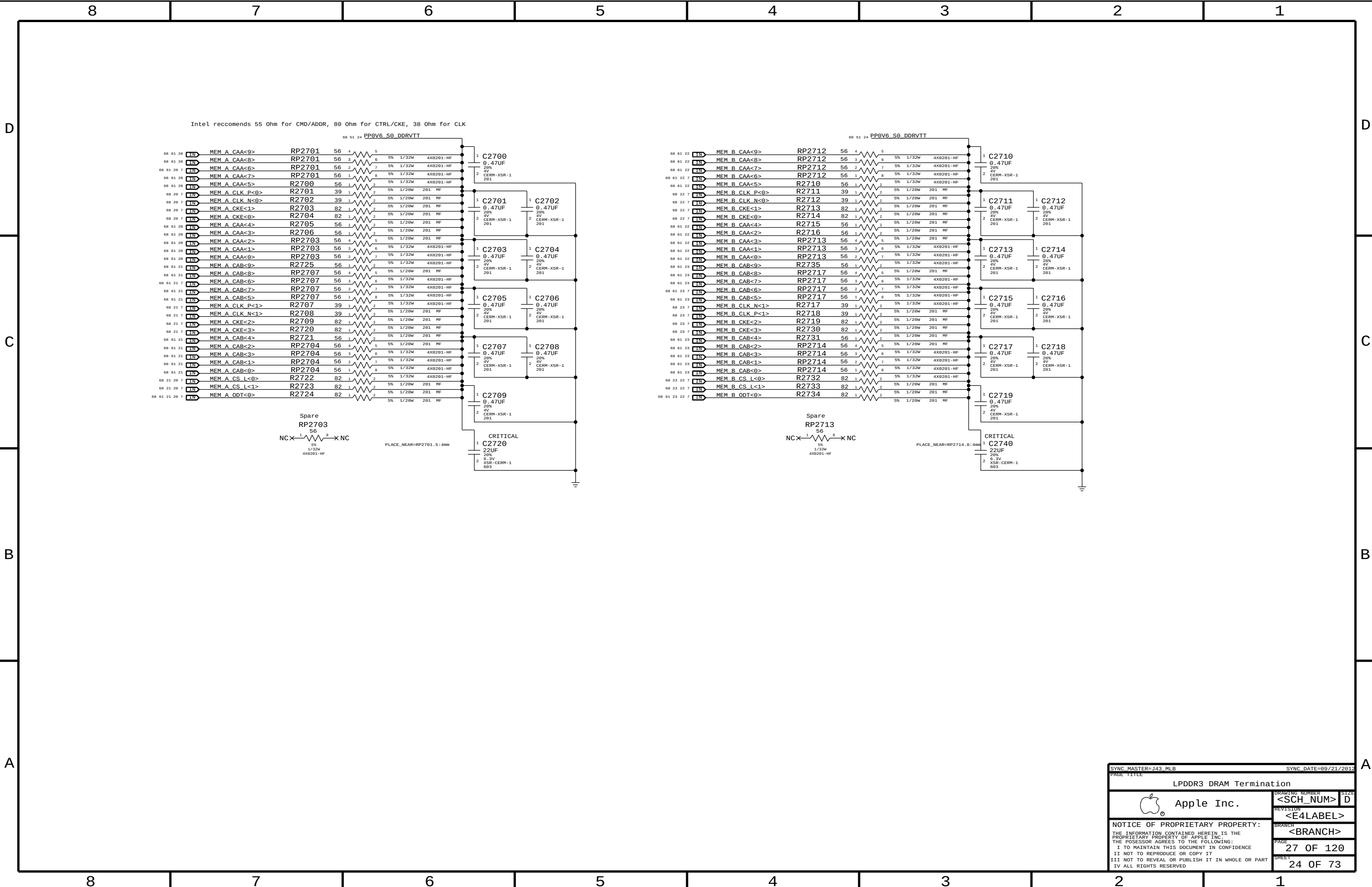
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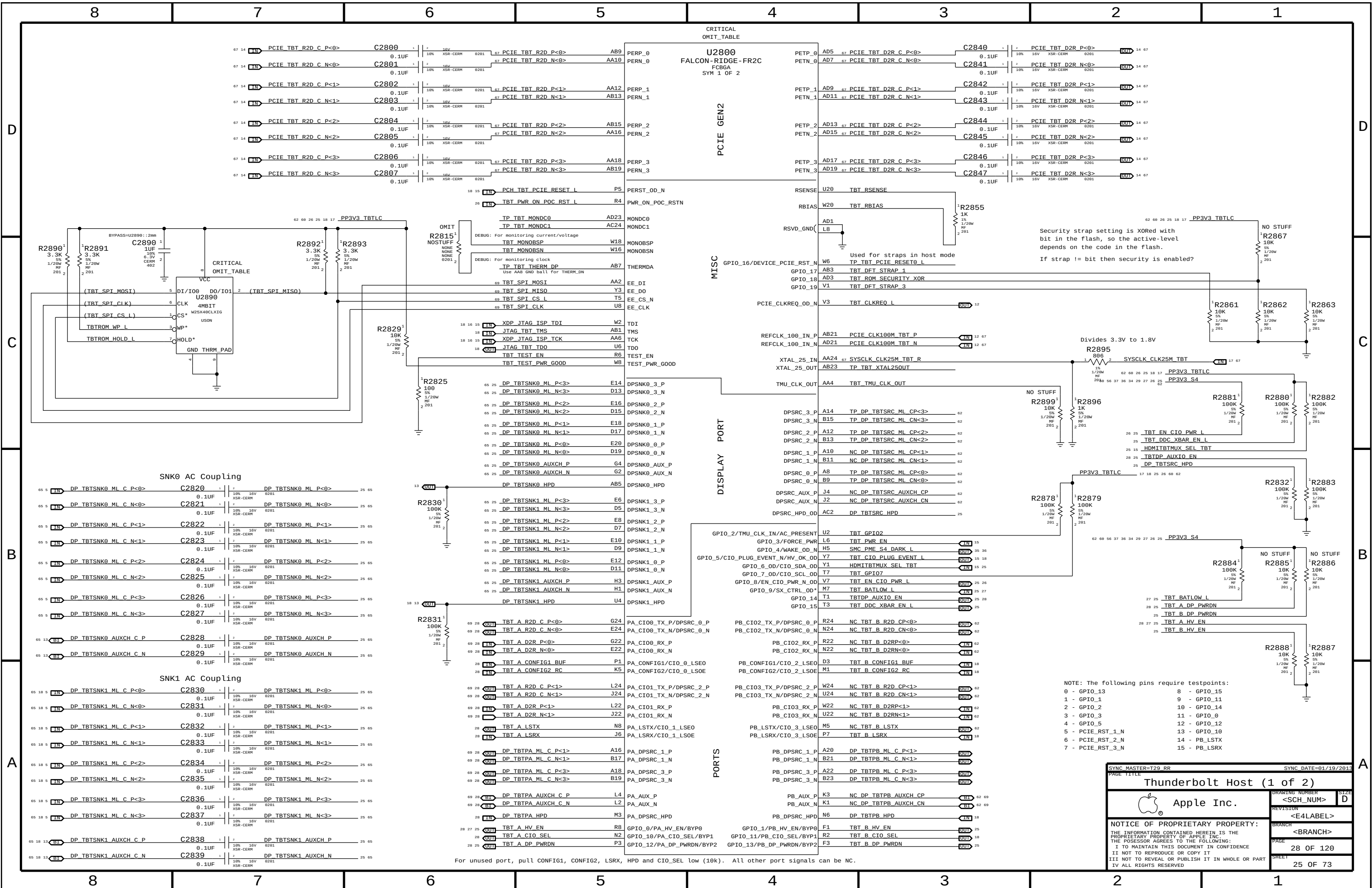


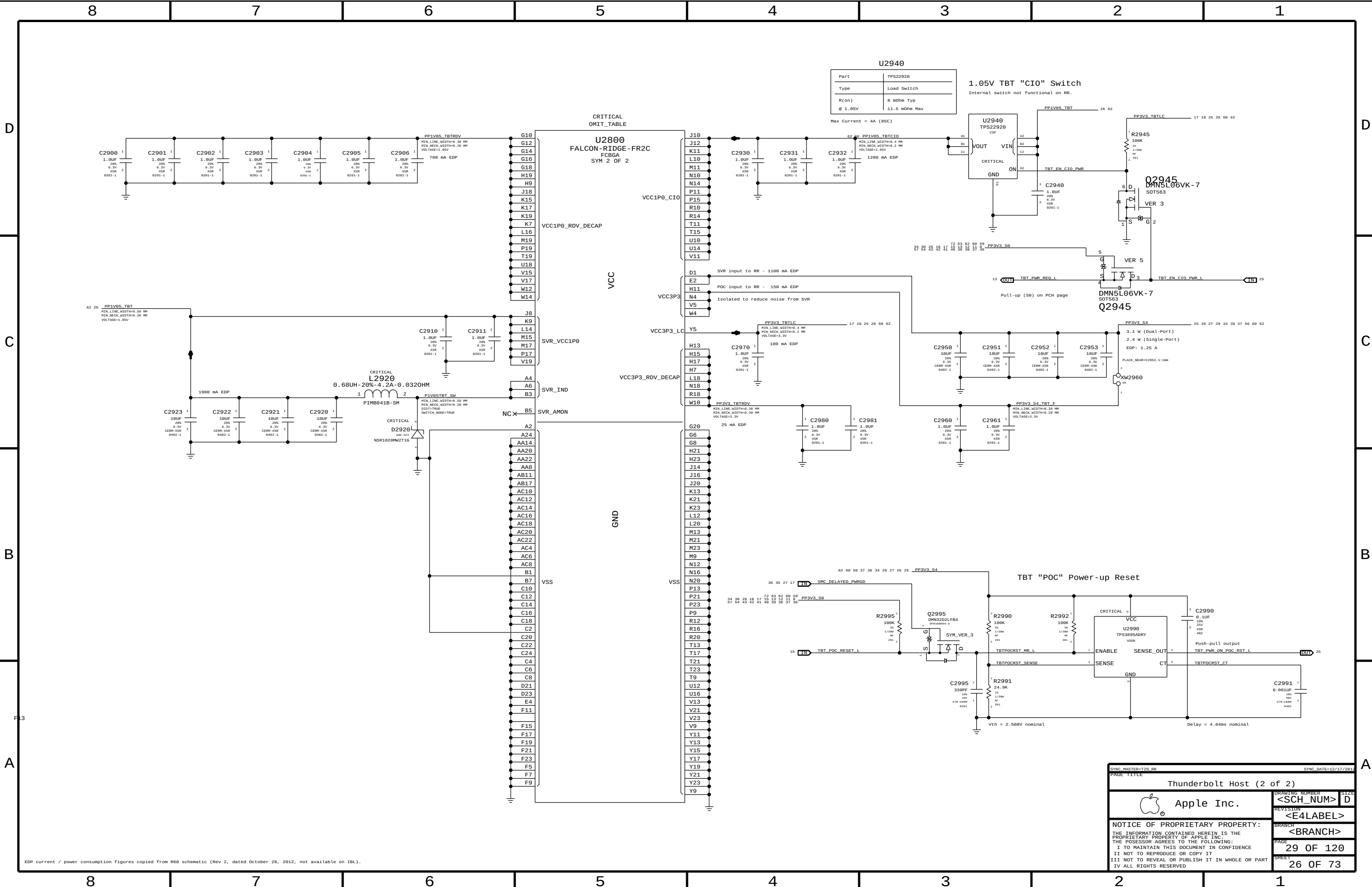
A

DCB

Δ







EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

SYNC MASTER=T20_RR

SYNC DATE=12/17/2015

Thunderbolt Host (2 of 2)

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PAGE29 OF 120

SHEET26 OF 73

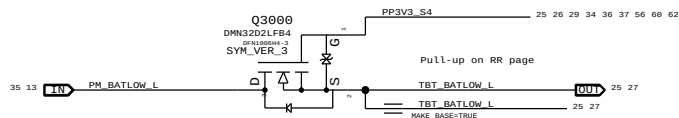
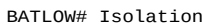
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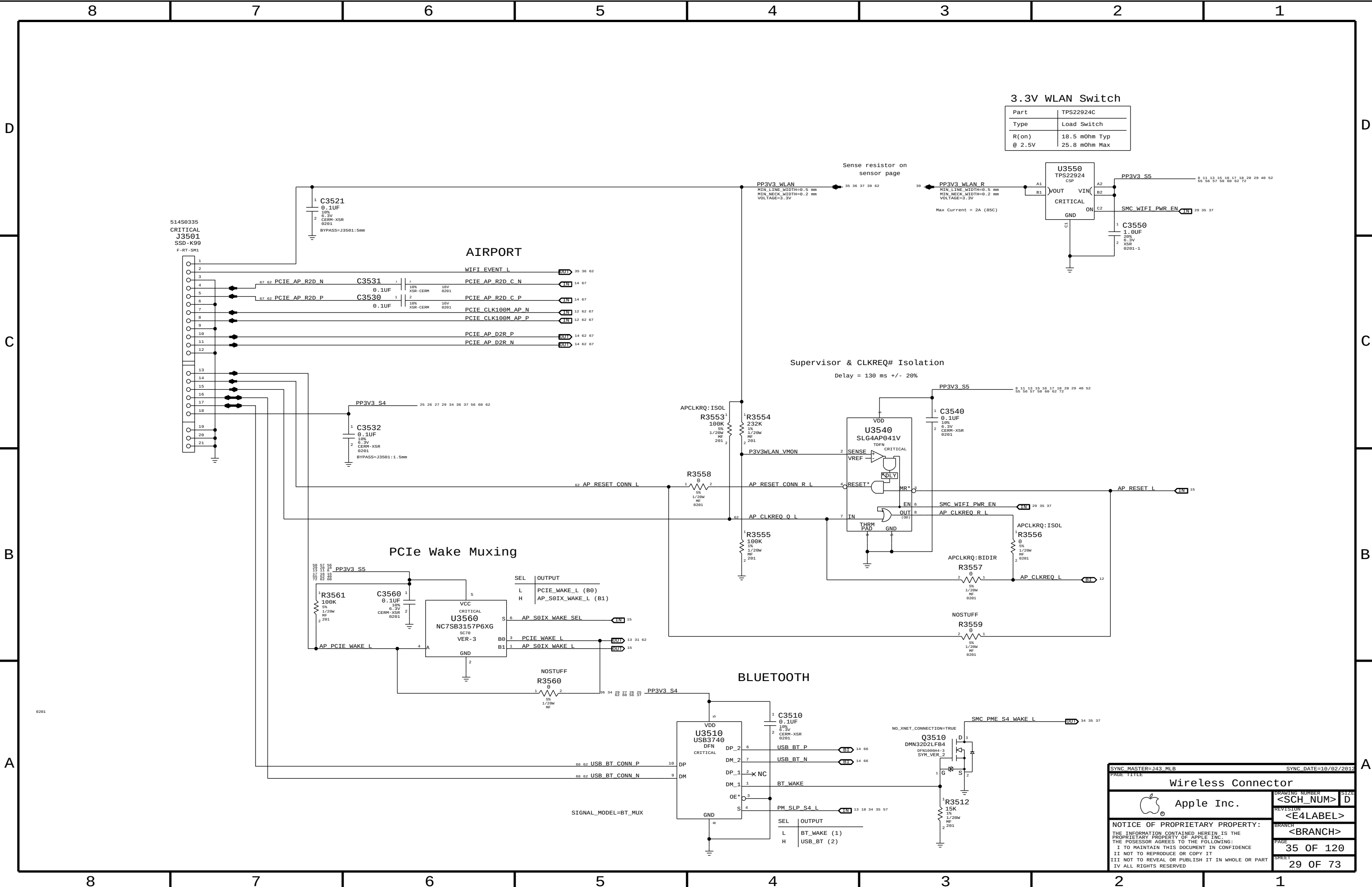
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3.3V WLAN Switch

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max

SYNC MASTER=J43 MLB

SYNC DATE=10/02/2012

Wireless Connector

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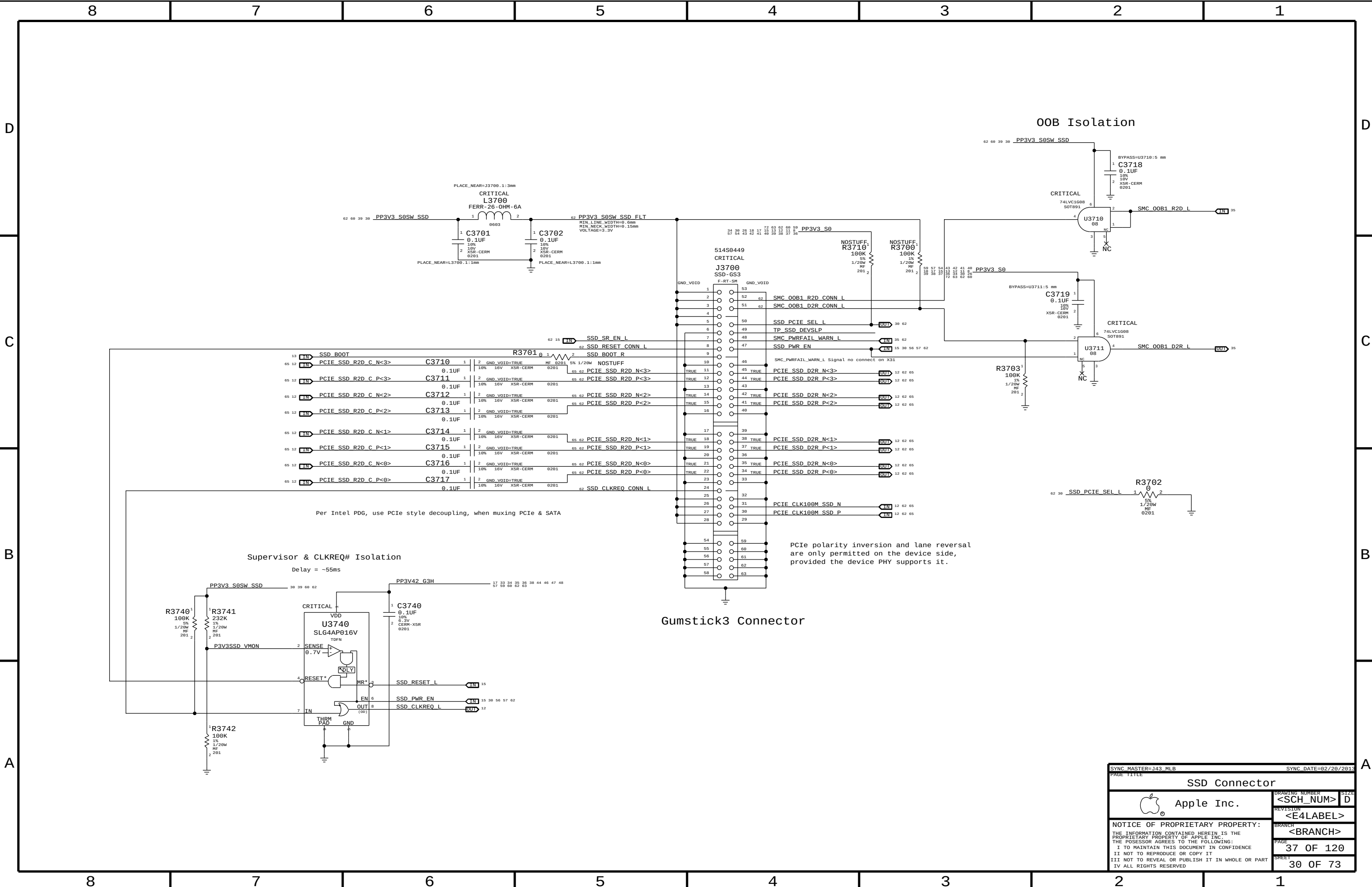
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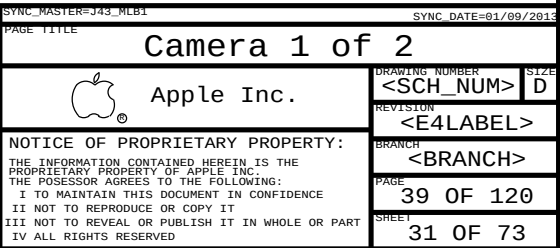
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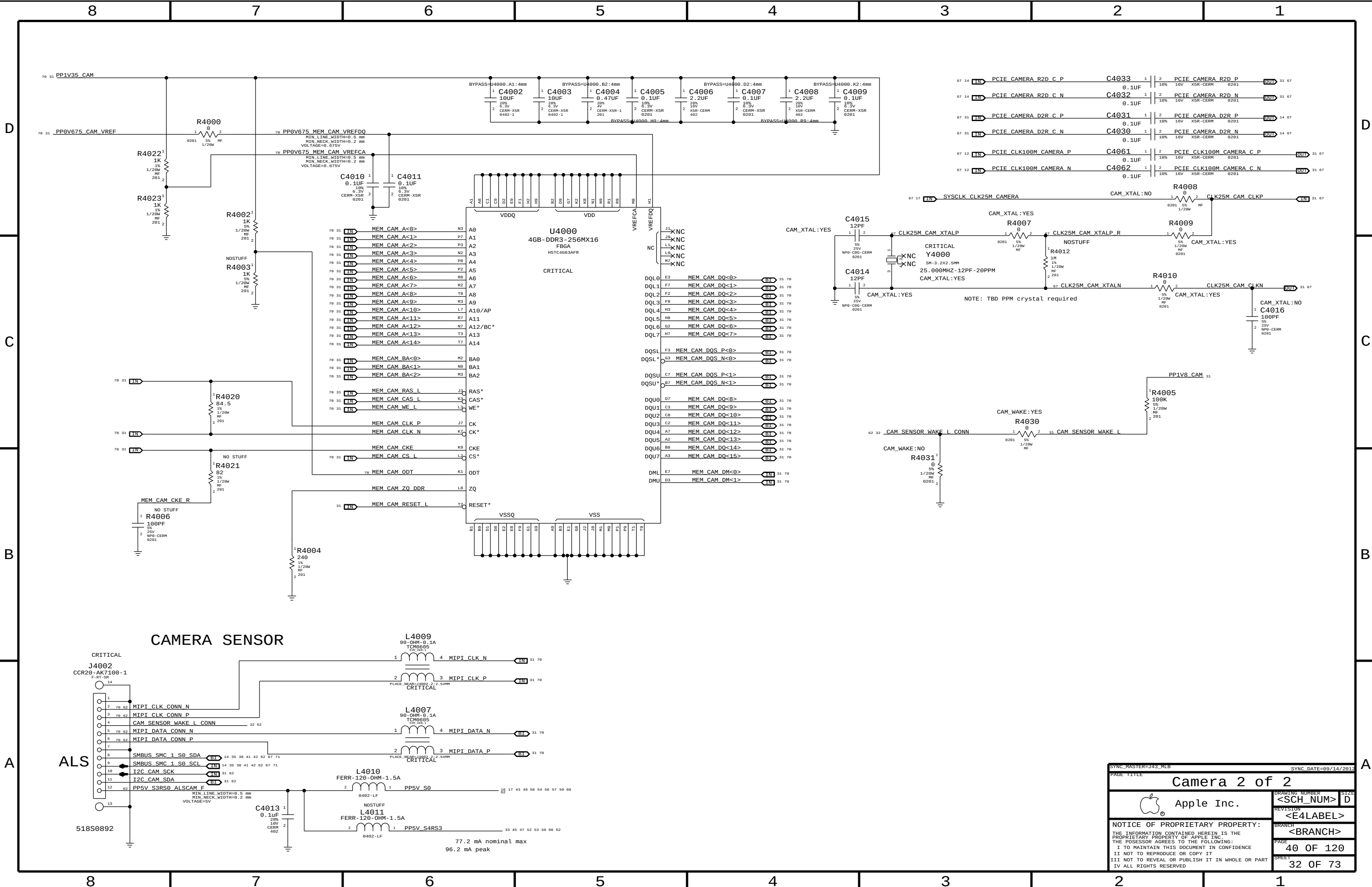
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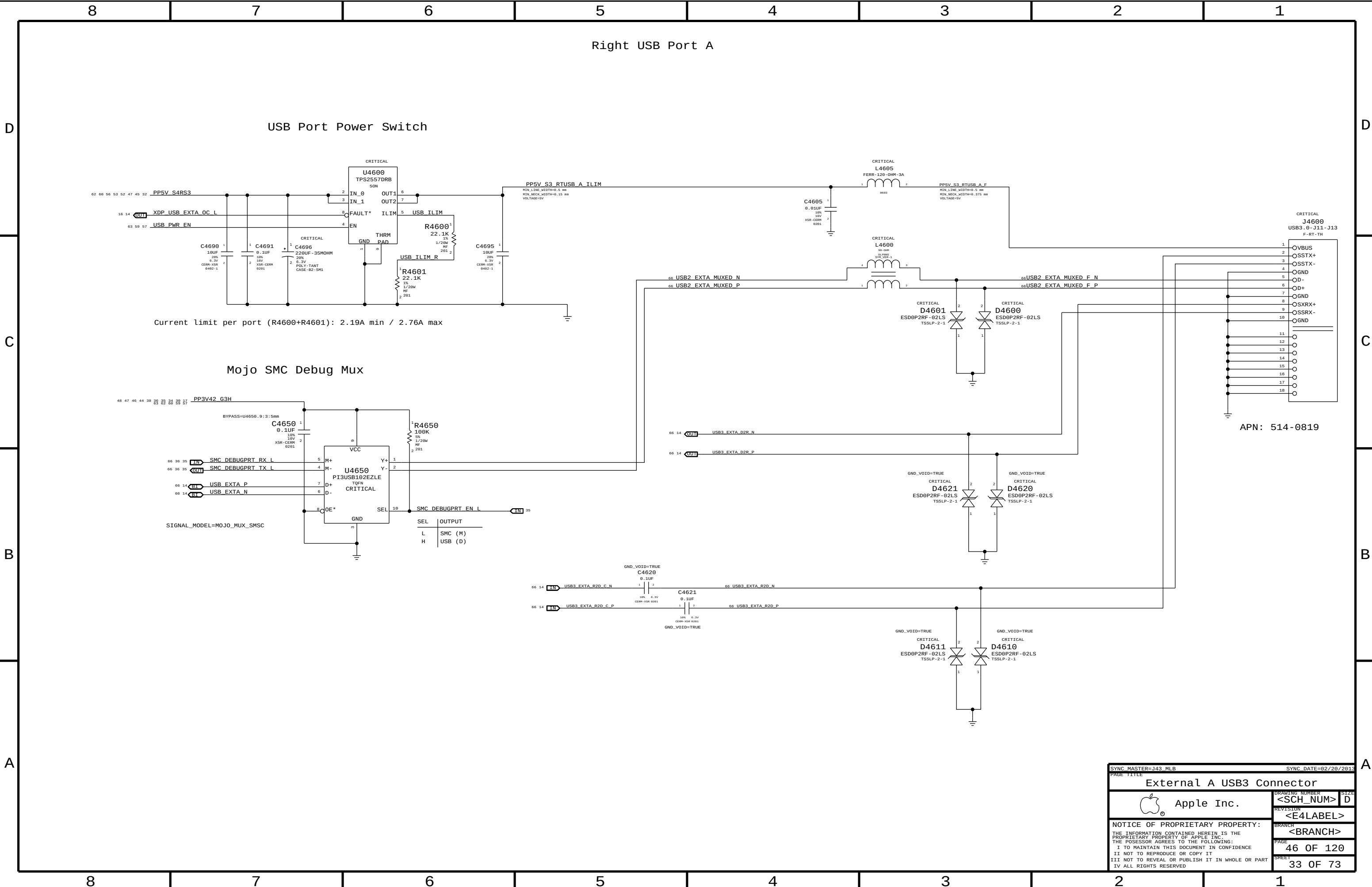
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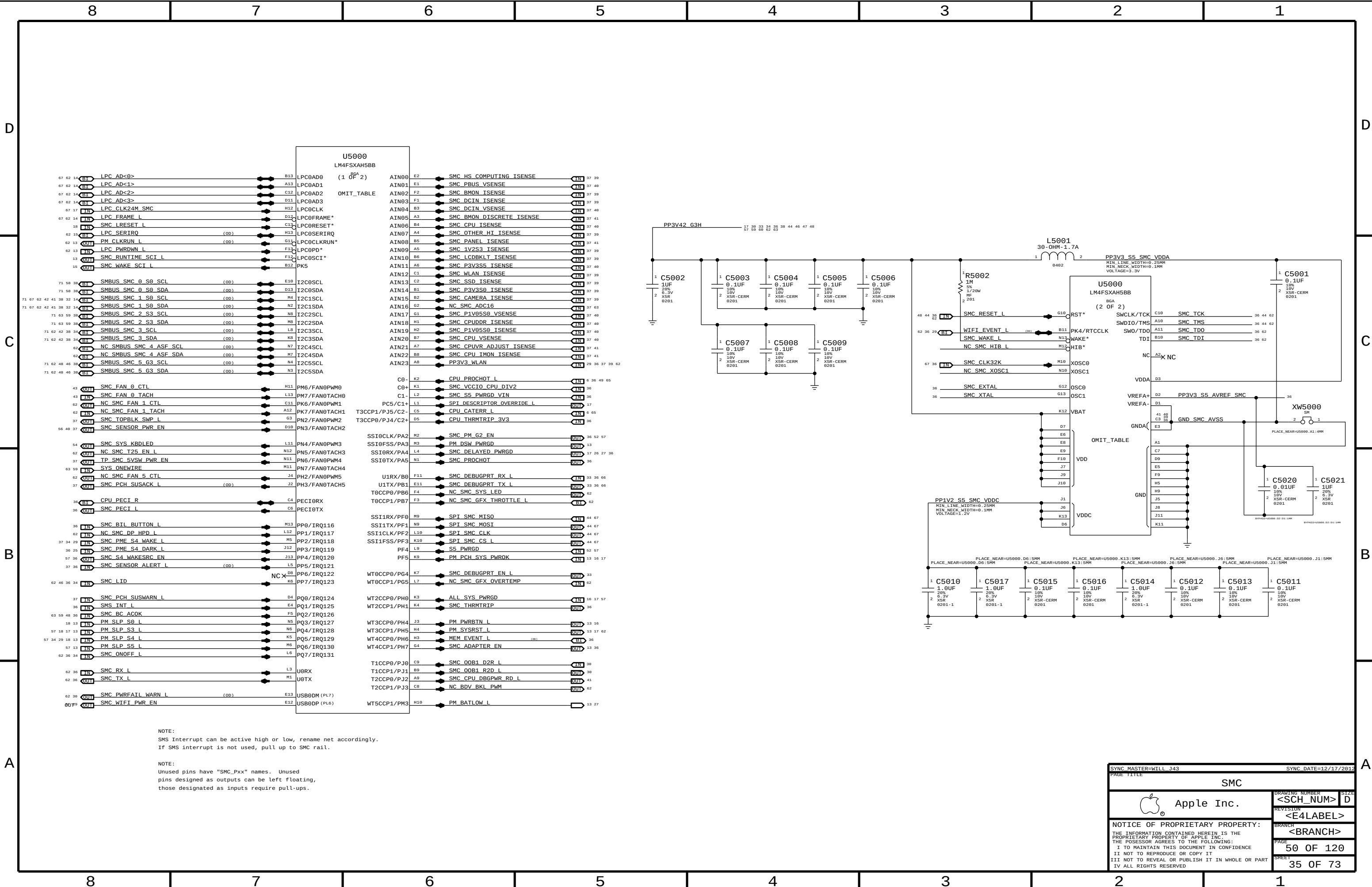
29 OF 73











NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

SYNC MASTER=WILL_343

SYNC DATE=12/17/2012

PAGE TITLE

SMC

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<SCH_NUM>

REVISION

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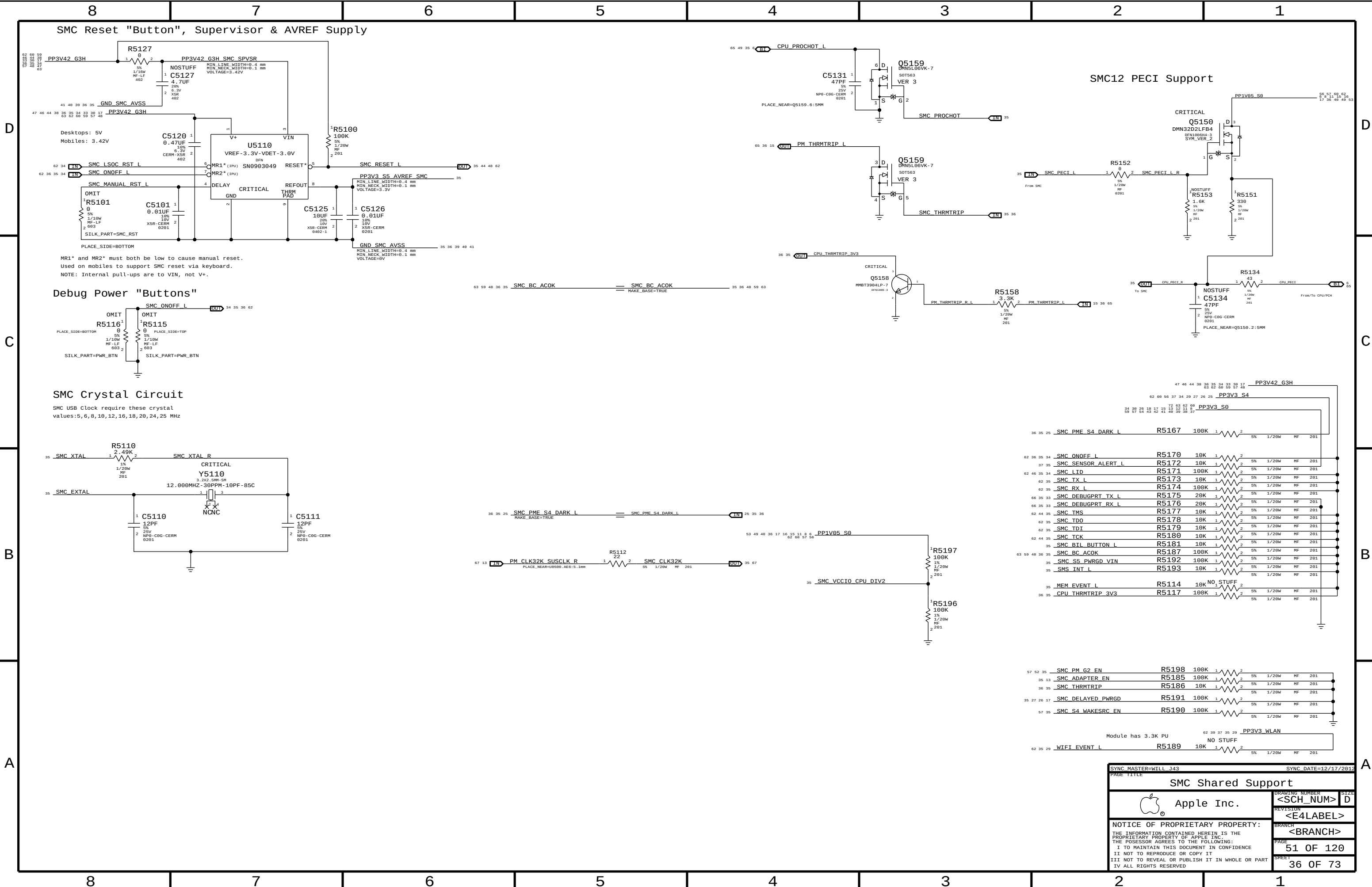
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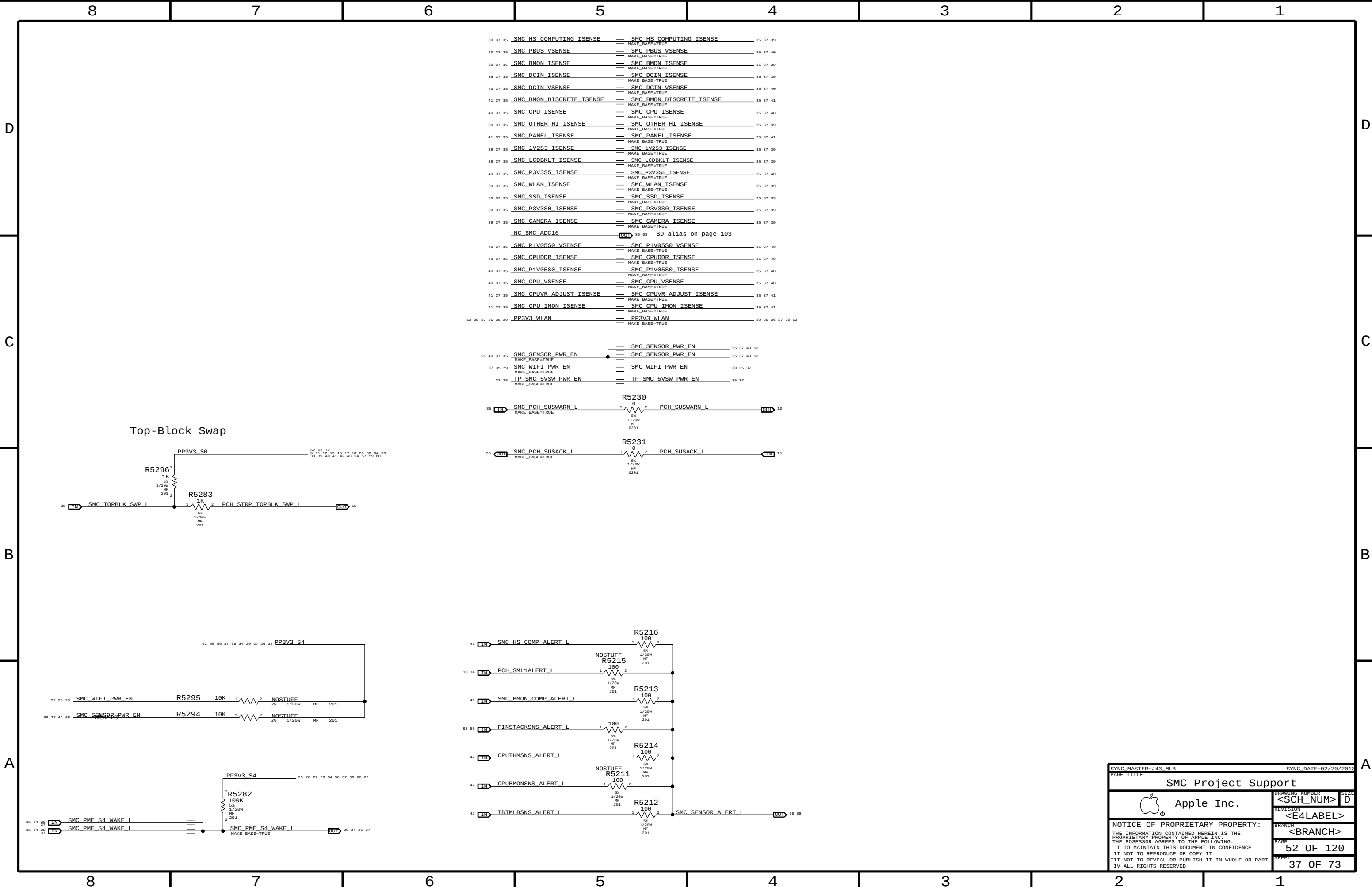
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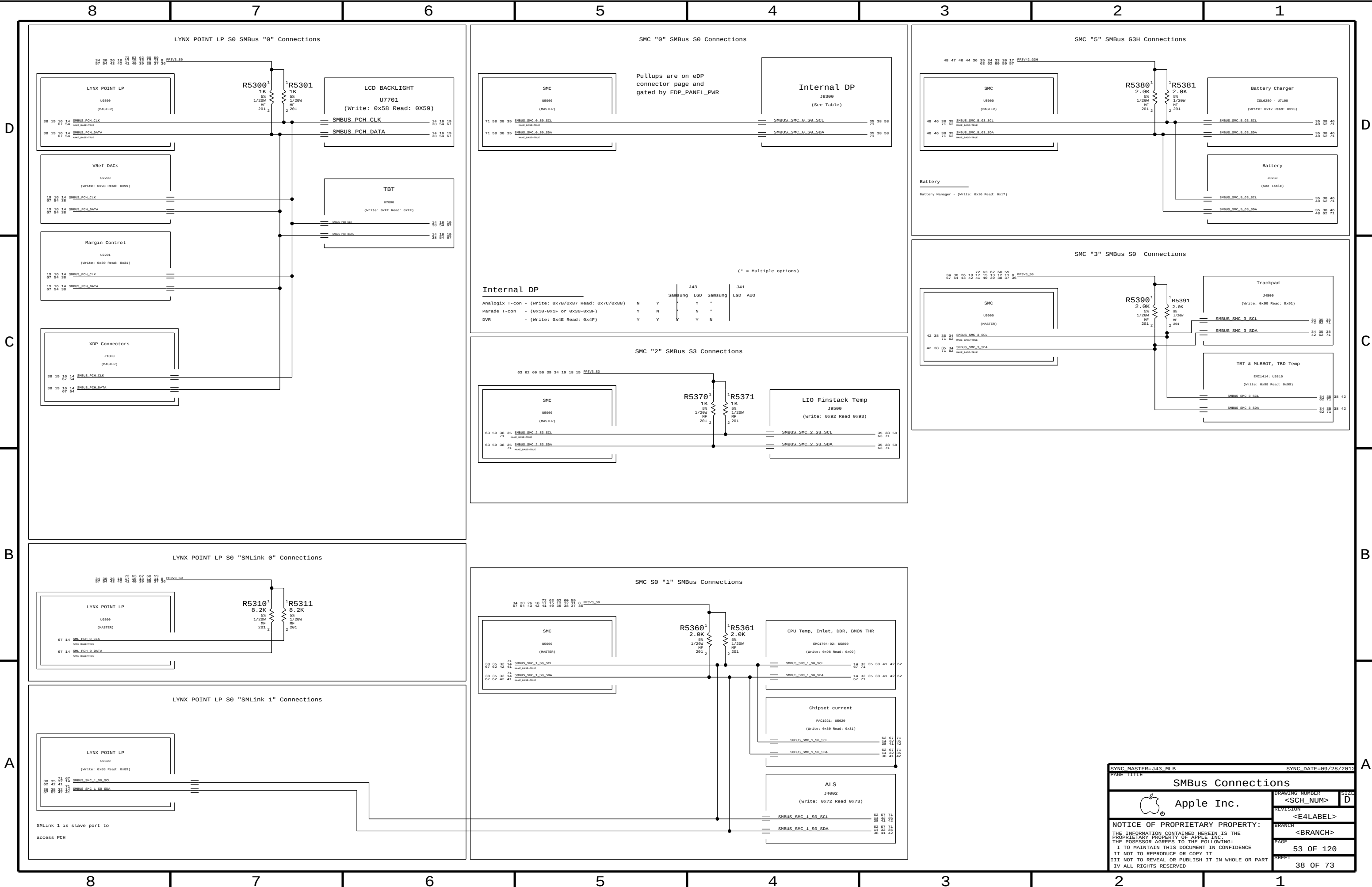
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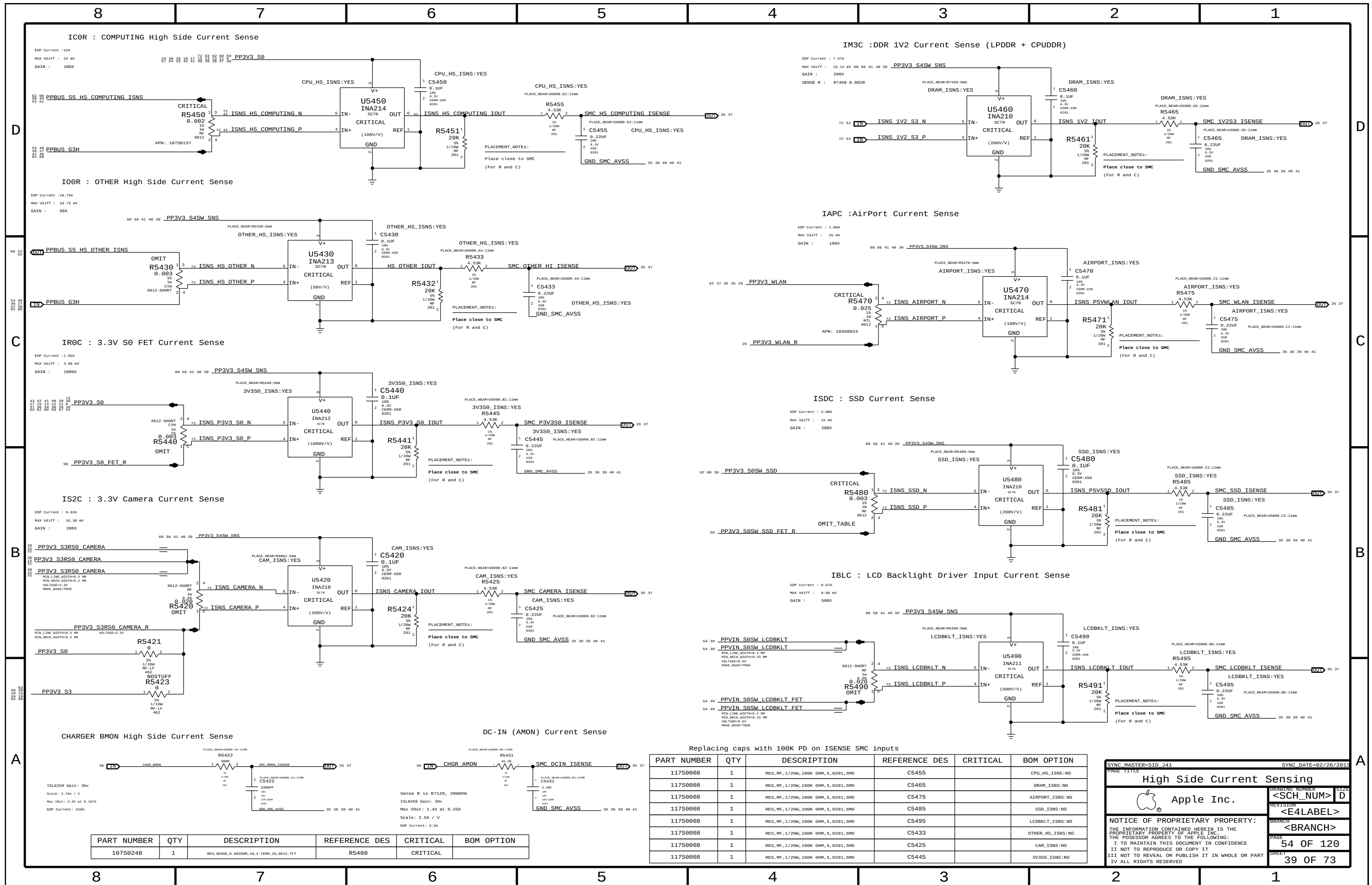
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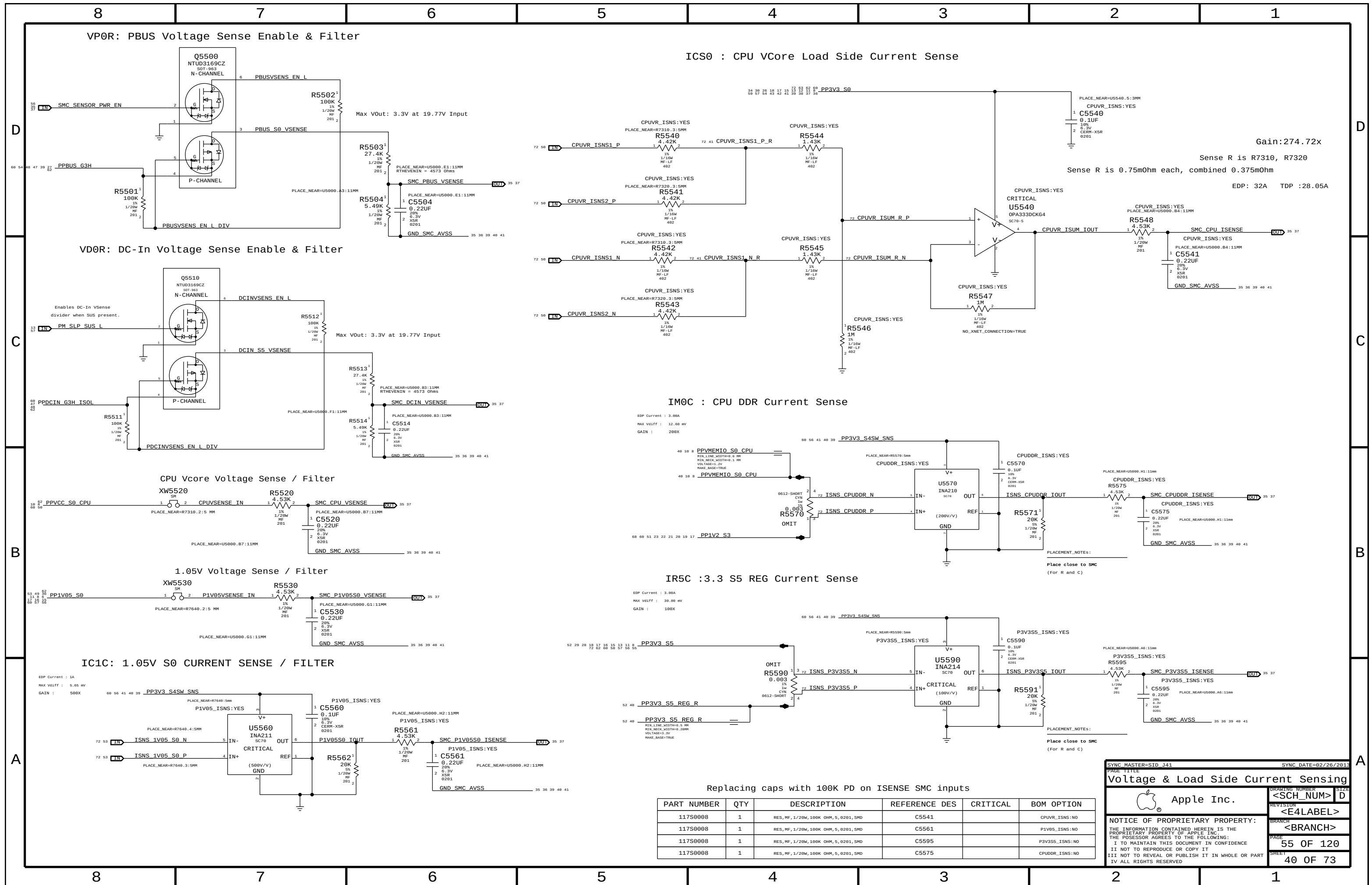
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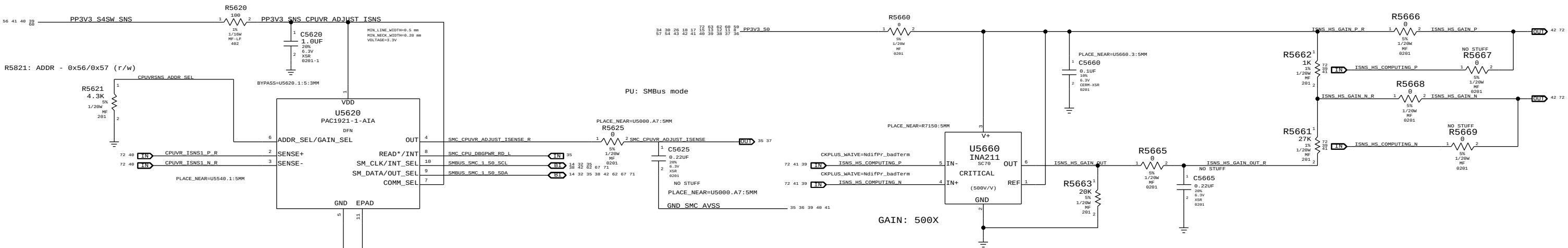




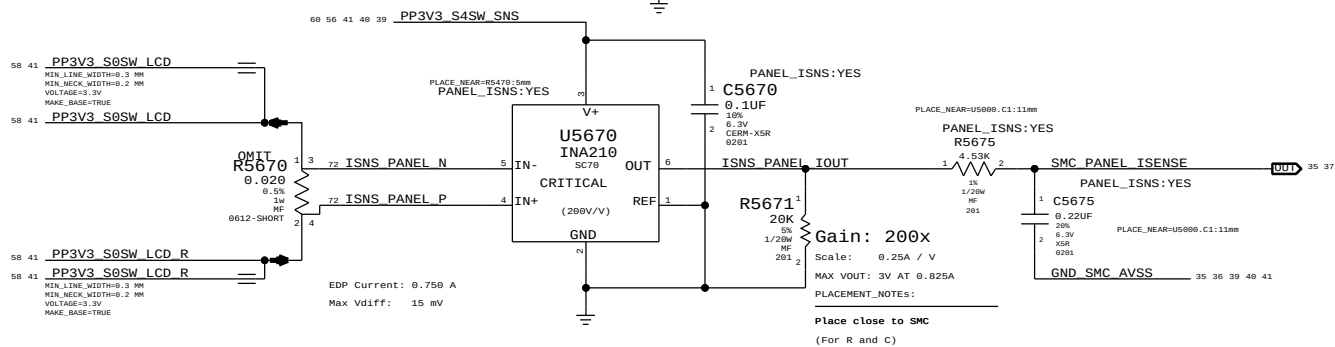


ICS3 : Adjustable Gain CPU VR Current

Sense Pins gain stage for U5800 (EMC1704)



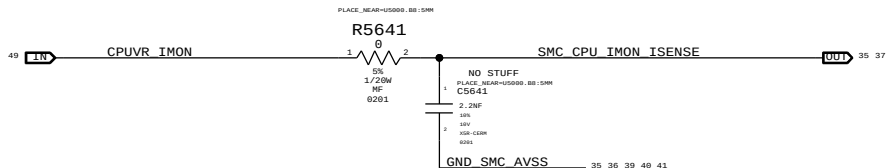
ILDC :LCD Panel Current Sense / Filter



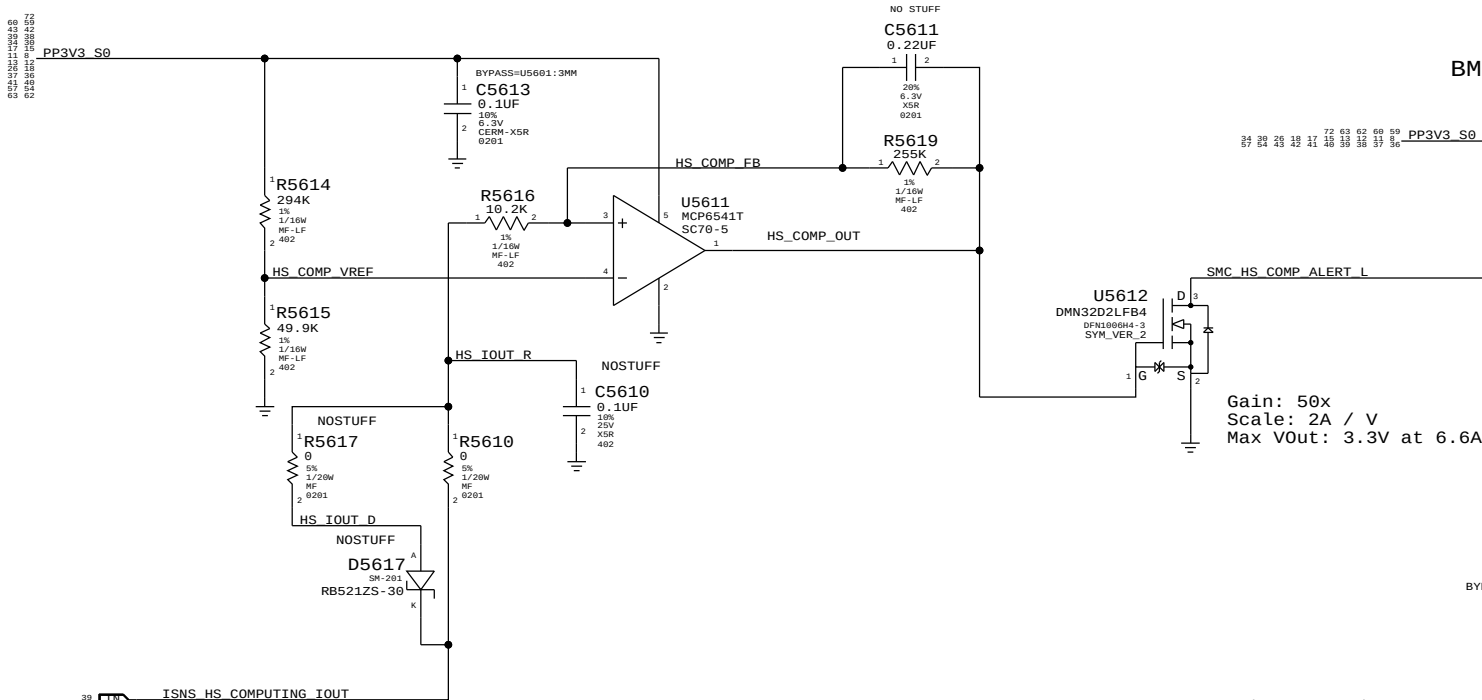
In battery discharge scenario negative voltage will be present on IN+/- pins with INA output voltage decreasing from 3.3V with increasing discharge current.

With 100mA battery current, Will have 10.2mV difference going into sense pins of U5800. This will set the minimum current threshold at 0.100mA

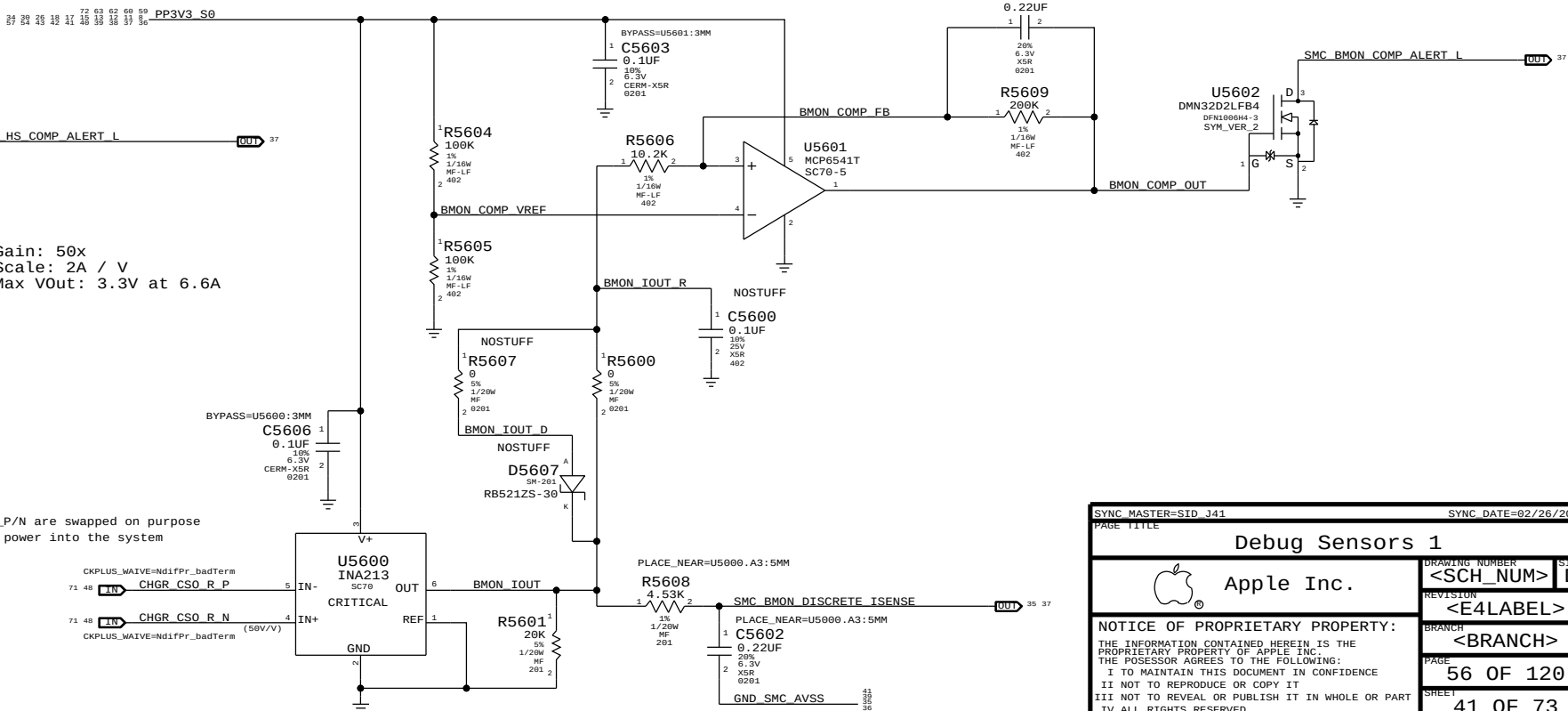
VR IMON Current Sense Filter



Discrete High side Current threshold



BMON : Discrete BMON Current Sense / Filter



Vref = 0.406mV Vth = 0.442 = 1A from Battery
Vtl = 0.290mV = 0.687A from battery
Hysteresis TBD based on RC value changes

Replacing caps with 100K PD on ISENSE SMC inputs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MF, 1/20W, 100K OHM, 5, 0201, SMD	C5675		PANEL_ISNS:NO

SYNC MASTER=SID J41

SYNC DATE=02/26/2013

Apple Inc.

Apple logo

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PAGE

SHEET

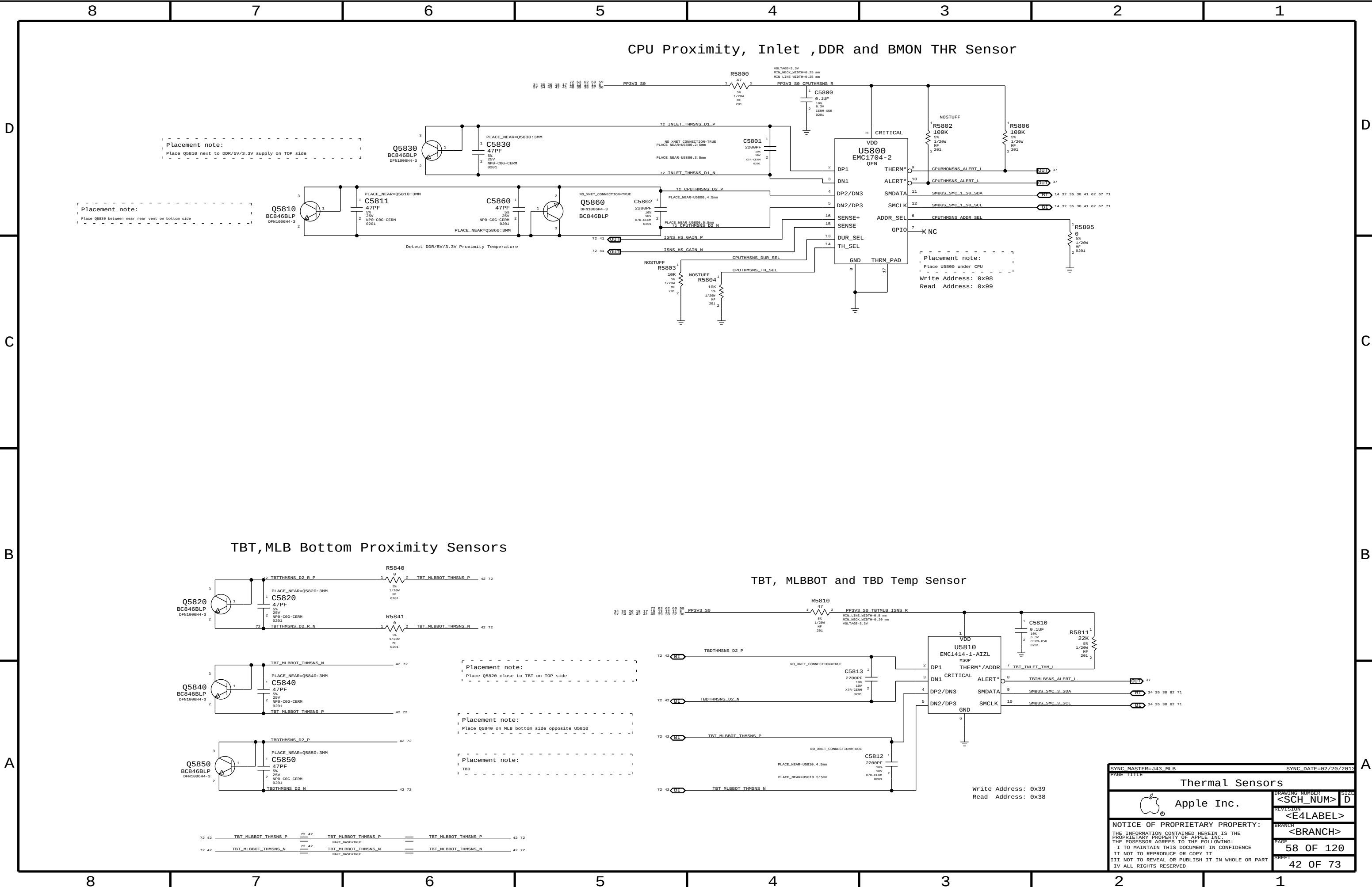
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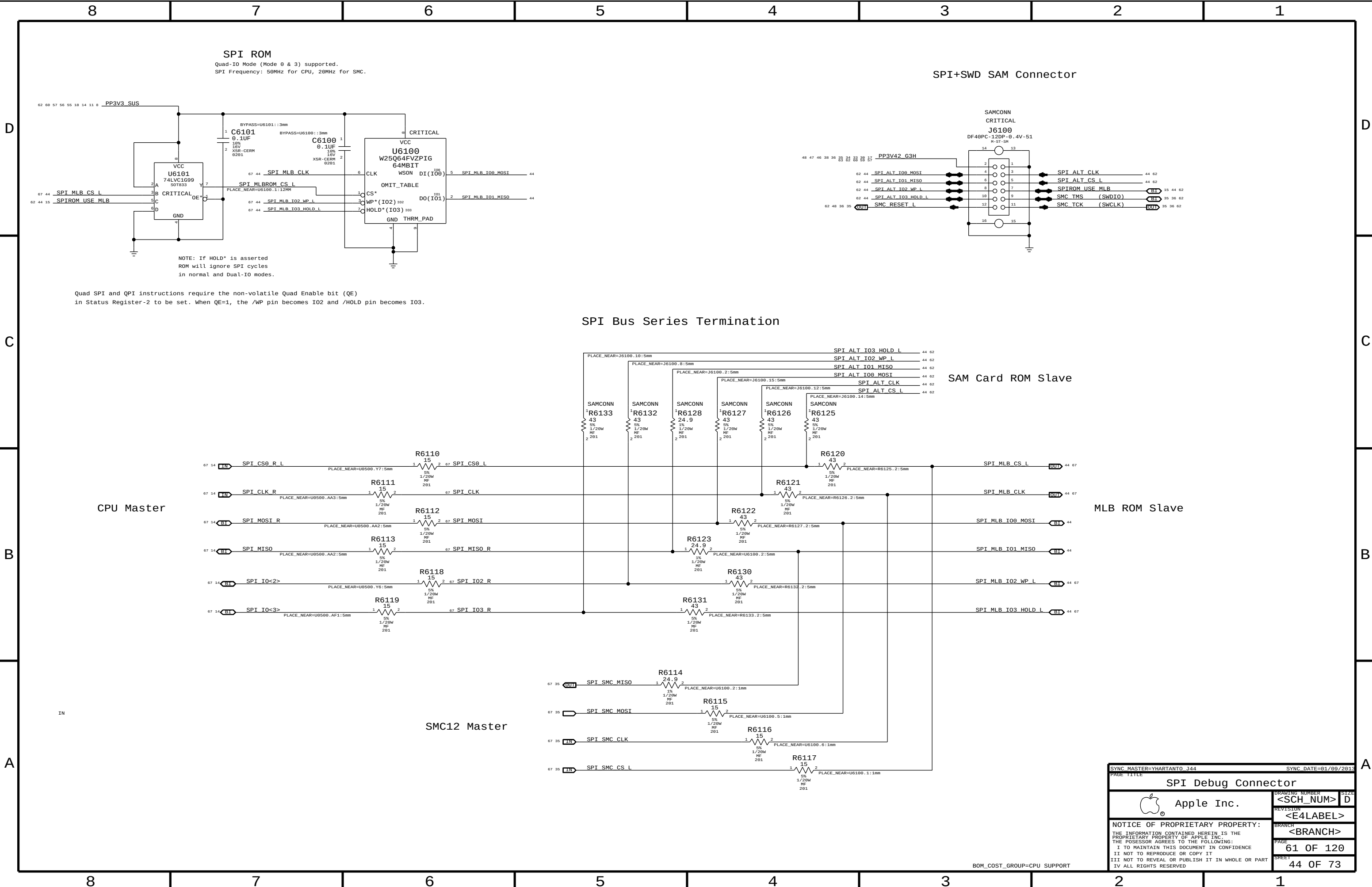
56 OF 120

41 OF 73



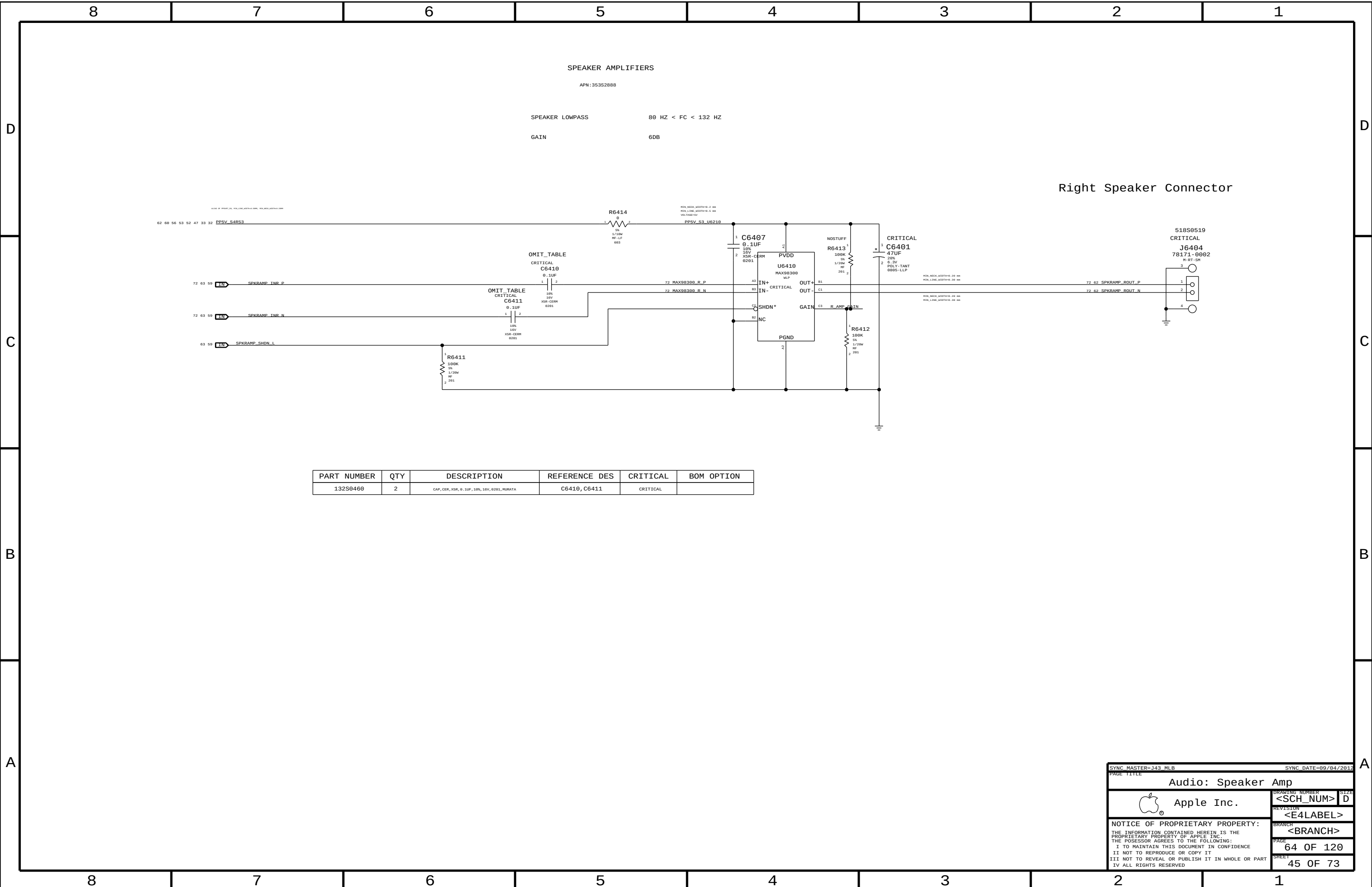
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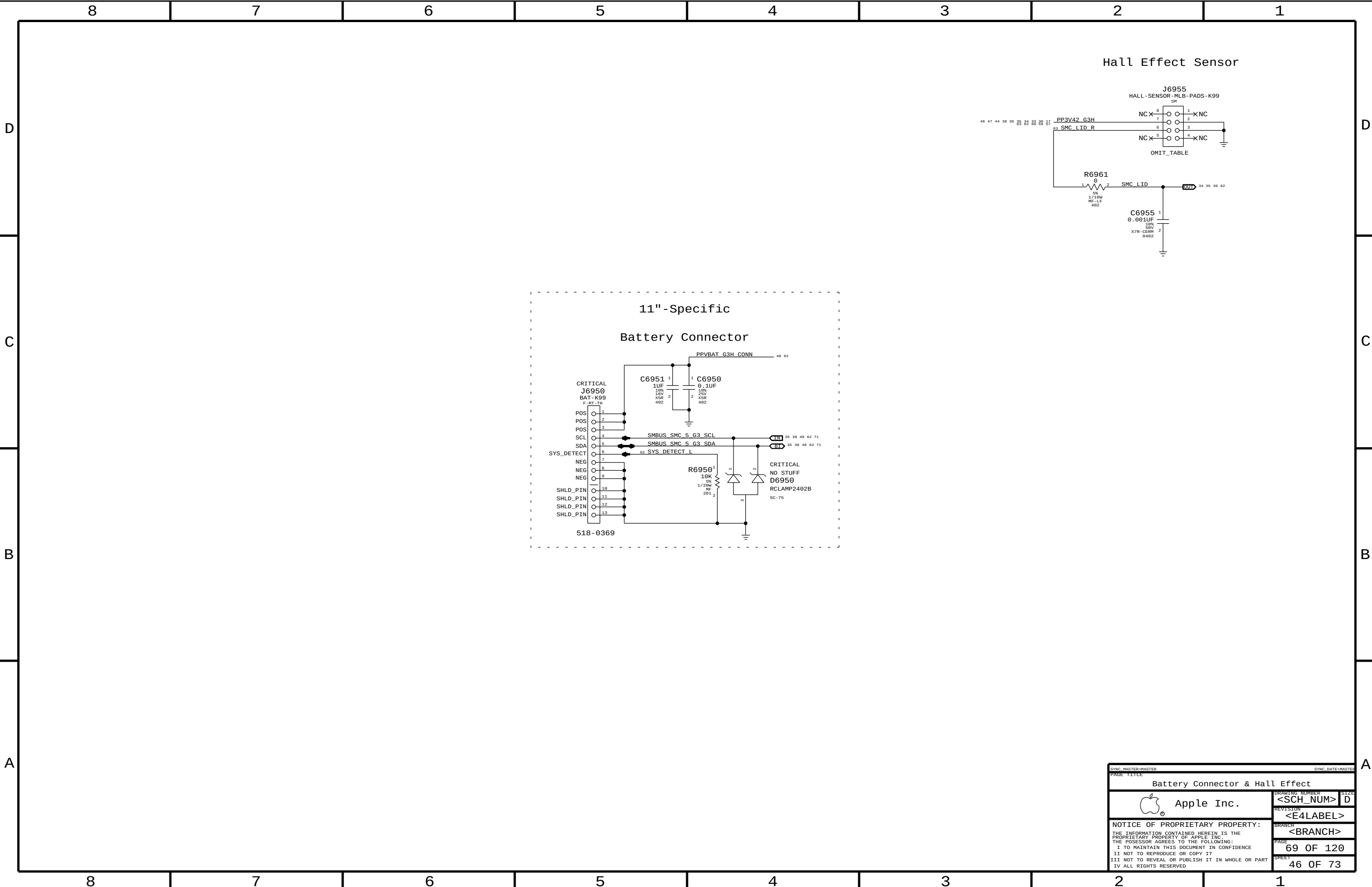
DCBA

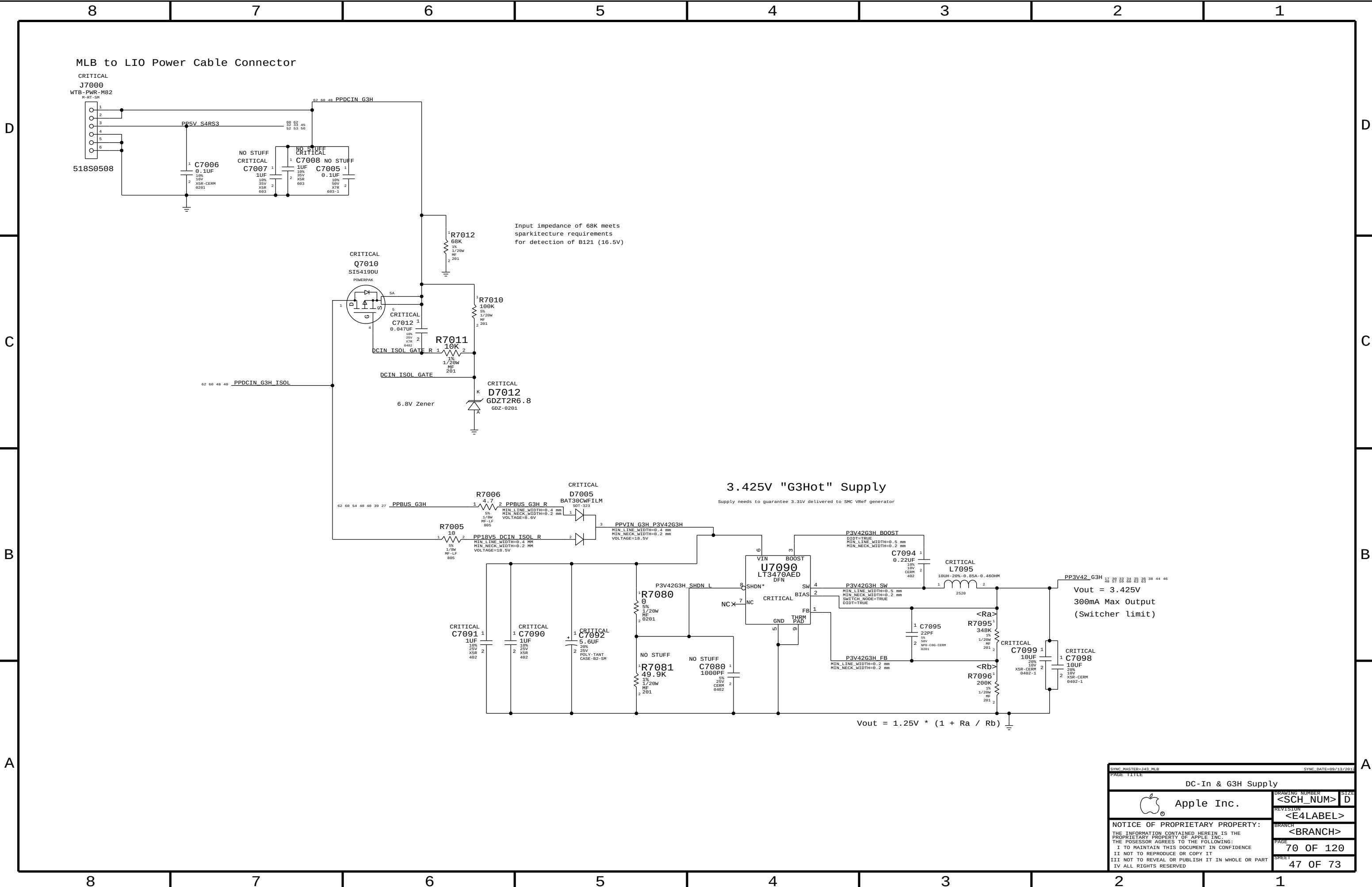


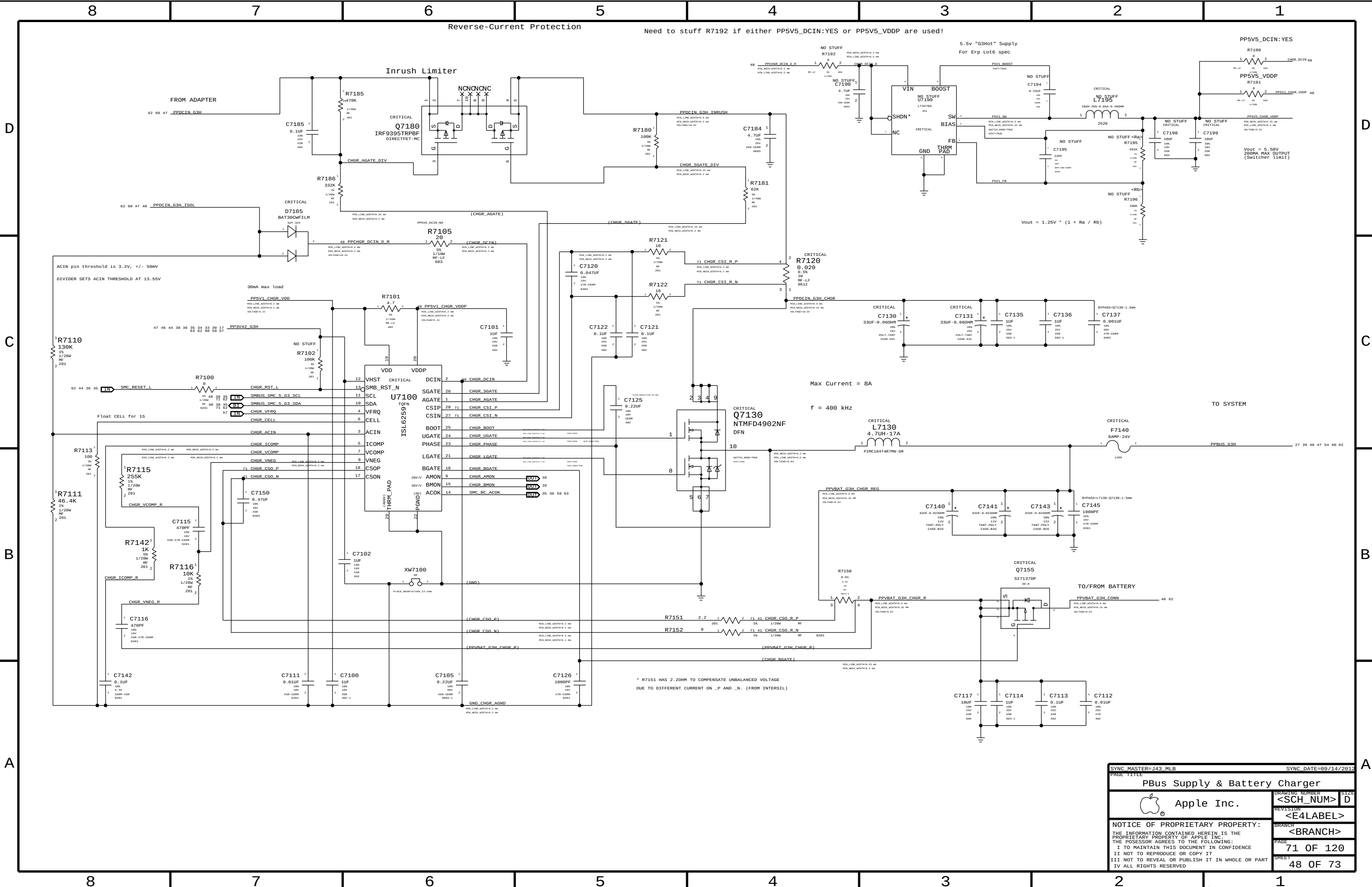
SYNC MASTER=YHARTANTO_344		SYNC DATE=01/09/2013	
PAGE TITLE			
SPI Debug Connector			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	SIZE D
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		BRANCH <BRANCH>	
		PAGE 61 OF 120	
		SHEET 44 OF 73	

BOM_COST_GROUP=CPU_SUPPORT









SYNC MASTER=J43 MLB		SYNC DATE=89/14/2012	
PAGE TITLE			
PBus Supply & Battery Charger		DRAWING Number	
 Apple Inc.		<SCH_NUM>	D
		REVISION	
		<E4LABEL>	
		BRANCH	
		<BRANCH>	
		PAGE	71 OF 120
		SHEET	48 OF 73
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D

C

B

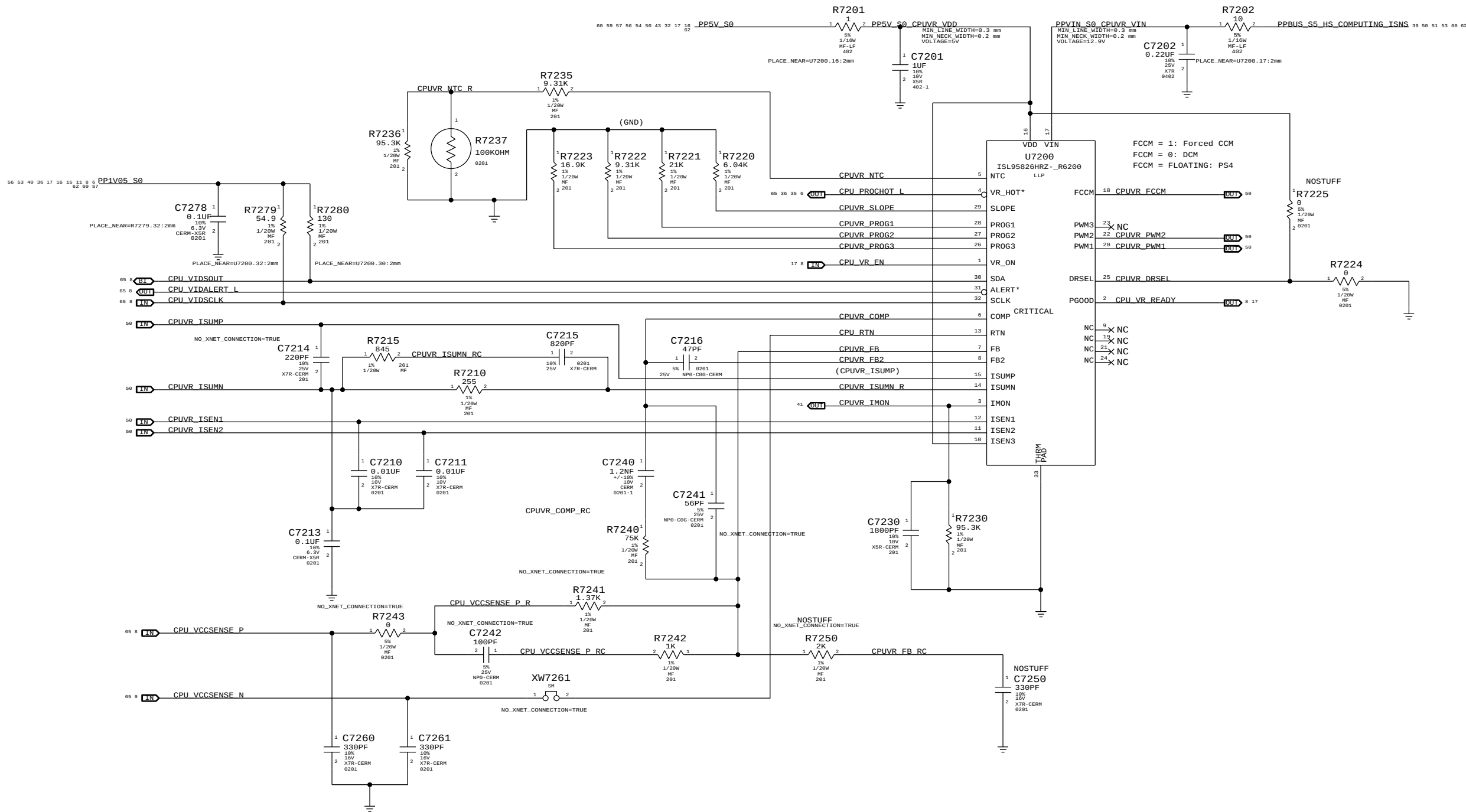
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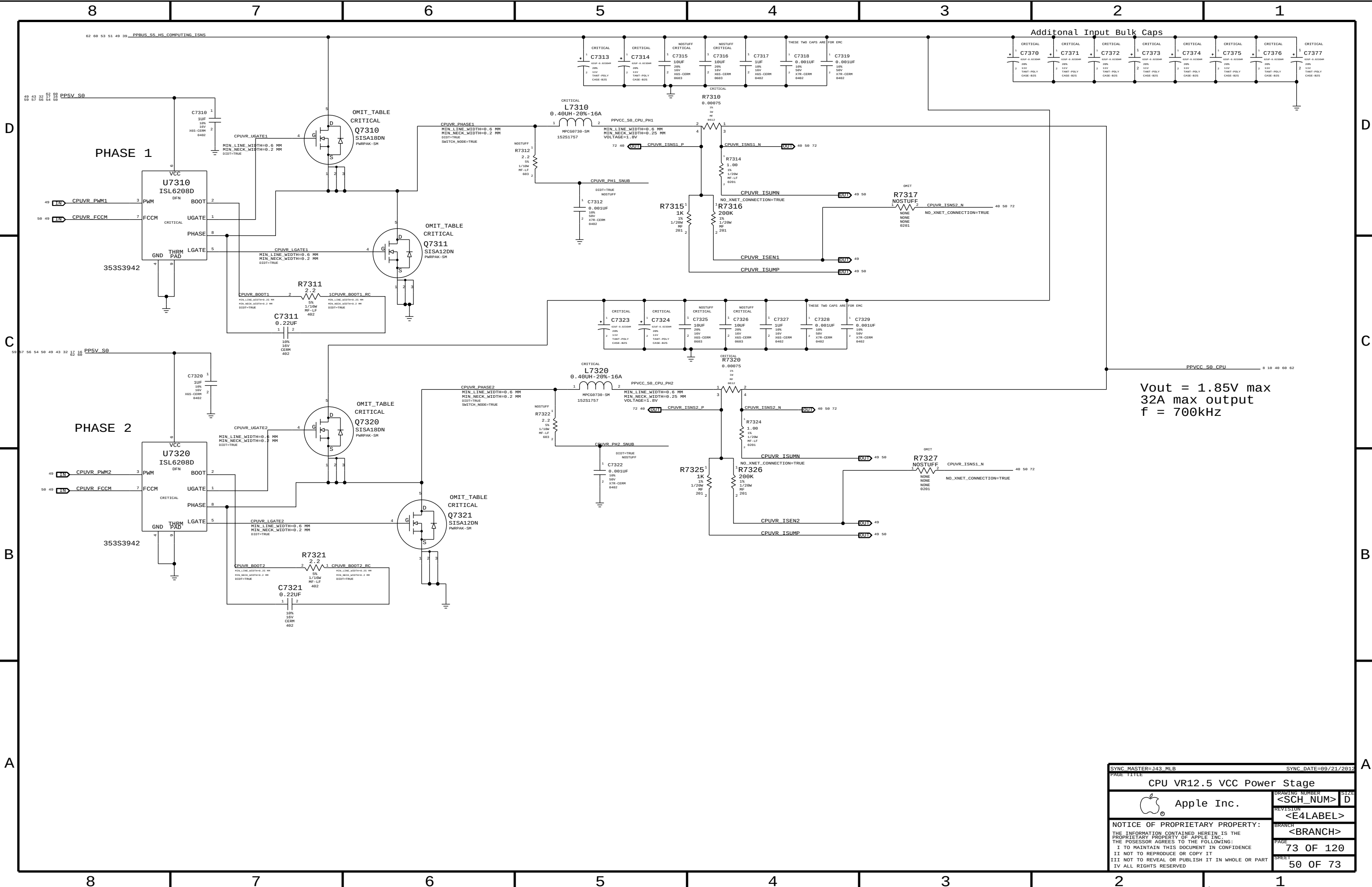
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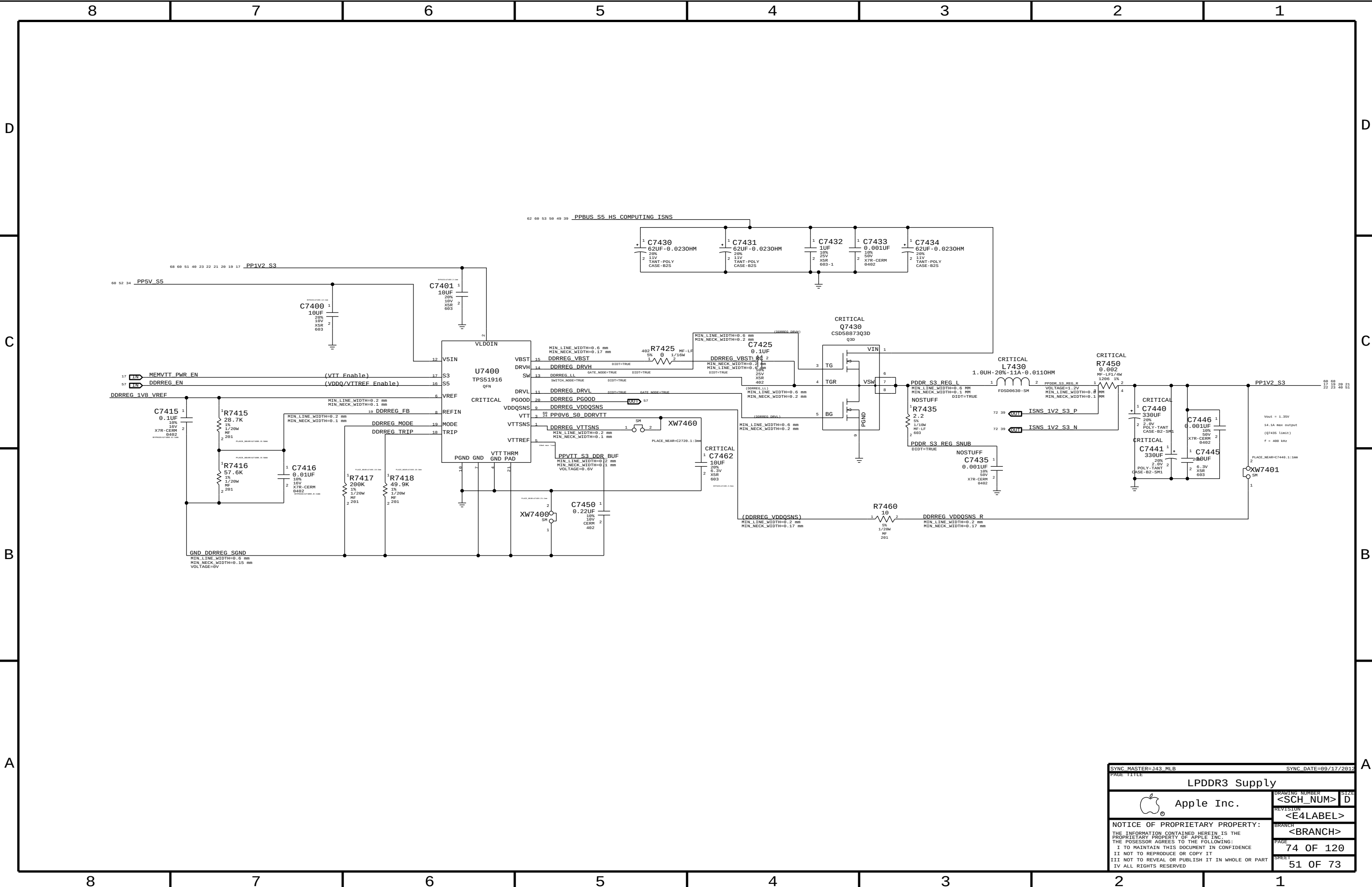


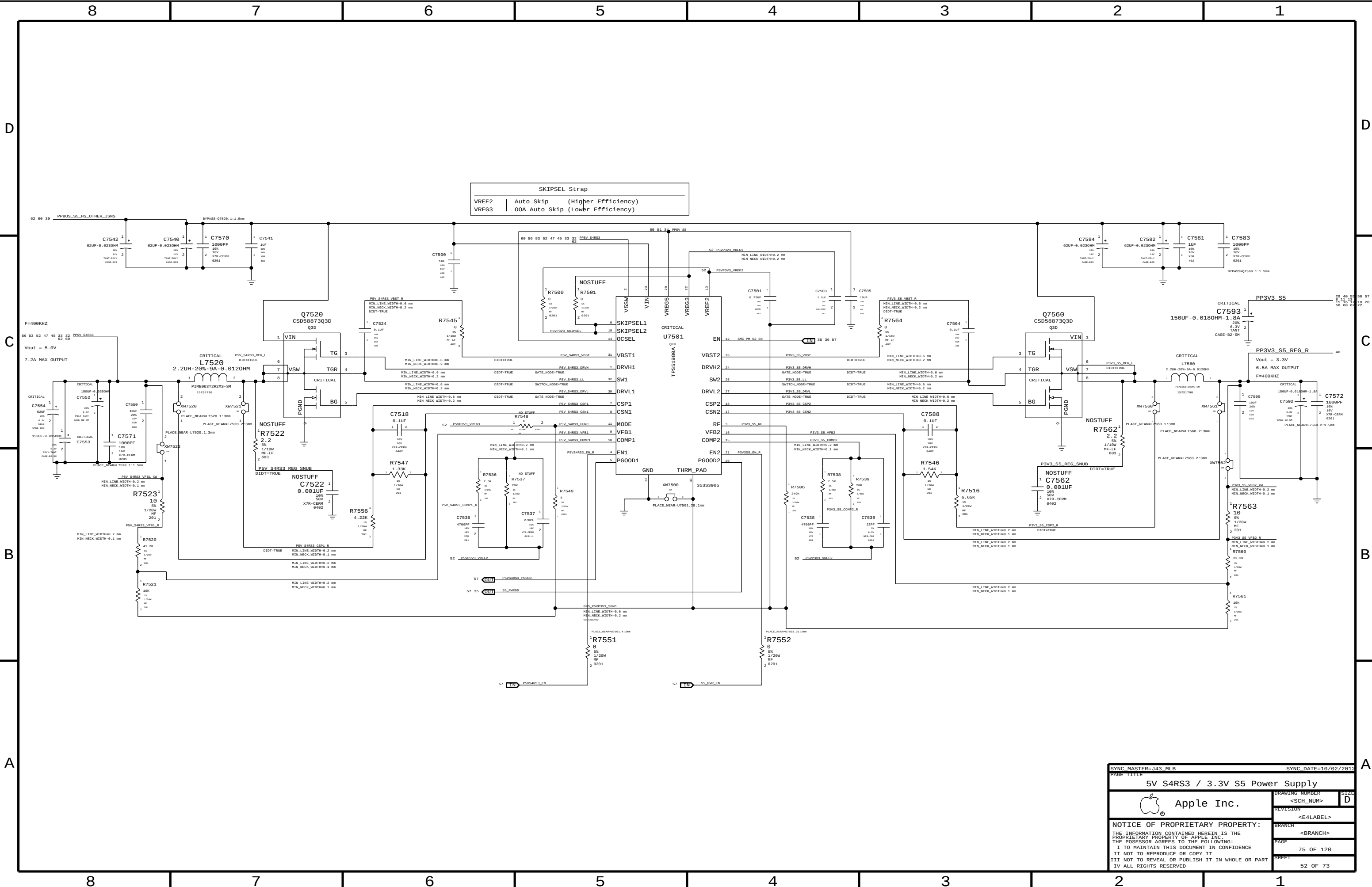
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PAGE TITLE			
CPU VR12.6 VCC Regulator IC			
 Apple Inc.		DRAWING NUMBER	SIZE
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		<BRANCH>	
		PAGE	72 OF 120
		SHEET	49 OF 73



Vout = 1.85V max
32A max output
f = 700kHz

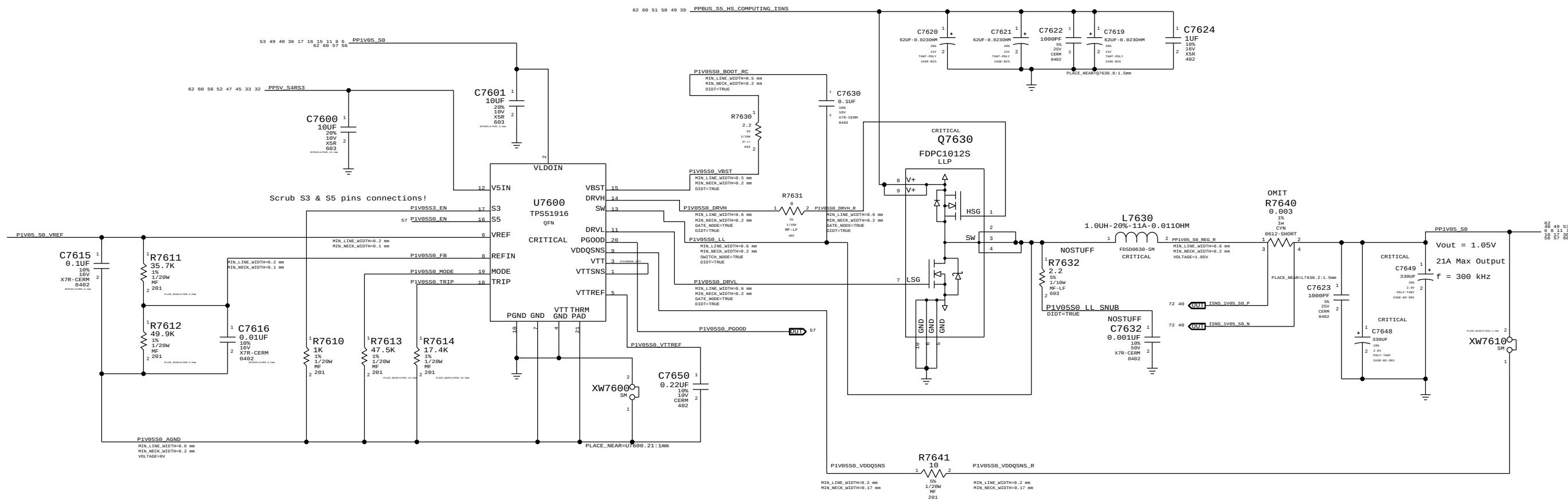
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CPU VR12.5 VCC Power Stage			
 Apple Inc.		DRAWING NUMBER	SIZE
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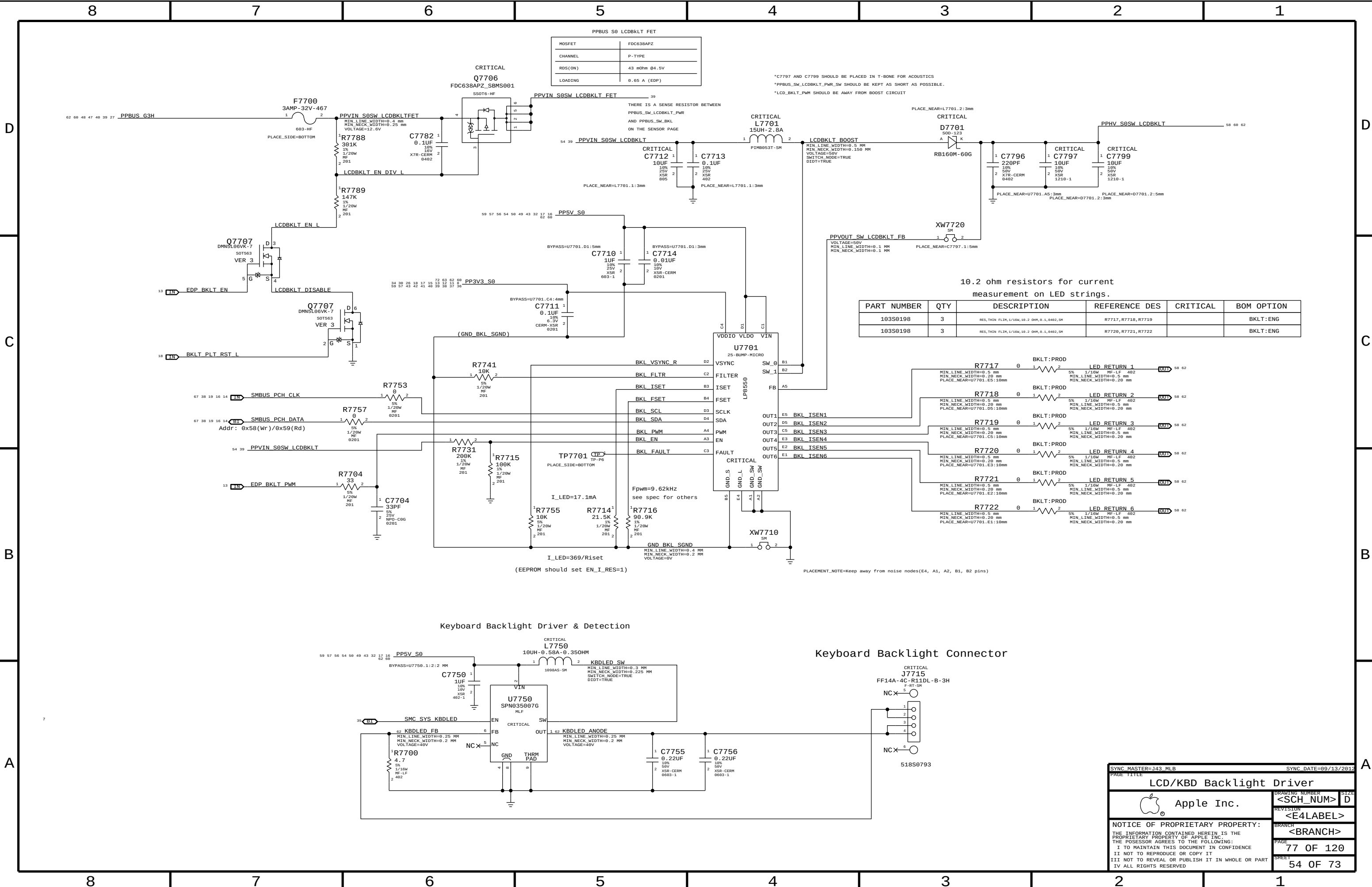


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PAGE TITLE			
5V S4RS3 / 3.3V S5 Power Supply			
 Apple Inc.	DRAWING NUMBER	SIZE	
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	BRANCH	<BRANCH>	
	PAGE	75 OF 120	
	SHEET	52 OF 73	

1.05V S0 Regulator



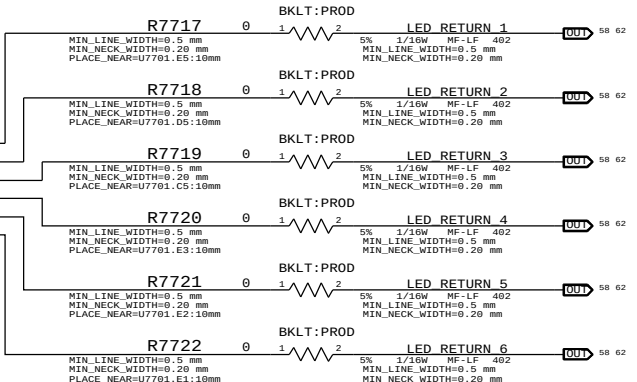
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1.05V S0 Power Supply			
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Apple Inc.	REVISION		
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	76 OF 120		
	SHEET		
	53 OF 73		



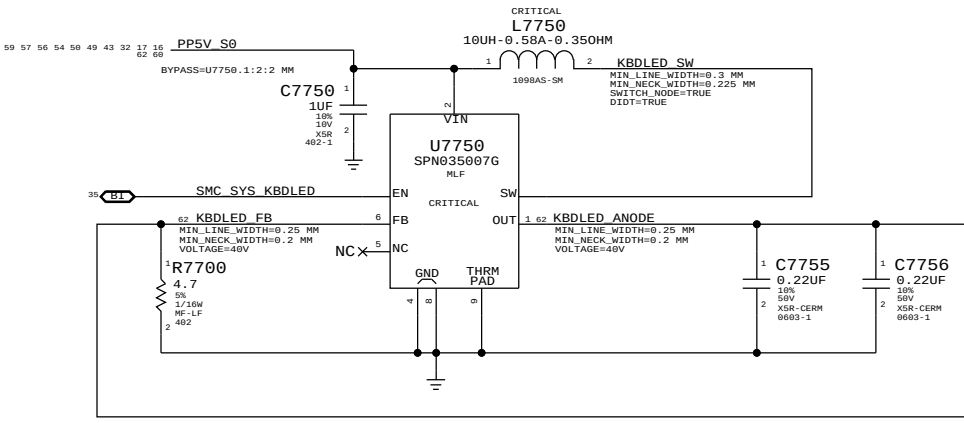
PPBUS S0 LCDBKLT FET	
MOSFET	FDC638APZ
CHANNEL	P-TYPE
RDS(ON)	43 mOhm @4.5V
LOADING	0.65 A (EDP)

*C7797 AND C7799 SHOULD BE PLACED IN T-BONE FOR ACOUSTICS
*PPBUS_SW_LCDBKLT_PWR_SW SHOULD BE KEPT AS SHORT AS POSSIBLE.
*LCD_BKLT_PWM SHOULD BE AWAY FROM BOOST CIRCUIT

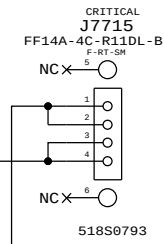
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES,THIN FLTM,1/16W,10.2 OHM,0.1,0402,SM	R7717,R7718,R7719		BKLT:ENG
103S0198	3	RES,THIN FLTM,1/16W,10.2 OHM,0.1,0402,SM	R7720,R7721,R7722		BKLT:ENG



Keyboard Backlight Driver & Detection



Keyboard Backlight Connector



SYNC MASTER=143 MLB

SYNC DATE=09/13/2012

LCD/KBD Backlight Driver

Apple Inc.

<SCH_NUM> D

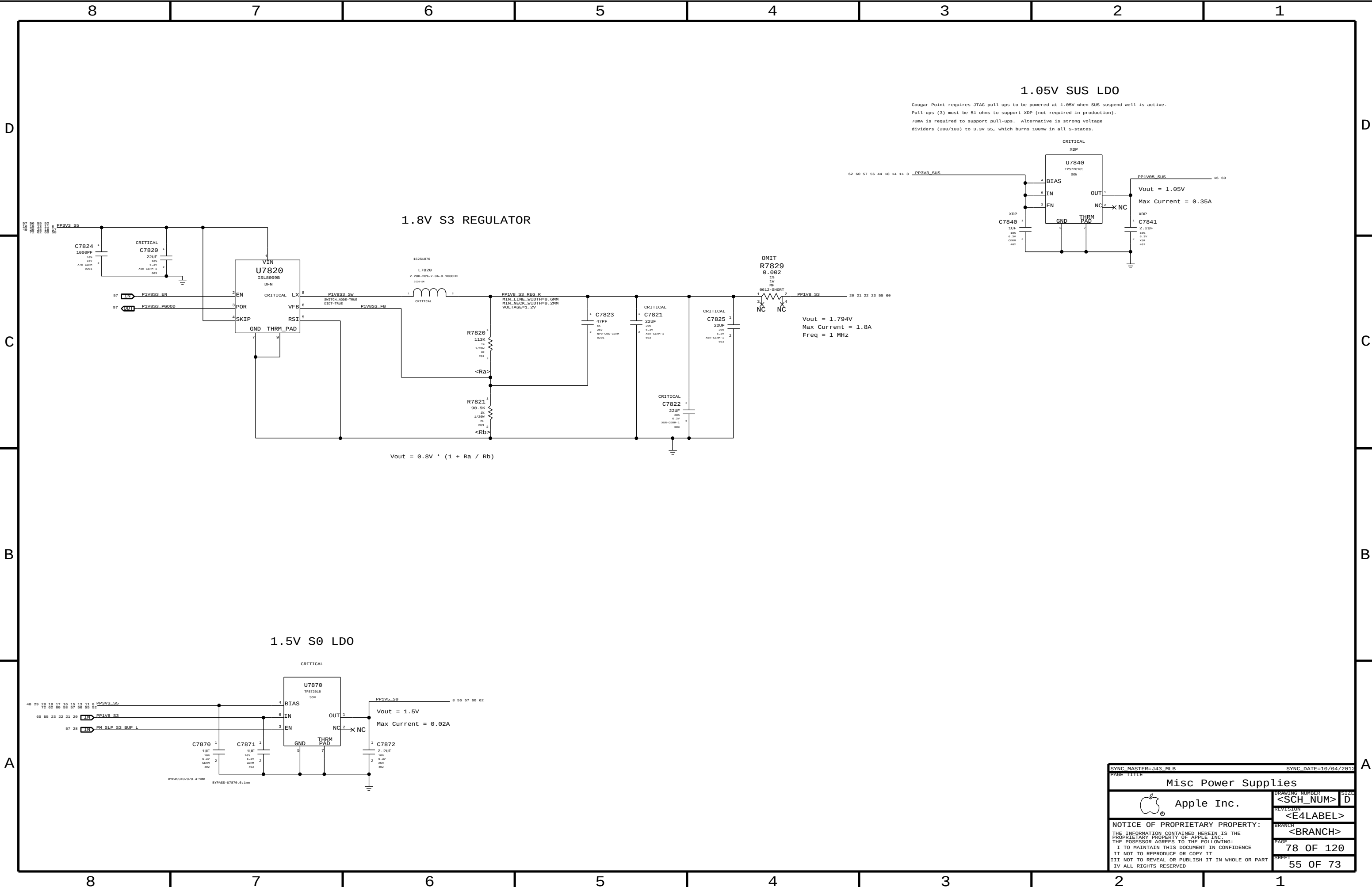
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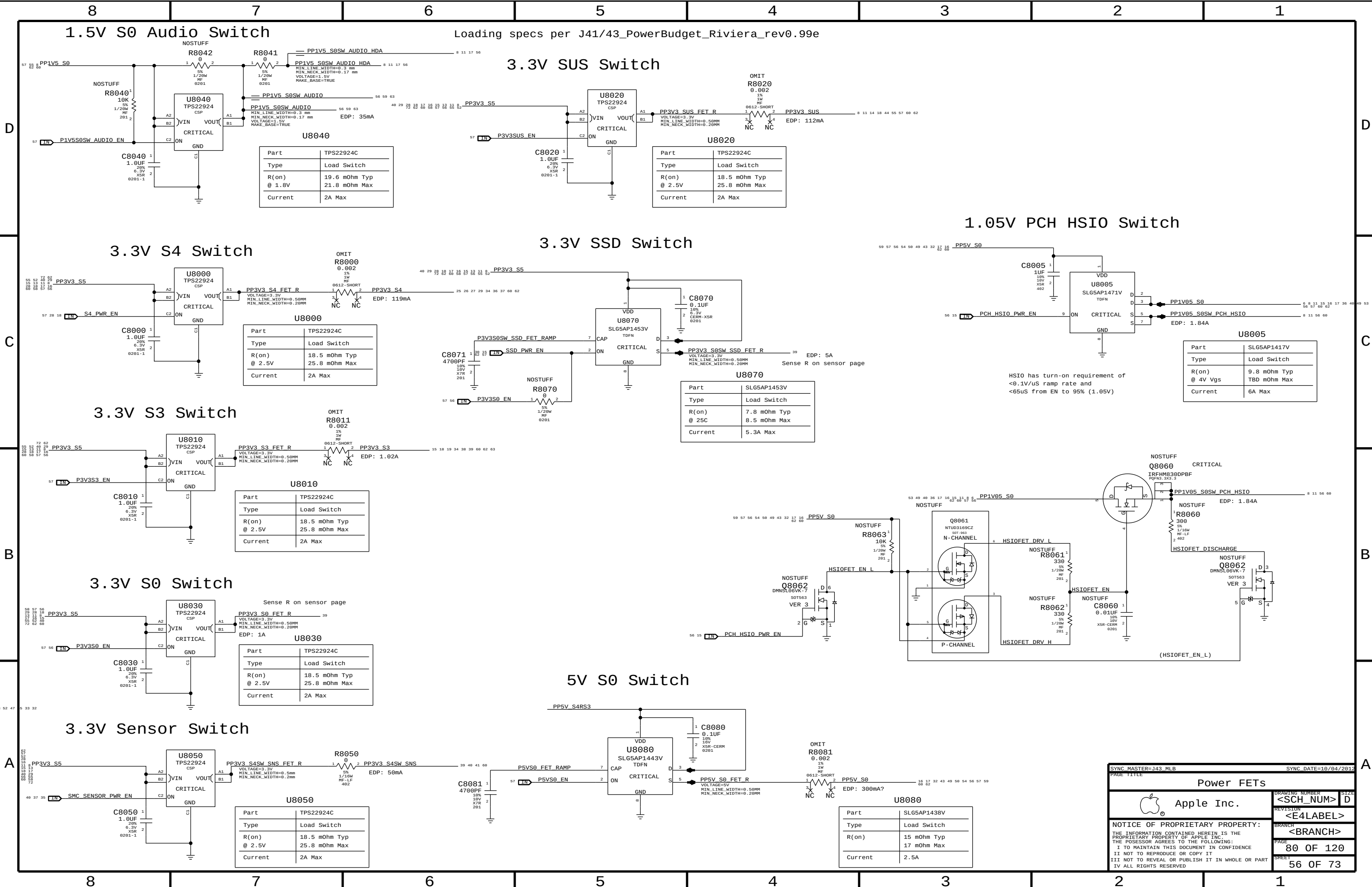
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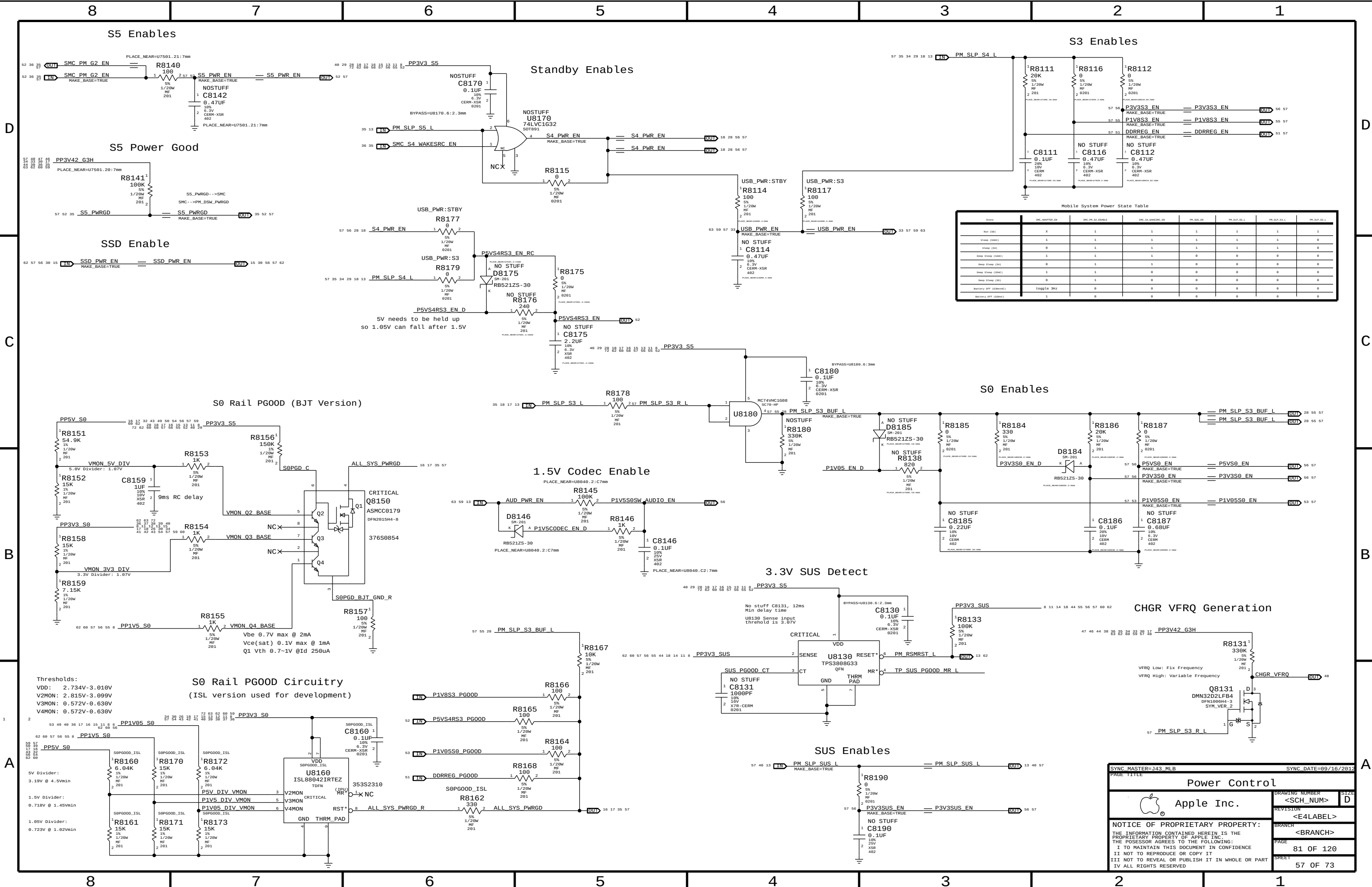
PAGE 77 OF 120

SHEET 54 OF 73



SYNC MASTER=J43 MLB		SYNC DATE=10/04/2012	
PAGE TITLE			
Misc Power Supplies			
 Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
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		PAGE	78 OF 120
		SHEET	55 OF 73
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Mobile System Power State Table							
State	SMC_ADAPTER_EN	SMC_PM_S2_ENABLE	SMC_S4_WAKE_SRC_EN	PM_SLP_EN	PM_SLP_S3_L	PM_SLP_S4_L	PM_SLP_S3_L
Run (S0)	X	1	1	1	1	1	1
Standby (S3AC)	1	1	1	1	1	1	0
Sleep (S3)	0	1	1	1	1	1	0
Deep Sleep (S4AC)	1	1	1	0	0	0	0
Deep Sleep (S4)	0	1	1	0	0	0	0
Deep Sleep (S3AC)	1	1	0	0	0	0	0
Deep Sleep (S3)	0	1	0	0	0	0	0
Battery Off (S400AC)	toggle SHZ	0	0	0	0	0	0
Battery Off (S400)	1	0	0	0	0	0	0

Thresholds:
VDD: 2.734V-3.010V
V2MON: 2.815V-3.099V
V3MON: 0.572V-0.630V
V4MON: 0.572V-0.630V

SYNC MASTER=143 MLB

SYNC DATE=09/16/2012

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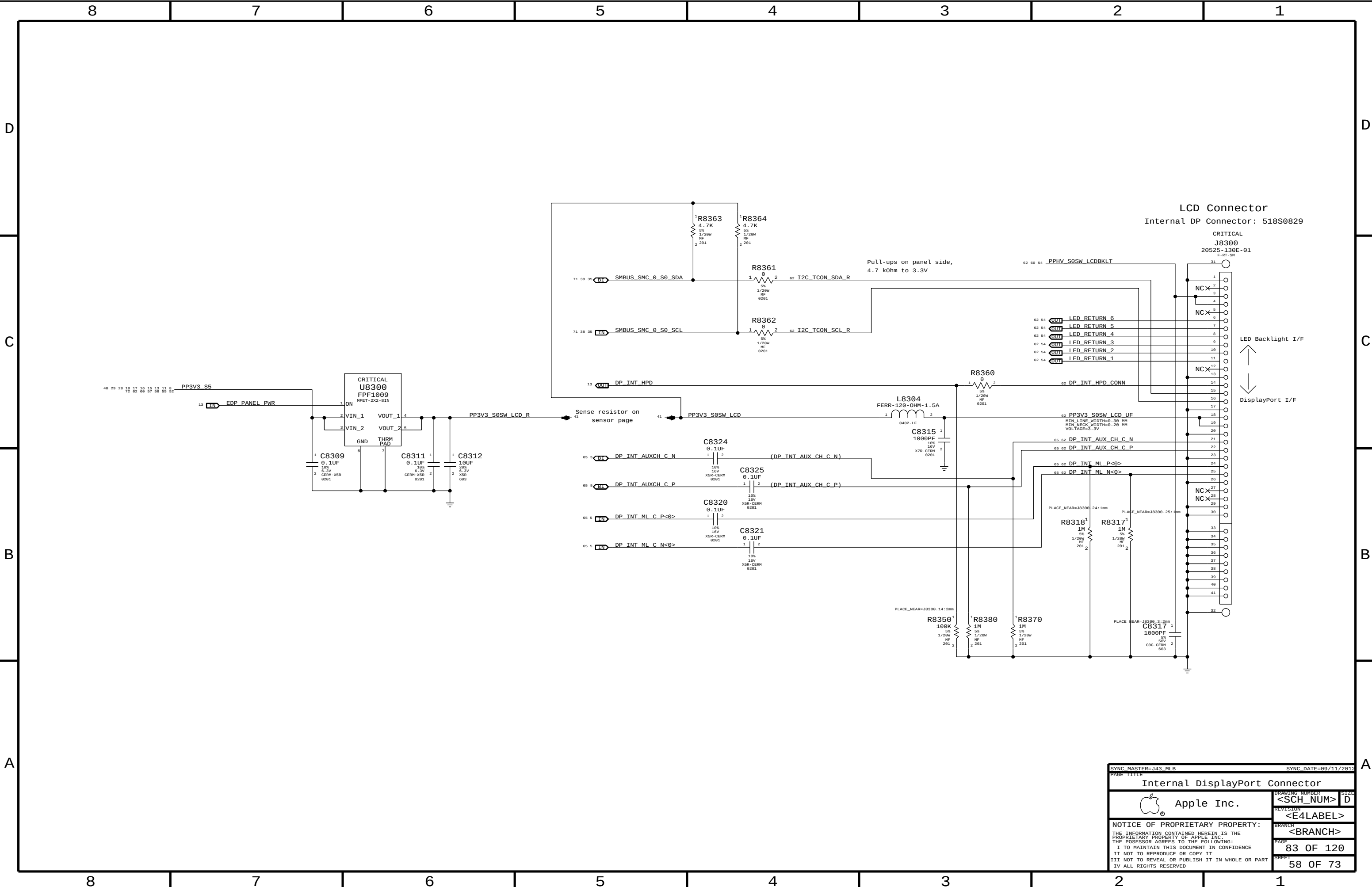
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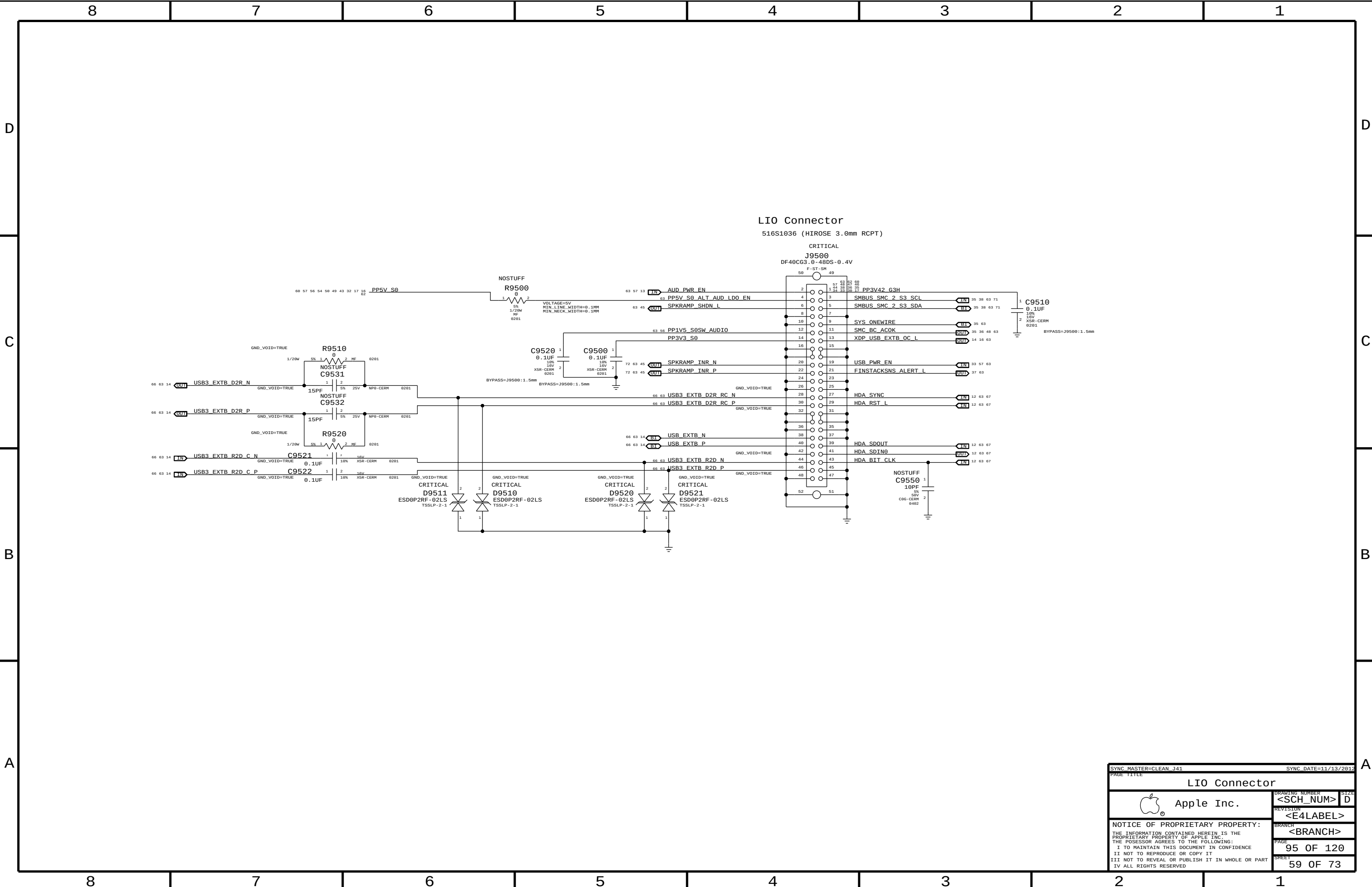
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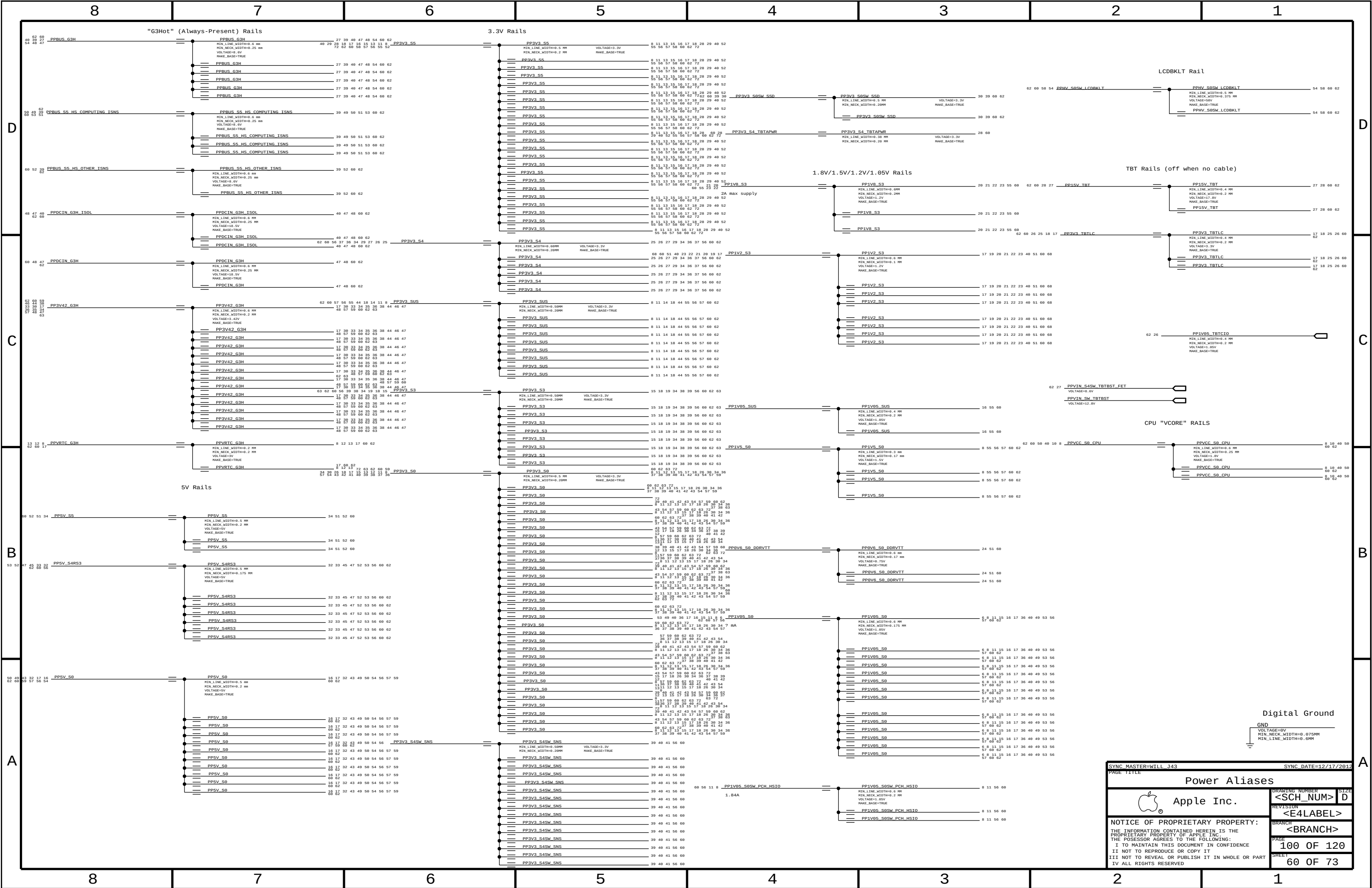
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PAGE
81 OF 120

SHEET
57 OF 73

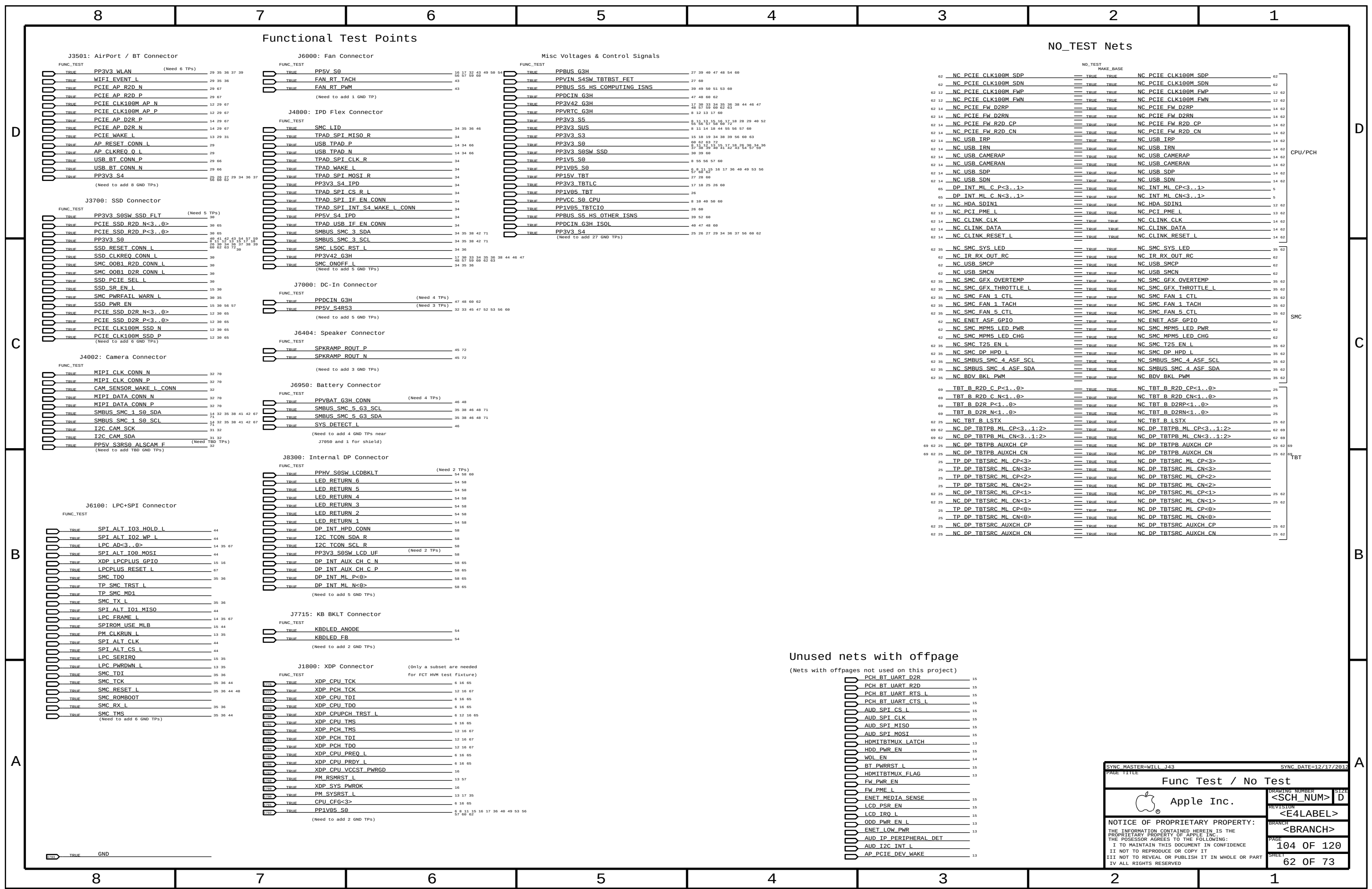




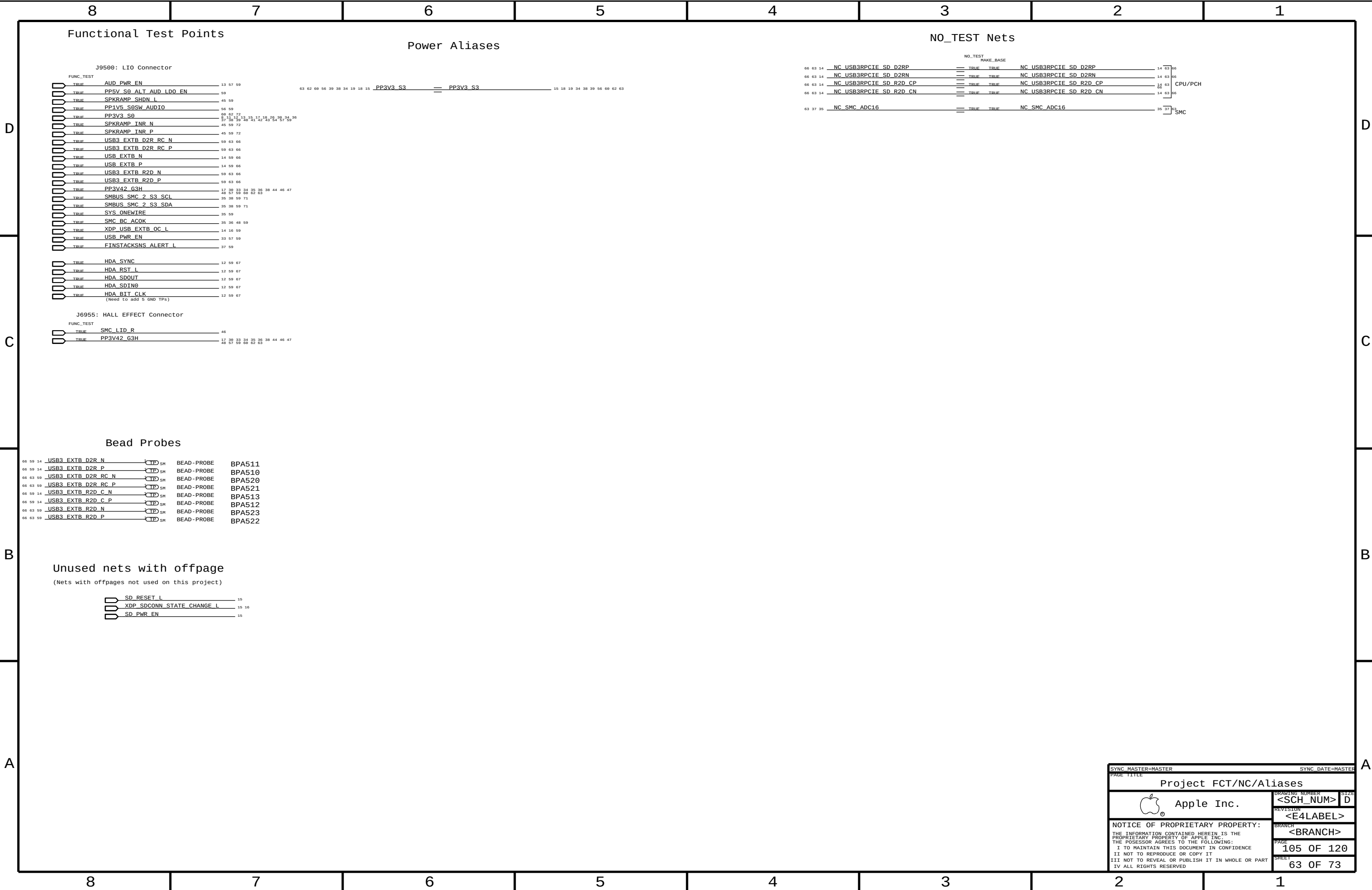


Functional Test Points

NO_TEST Nets



SYNC MASTER=WILL J43		SYNC DATE=12/17/2012	
PAGE TITLE		PAGE	
Func Test / No Test		DRAWING NUMBER	
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		SHEET	
		62 OF 73	



8		7		6		5		4		3		2		1									
J41/J43 Board-Specific Spacing & Physical Constraints																							
BOARD LAYERS								BOARD AREAS				BOARD UNITS (ALL OF MM)		ALLEGRO VERSION									
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM								NO_TYPE, BGA, MEM_TERM				MM		16.2									
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
DEFAULT	TOP, BOTTOM	Y	=50_OHM_SE	=50_OHM_SE																			
DEFAULT	ISL2, ISL11	Y	=45_OHM_SE	=45_OHM_SE																			
DEFAULT	ISL3, ISL10	Y	=45_OHM_SE	=45_OHM_SE																			
DEFAULT	ISL4, ISL9	Y	=45_OHM_SE	=45_OHM_SE																			
DEFAULT	*	N	100 MM	100 MM	10 MM	0 MM	0 MM																
STANDARD	*	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT																
Single-ended Physical Constraints																							
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
27P4_OHM_SE	TOP, BOTTOM	Y	0.310 MM	0.310 MM																			
27P4_OHM_SE	ISL2, ISL11	Y	0.182 MM	0.182 MM																			
27P4_OHM_SE	ISL3, ISL10	Y	0.182 MM	0.182 MM																			
27P4_OHM_SE	ISL4, ISL9	Y	0.182 MM	0.182 MM																			
27P4_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
35_OHM_SE	TOP, BOTTOM	Y	0.195 MM	0.195 MM																			
35_OHM_SE	ISL2, ISL11	Y	0.125 MM	0.125 MM																			
35_OHM_SE	ISL3, ISL10	Y	0.125 MM	0.125 MM																			
35_OHM_SE	ISL4, ISL9	Y	0.125 MM	0.125 MM																			
35_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
40_OHM_SE	TOP, BOTTOM	Y	0.170 MM	0.170 MM																			
40_OHM_SE	ISL2, ISL11	Y	0.096 MM	0.096 MM																			
40_OHM_SE	ISL3, ISL10	Y	0.096 MM	0.096 MM																			
40_OHM_SE	ISL4, ISL9	Y	0.099 MM	0.099 MM																			
40_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
45_OHM_SE	TOP, BOTTOM	Y	0.135 MM	0.135 MM																			
45_OHM_SE	ISL2, ISL11	Y	0.075 MM	0.075 MM																			
45_OHM_SE	ISL3, ISL10	Y	0.075 MM	0.075 MM																			
45_OHM_SE	ISL4, ISL9	Y	0.080 MM	0.080 MM																			
45_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
50_OHM_SE	TOP, BOTTOM	Y	0.110 MM	0.110 MM																			
50_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM																			
55_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
Differential Pair Physical Constraints																							
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
70_OHM_DIFF	TOP, BOTTOM	Y	0.165 MM	0.165 MM		0.110 MM	0.110 MM																
70_OHM_DIFF	ISL2, ISL11	Y	0.105 MM	0.105 MM		0.100 MM	0.100 MM																
70_OHM_DIFF	ISL3, ISL10	Y	0.105 MM	0.105 MM		0.100 MM	0.100 MM																
70_OHM_DIFF	ISL4, ISL9	Y	0.110 MM	0.110MM		0.095 MM	0.095 MM																
70_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
80_OHM_DIFF	TOP, BOTTOM	Y	0.132 MM	0.132 MM		0.130 MM	0.130 MM																
80_OHM_DIFF	ISL2, ISL11	Y	0.081 MM	0.081 MM		0.115 MM	0.115 MM																
80_OHM_DIFF	ISL3, ISL10	Y	0.081 MM	0.081 MM		0.115 MM	0.115 MM																
80_OHM_DIFF	ISL4, ISL9	Y	0.088 MM	0.088 MM		0.110 MM	0.110 MM																
80_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.115 MM		0.200 MM	0.200 MM																
90_OHM_DIFF	ISL2, ISL11	Y	0.070 MM	0.070 MM		0.180 MM	0.180 MM																
90_OHM_DIFF	ISL3, ISL10	Y	0.070 MM	0.070 MM		0.180 MM	0.180 MM																
90_OHM_DIFF	ISL4, ISL9	Y	0.076 MM	0.076 MM		0.180 MM	0.180 MM																
90_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
73_OHM_DIFF	TOP, BOTTOM	Y	0.105 MM	0.105 MM		0.150 MM	0.150 MM																
73_OHM_DIFF	ISL2, ISL11	Y	0.100 MM	0.100 MM		0.150 MM	0.150 MM																
73_OHM_DIFF	ISL3, ISL10	Y	0.100 MM	0.100 MM		0.150 MM	0.150 MM																
73_OHM_DIFF	ISL4, ISL9	Y	0.110 MM	0.110 MM		0.150 MM	0.150 MM																
73_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
85_OHM_DIFF	TOP, BOTTOM	Y	0.120 MM	0.120 MM		0.150 MM	0.150 MM																
85_OHM_DIFF	ISL2, ISL11	Y	0.078 MM	0.078 MM		0.160 MM	0.160 MM																
85_OHM_DIFF	ISL3, ISL10	Y	0.078 MM	0.078 MM		0.160 MM	0.160 MM																
85_OHM_DIFF	ISL4, ISL9	Y	0.082 MM	0.082 MM		0.140 MM	0.140 MM																
85_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD																
PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP																
90_OHM_DIFF	TOP, BOTTOM	Y	0.115 MM	0.115 MM		0.200 MM	0.200 MM																
90_OHM_DIFF	ISL2, ISL11	Y	0.070 MM	0.070 MM		0.180 MM	0.180 MM																
90_OHM_DIFF	ISL3, ISL10	Y	0.070 MM	0.070 MM		0.180 MM	0																

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_B0D	*	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ICOMP	*	=4x_DIELECTRIC	?

SOURCE: 471984_Chief_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

UART Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
UART_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
UART	*	=2x_DIELECTRIC	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_BIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_80D	*	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ixex Peak M (DG-398905-398905_v1.5), Section 3.8

USB 3.0 Interface Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_PCH_TX	USB3_PCH_TX	*	USB3_TX2TX	USB3_TX2TX	TOP, BOTTOM	=5X_DIELECTRIC	?
USB3_PCH_RX	USB3_PCH_RX	*	USB3_RX2RX	USB3_RX2RX	TOP, BOTTOM	=5X_DIELECTRIC	?
USB3_PCH_TX	*_PCH_TX	*	USB3_TX20THERTX	USB3_TX20THERTX	TOP, BOTTOM	=5X_DIELECTRIC	?
USB3_PCH_RX	*_PCH_RX	*	USB3_RX20THERRX	USB3_RX20THERRX	TOP, BOTTOM	=5X_DIELECTRIC	?
USB3_PCH_TX	*_PCH_RX	*	USB3_TX2RX	USB3_TX2RX	TOP, BOTTOM	=7X_DIELECTRIC	?
USB3_PCH_RX	*_PCH_TX	*	USB3_RX2TX	USB3_RX2TX	TOP, BOTTOM	=7X_DIELECTRIC	?
USB3_PCH_TX	*_TX	*	USB3_20THERHS	USB3_20THERHS	TOP, BOTTOM	=6X_DIELECTRIC	?
USB3_PCH_RX	*_TX	*	USB3_20THERHS	USB3_20THER	TOP, BOTTOM	=5X_DIELECTRIC	?
USB3_PCH_TX	*_RX	*	USB3_20THERHS				
USB3_PCH_RX	*_RX	*	USB3_20THERHS				
USB3_PCH_TX	*	*	USB3_20THER				
USB3_PCH_RX	*	*	USB3_20THER				

SOURCE: 471984_Cheif_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

PCH Net Properties

[illegible]

USB Hucopyb nets

TP SPI nets

USB EXTA nets (Right USB port)

USB EXTB nets (Left USB port)

SYNC MASTER=CLEAN J41		SYNC DATE=11/13/2012	
PAGE TITLE			
PCH Constraints 1			
 Apple Inc.		DRAWING NUMBER <SCH_NUM>	
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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=3x_DIELECTRIC	?
CLK_LPC	*	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ixex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_455_R_505	TOP, BOTTOM	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE		
SMB_455_R_505	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_455	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ixex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=4x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_455	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=4x_DIELECTRIC	?

XDP Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_455	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_ITP	*	=2:1_SPACING	?

DisplayPort

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_B0D	*	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF	=B0_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2DP	*	=3x_DIELECTRIC	?	DP_2DP	TOP, BOTTOM	=4x_DIELECTRIC	?
DP_20THERHS	*	=4x_DIELECTRIC	?	DP_20THERHS	TOP, BOTTOM	=6x_DIELECTRIC	?
DP_20THER	*	=3x_DIELECTRIC	?	DP_20THER	TOP, BOTTOM	=4x_DIELECTRIC	?
DP_AUX	*	=3x_DIELECTRIC	?	DP_AUX	TOP, BOTTOM	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DP_TX	DP_TX	*	DP_2DP
DP_TX	*_TX	*	DP_2OTHERHS
DP_TX	*_RX	*	DP_2OTHERHS
DP_TX	*	*	DP_2OTHER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	14 35 62
	LPC_FRAME_L	LPC_45S	LPC	LPC_FRAME_L	14 35 62
		LPC_45S	LPC	LPCLPLUS_RESET_L	62
	LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC	17 35
		CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R	12 17
	LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCLPLUS	
		CLK_LPC_45S	CLK_LPC	LPC_CLK24M_LPCLPLUS_R	

	SMbus_PCH_CLK	SMR_455_P_50S	SMR	SMbus_PCH_CLK	14 16 19
	SMbus_PCH_DATA	SMR_455_P_50S	SMR	SMbus_PCH_DATA	14 16 19
	SMbus_PCH_0_CLK	SMR_455_P_50S	SMR	SML_PCH_0_CLK	14 38
	SMbus_PCH_0_DATA	SMR_455_P_50S	SMR	SML_PCH_0_DATA	14 38
	SMbus_SMC_1_S0_SCL	SMR_455_P_50S	SMR	SMbus_SMC_1_S0_SCL	14 32 35
	SMbus_SMC_1_S0_SDA	SMR_455_P_50S	SMR	SMbus_SMC_1_S0_SDA	14 32 35

	HDA_BTT_CLK	HDA_45S	HDA	HDA_BTT_CLK	12	59	63
		HDA_45S	HDA	HDA_BTT_CLK_R	12		
	HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12	59	63
		HDA_45S	HDA	HDA_SYNC_R	12		
	HDA_RST_L	HDA_45S	HDA	HDA_RST_R_L	12		
		HDA_45S	HDA	HDA_RST_L	12	59	63
	HDA_SDIN0	HDA_45S	HDA	HDA_SDIN0	12	59	63
	HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	12	59	63
		HDA_45S	HDA	HDA_SDOUT_R	12	17	

PM SUS CLK	CLK SLOW 45S	CLK SLOW	PM CLK32K SUSCLK R	13 36
	CLK SLOW 45S	CLK SLOW	SMC CLK32K	35 36
SPI CLK	SPT 45S	SPT	SPI CLK R	14 44
	SPT 45S	SPT	SPI CLK	44
SPI MOSI	SPT 45S	SPT	SPI MOSI R	14 44
	SPT 45S	SPT	SPI MOSI	44
SPI MISO	SPT 45S	SPT	SPI MISO	14 44
	SPT 45S	SPT	SPI MISO R	44
SPI CS0	SPT 45S	SPT	SPI CS0 R L	14 44
	SPT 45S	SPT	SPI CS0 L	44
	SPT 45S	SPT	SPI SMC CLK	35 44
	SPT 45S	SPT	SPI SMC MOSI	35 44
	SPT 45S	SPT	SPI SMC MISO	35 44
	SPT 45S	SPT	SPI SMC CS L	35 44
	SPT 45S	SPT	SPI MLB CLK	44
	SPT 45S	SPT	SPI MLB I02 WP L	44
	SPT 45S	SPT	SPI MLB I03 HOLD L	44
	SPT 45S	SPT	SPI MLB CS L	44
	SPT 45S	SPT	SPI I0<2> R	14 44
	SPT 45S	SPT	SPI I02 R	44
	SPT 45S	SPT	SPI I0<3> R	14 44
	SPT 45S	SPT	SPI I03 R	44

	PCIE AP R2D	PCIE R80	PCIE PCH TX	PCIE AP R2D P	29 62
	PCIE AP R2D	PCIE R80	PCIE PCH TX	PCIE AP R2D N	29 62
		PCIE R80	PCIE PCH TX	PCIE AP R2D C P	14 29
		PCIE R80	PCIE PCH TX	PCIE AP R2D C N	14 29
	PCIE AP D2R	PCIE R80	PCIE PCH RX	PCIE AP D2R P	14 29 62
	PCIE AP D2R	PCIE R80	PCIE PCH RX	PCIE AP D2R N	14 29 62
	PCIE CLK100M AP	CLK PCIE R80	CLK PCIE	PCIE CLK100M AP P	12 29 62
	PCIE CLK100M AP	CLK PCIE R80	CLK PCIE	PCIE CLK100M AP N	12 29 62

	PCIE TBT R2D	PCIE R80	PCIE PCH_TX	PCIE TBT R2D P<3...0>	25
	PCIE TBT R2D	PCIE PCH_TX	PCIE TBT R2D N<3...0>		
		PCIE R80	PCIE PCH_TX	PCIE TBT R2D C P<3...0>	14 25
		PCIE R80	PCIE PCH_TX	PCIE TBT R2D C N<3...0>	14 25
	PCIE TBT D2R	PCIE R80	PCIE PCH_RX	PCIE TBT D2R P<3...0>	14 25
	PCIE TBT D2R	PCIE R80	PCIE PCH_RX	PCIE TBT D2R N<3...0>	14 25
		PCIE R80	PCIE PCH_RX	PCIE TBT D2R C P<3...0>	25
		PCIE R80	PCIE PCH_RX	PCIE TBT D2R C N<3...0>	25
	PCIE CLK100M TBT	CLK_PCIE_R80	CLK_PCIE	PCIE CLK100M TBT P	12 25
	PCIE CLK100M TBT	CLK_PCIE_R80	CLK_PCIE	PCIE CLK100M TBT N	12 25

	CLK_PCTF_B00	CLK_PCTF	PEG CLK100M P
	CLK_PCTF_B00	CLK_PCTF	PEG CLK100M N

XDP TDO	PCH 45S	PCH TTP	XDP PCH TDO	12 16 62
XDP TMS	PCH 45S	PCH TTP	XDP PCH TMS	12 16 62
XDP TCK	PCH 45S	PCH TTP	XDP PCH TCK	12 16 62

	PCIE_CAM	PCIE_800	PCIE_PCH_TX	PCIE CAMERA R2D P	31	32
	PCIE_CAM	PCIE_800	PCIE_PCH_TX	PCIE CAMERA R2D N	31	32
		PCIE_800	PCIE_PCH_TX	PCIE CAMERA R2D C P	31	32
		PCIE_800	PCIE_PCH_TX	PCIE CAMERA R2D C N	31	32
	PCIE_CAM	PCIE_800	PCIE_PCH_RX	PCIE CAMERA D2R P	31	32
	PCIE_CAM	PCIE_800	PCIE_PCH_RX	PCIE CAMERA D2R N	31	32
		PCIE_800	PCIE_PCH_RX	PCIE CAMERA D2R C P	31	32
		PCIE_800	PCIE_PCH_RX	PCIE CAMERA D2R C N	31	32
	PCIE_CLK100M_CAM	CLK_PCIE_800	CLK_PCIE	PCIE CLK100M CAMERA P	31	32
	PCIE_CLK100M_CAM	CLK_PCIE_800	CLK_PCIE	PCIE CLK100M CAMERA N	31	32
		CLK_PCIE_800	CLK_PCIE	PCIE CLK100M CAMERA C P	31	32
		CLK_PCIE_800	CLK_PCIE	PCIE CLK100M CAMERA C N	31	32

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
	SYSCLK CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK CLK32K_RTCX1
	SYSCLK CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK CLK25M_CAMERA
		CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP
		CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R
		CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP
		CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN
		CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN
	SYSCLK CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK CLK25M_TBT
		CLK_25M_45S	CLK_25M	SYSCLK CLK25M_TBT_R
	SYSCLK CLK25M_XTAL	CLK_25M_45S	CLK_25M	SYSCLK CLK25M_X1
		CLK_25M_45S	CLK_25M	SYSCLK CLK25M_X2
		CLK_25M_45S	CLK_25M	SYSCLK CLK25M_X2_R
		CLK_25M_45S	CLK_25M	SDCLK CLK25M_X2
		CLK_25M_45S	CLK_25M	SDCLK CLK25M_X2_R
		CLK_25M_45S	CLK_25M	SDCLK CLK25M_X1

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_73D	*	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2ASELF	*	=2X_DIELECTRIC	?
MEM_DATA2OTHERMEM	*	=8X_DIELECTRIC	?
MEM_QDS2OWNDATA	*	=3X_DIELECTRIC	?
MEM_CMD2CMD	*	=3X_DIELECTRIC	?
MEM_CMD2CTRL	*	=3X_DIELECTRIC	?
MEM_CTRL2CTRL	*	=3X_DIELECTRIC	?
MEM_CLK2CLK	*	=6X_DIELECTRIC	?
MEM_2OTHERMEM	*	=4X_DIELECTRIC	?
MEM_2PWR	*	=2X_DIELECTRIC	10000
MEM_2GND	*	=2X_DIELECTRIC	10000
MEM_2OTHER	*	=6X_DIELECTRIC	?

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_70D	MEM_TERM	MEM_73D
MEM_40S	MEM_TERM	MEM_50S

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM *	*	MEM_2GND

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_DQS20WINDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_DQS20WINDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_DQS20WINDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_DQS20WINDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_DQS20WINDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_DQS20WINDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_DQS20WINDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_DQS20WINDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_DQS20WINDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_DQS20WINDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_DQS20WINDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_DQS20WINDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_DQS20WINDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_DQS20WINDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_DQS20WINDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_DQS20WINDATA

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	*	*	MEM_20THER
MEM_A_DQS_1	*	*	MEM_20THER
MEM_A_DQS_2	*	*	MEM_20THER
MEM_A_DQS_3	*	*	MEM_20THER
MEM_A_DQS_4	*	*	MEM_20THER
MEM_A_DQS_5	*	*	MEM_20THER
MEM_A_DQS_6	*	*	MEM_20THER
MEM_A_DQS_7	*	*	MEM_20THER
MEM_B_DQS_0	*	*	MEM_20THER
MEM_B_DQS_1	*	*	MEM_20THER
MEM_B_DQS_2	*	*	MEM_20THER
MEM_B_DQS_3	*	*	MEM_20THER
MEM_B_DQS_4	*	*	MEM_20THER
MEM_B_DQS_5	*	*	MEM_20THER
MEM_B_DQS_6	*	*	MEM_20THER
MEM_B_DQS_7	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	MEM_*	*	MEM_DATA20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*	MEM_*	*	MEM_20THERMEM

MEM_A_DATA_0	*	*	MEM_20OTHER
MEM_A_DATA_1	*	*	MEM_20OTHER
MEM_A_DATA_2	*	*	MEM_20OTHER
MEM_A_DATA_3	*	*	MEM_20OTHER
MEM_A_DATA_4	*	*	MEM_20OTHER
MEM_A_DATA_5	*	*	MEM_20OTHER
MEM_A_DATA_6	*	*	MEM_20OTHER
MEM_A_DATA_7	*	*	MEM_20OTHER
MEM_B_DATA_0	*	*	MEM_20OTHER
MEM_B_DATA_1	*	*	MEM_20OTHER
MEM_B_DATA_2	*	*	MEM_20OTHER
MEM_B_DATA_3	*	*	MEM_20OTHER
MEM_B_DATA_4	*	*	MEM_20OTHER
MEM_B_DATA_5	*	*	MEM_20OTHER
MEM_B_DATA_6	*	*	MEM_20OTHER
MEM_B_DATA_7	*	*	MEM_20OTHER
MEM_CMD	*	*	MEM_20OTHER
MEM_CTRL	*	*	MEM_20OTHER
MEM_CLK	*	*	MEM_20OTHER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DATA_0	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_1	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_2	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_3	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_4	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_5	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_6	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_7	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_0	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_1	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_2	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_3	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_4	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_5	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_6	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_7	MEM_*_DATA_*	*	MEM_20THERMEM

Memory Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
	PHYSICAL	SPACING		
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM A CLK P<0>	7 20 24
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM A CLK N<0>	7 20 24
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM A CLK P<1>	7 21 24
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM A CLK N<1>	7 21 24
MEM_A_CTRL	MEM_40S	MEM_CTRL	MEM A CS I<1..0>	7 20 21 24
MEM_A_CTRL	MEM_40S	MEM_CTRL	MEM A ODT<0>	7 20 21 24 61
MEM_A_CKE0	MEM_40S	MEM_CMD	MEM A CKE<1..0>	7 20 24
MEM_A_CKE1	MEM_40S	MEM_CMD	MEM A CKE<3..2>	7 21 24
MEM_A_CMD0	MEM_40S	MEM_CMD	MEM A CAA<9..0>	7 20 24 61
MEM_A_CMD1	MEM_40S	MEM_CMD	MEM A CAB<9..0>	7 21 24 61
MEM_A_DQ_BYTE0	MEM_40S	MEM_A_DATA_0	MEM A DQ<7..0>	7 61
MEM_A_DQ_BYTE1	MEM_40S	MEM_A_DATA_1	MEM A DQ<15..8>	7 61
MEM_A_DQ_BYTE2	MEM_40S	MEM_A_DATA_2	MEM A DQ<23..16>	7 61
MEM_A_DQ_BYTE3	MEM_40S	MEM_A_DATA_3	MEM A DQ<31..24>	7 61
MEM_A_DQ_BYTE4	MEM_40S	MEM_A_DATA_4	MEM A DQ<39..32>	7 21 61
MEM_A_DQ_BYTE5	MEM_40S	MEM_A_DATA_5	MEM A DQ<47..40>	7 61
MEM_A_DQ_BYTE6	MEM_40S	MEM_A_DATA_6	MEM A DQ<55..48>	7 61
MEM_A_DQ_BYTE7	MEM_40S	MEM_A_DATA_7	MEM A DQ<63..56>	7 61
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM A DQS P<0>	7 61
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM A DQS N<0>	7 61
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM A DQS P<1>	7 61
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM A DQS N<1>	7 61
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM A DQS P<2>	7 61
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM A DQS N<2>	7 61
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM A DQS P<3>	7 61
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM A DQS N<3>	7 61
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM A DQS P<4>	7 61
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM A DQS N<4>	7 61
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM A DQS P<5>	7 61
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM A DQS N<5>	7 61
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM A DQS P<6>	7 21 61
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM A DQS N<6>	7 21 61
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM A DQS P<7>	7 61
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM A DQS N<7>	7 61
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM B CLK P<0>	7 22 24
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM B CLK N<0>	7 22 24
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM B CLK P<1>	7 23 24
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM B CLK N<1>	7 23 24
MEM_B_CTRL	MEM_40S	MEM_CTRL	MEM B CS I<1..0>	7 22 23 24
MEM_B_CTRL	MEM_40S	MEM_CTRL	MEM B ODT<0>	7 22 23 24 61
MEM_B_CKE0	MEM_40S	MEM_CMD	MEM B CKE<1..0>	7 22 24
MEM_B_CKE1	MEM_40S	MEM_CMD	MEM B CKE<3..2>	7 23 24
MEM_B_CMD0	MEM_40S	MEM_CMD	MEM B CAA<9..0>	7 22 24 61
MEM_B_CMD1	MEM_40S	MEM_CMD	MEM B CAB<9..0>	7 23 24 61
MEM_B_DQ_BYTE0	MEM_40S	MEM_B_DATA_0	MEM B DQ<7..0>	7 61
MEM_B_DQ_BYTE1	MEM_40S	MEM_B_DATA_1	MEM B DQ<15..8>	7 61
MEM_B_DQ_BYTE2	MEM_40S	MEM_B_DATA_2	MEM B DQ<23..16>	7 61
MEM_B_DQ_BYTE3	MEM_40S	MEM_B_DATA_3	MEM B DQ<31..24>	7 61
MEM_B_DQ_BYTE4	MEM_40S	MEM_B_DATA_4	MEM B DQ<39..32>	7 23 61
MEM_B_DQ_BYTE5	MEM_40S	MEM_B_DATA_5	MEM B DQ<47..40>	7 61
MEM_B_DQ_BYTE6	MEM_40S	MEM_B_DATA_6	MEM B DQ<55..48>	7 61
MEM_B_DQ_BYTE7	MEM_40S	MEM_B_DATA_7	MEM B DQ<63..56>	7 61
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM B DQS P<0>	7 61
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM B DQS N<0>	7 61
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM B DQS P<1>	7 61
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM B DQS N<1>	7 61
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM B DQS P<2>	7 61
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM B DQS N<2>	7 61
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM B DQS P<3>	7 61
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM B DQS N<3>	7 61

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		PAGE	
		114 OF 120	
		SHEET	
		68 OF 73	

DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_4SS	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTPD_800	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTD _{DP} _TX	TBTD _{DP} _TX	*	TBTD _{DP} _TX2TX	TBTD _{DP} _TX2TX	TOP, BOTTOM	=6x_DIELECTRIC	?
TBTD _{DP} _RX	TBTD _{DP} _RX	*	TBTD _{DP} _RX2RX	TBTD _{DP} _RX2RX	TOP, BOTTOM	=6x_DIELECTRIC	?
TBTD _{DP} _TX	TBTD _{DP} _RX	*	TBTD _{DP} _TX2RX	TBTD _{DP} _TX2RX	TOP, BOTTOM	=10x_DIELECTRIC	?
TBTD _{DP} _RX	TBTD _{DP} _TX	*	TBTD _{DP} _TX2RX	TBTD _{DP} _20THERHS	TOP, BOTTOM	=10x_DIELECTRIC	?
TBTD _{DP} _TX	*_TX	*	TBTD _{DP} _20THERHS	TBTD _{DP} _20THER	TOP, BOTTOM	=6x_DIELECTRIC	?
TBTD _{DP} _RX	*_TX	*	TBTD _{DP} _20THERHS				
TBTD _{DP} _TX	*_RX	*	TBTD _{DP} _20THERHS				
TBTD _{DP} _RX	*_RX	*	TBTD _{DP} _20THERHS				
TBTD _{DP} _TX	*	*	TBTD _{DP} _20THER				
TBTD _{DP} _RX	*	*	TBTD _{DP} _20THER				

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDOP_TX2TX	"	=4x_DIELECTRIC	?
TBTDOP_RX2RX	"	=4x_DIELECTRIC	?
TBTDOP_TX2RX	"	=6x_DIELECTRIC	?
TBTDOP_20THERHS	"	=6x_DIELECTRIC	?
TBTDOP_20THERL	"	=6x_DIELECTRIC	?
TBTDOP_20THER	"	=4x_DIELECTRIC	?

Thunderbolt/DP Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	TBT_A_R2D	TBTDP_R8D	TBTDP_TX	TBT A R2D C P<1..0>	25
	TBT_A_R2D	TBTDP_R8D	TBTDP_TX	TBT A R2D C N<1..0>	25
		TBTDP_R8D	TBTDP_TX	TBT A R2D P<1..0>	28
		TBTDP_R8D	TBTDP_TX	TBT A R2D N<1..0>	28
	DP TBTPA_ML1	DP_R8D	DP_TX	DP TBTPA_ML C P<1>	25
	DP TBTPA_ML1	DP_R8D	DP_TX	DP TBTPA_ML C N<1>	25
	DP TBTPA_ML3	DP_R8D	DP_TX	DP TBTPA_ML C P<3>	25
	DP TBTPA_ML3	DP_R8D	DP_TX	DP TBTPA_ML C N<3>	25
		DP_R8D	DP_TX	DP TBTPA_ML P<3..1:2>	28
		DP_R8D	DP_TX	DP TBTPA_ML N<3..1:2>	28
		DP_R8D	DP_TX	DP A LSX_ML P<1>	28
		DP_R8D	DP_TX	DP A LSX_ML N<1>	28
		TBTDP_R8D	TBTDP_RX	TBT A D2R C P<1..0>	28
		TBTDP_R8D	TBTDP_RX	TBT A D2R C N<1..0>	28
	TBT_A_D2R1	TBTDP_R8D	TBTDP_RX	TBT A D2R P<1>	25
	TBT_A_D2R1	TBTDP_R8D	TBTDP_RX	TBT A D2R N<1>	25
	TBT_A_D2R8	TBTDP_R8D	TBTDP_RX	TBT A D2R P<0>	25
	TBT_A_D2R8	TBTDP_R8D	TBTDP_RX	TBT A D2R N<0>	25
	TBT_A_AUXCH	DP_R8D	DP_AUX	DP TBTPA_AUXCH C P	25
	TBT_A_AUXCH	DP_R8D	DP_AUX	DP TBTPA_AUXCH C N	25
		DP_R8D	DP_AUX	DP TBTPA_AUXCH P	28
		DP_R8D	DP_AUX	DP TBTPA_AUXCH N	28
		DP_R8D	DP_AUX	DP A_AUXCH_DDC P	28
		DP_R8D	DP_AUX	DP A_AUXCH_DDC N	28
		TBTDP_R8D	TBTDP_RX	TBT A D2R1_AUXDDC P	28
		TBTDP_R8D	TBTDP_RX	TBT A D2R1_AUXDDC N	28
	TBT_B_R2D	TBTDP_R8D	TBTDP_TX	TBT B R2D C P<1..0>	62
	TBT_B_R2D	TBTDP_R8D	TBTDP_TX	TBT B R2D C N<1..0>	62
		TBTDP_R8D	TBTDP_TX	TBT B R2D P<1..0>	62
		TBTDP_R8D	TBTDP_TX	TBT B R2D N<1..0>	62
	DP TBTPB_ML	DP_R8D	DP_TX	NC DP TBTPB_ML CP<3..1:2>	62
	DP TBTPB_ML	DP_R8D	DP_TX	NC DP TBTPB_ML CN<3..1:2>	62
		DP_R8D	DP_TX	DP TBTPB_ML P<3..1:2>	62
		DP_R8D	DP_TX	DP TBTPB_ML N<3..1:2>	62
		DP_R8D	DP_TX	DP B LSX_ML P<1>	62
		DP_R8D	DP_TX	DP B LSX_ML N<1>	62
		TBTDP_R8D	TBTDP_RX	TBT B D2R C P<1..0>	62
		TBTDP_R8D	TBTDP_RX	TBT B D2R C N<1..0>	62
	TBT_B_D2R	TBTDP_R8D	TBTDP_RX	TBT B D2R P<1..0>	62
	TBT_B_D2R	TBTDP_R8D	TBTDP_RX	TBT B D2R N<1..0>	62
	TBT_B_AUXCH	DP_R8D	DP_AUX	NC DP TBTPB_AUXCH CP	25
	TBT_B_AUXCH	DP_R8D	DP_AUX	NC DP TBTPB_AUXCH CN	25
		DP_R8D	DP_AUX	DP TBTPB_AUXCH P	62
		DP_R8D	DP_AUX	DP TBTPB_AUXCH N	62
		DP_R8D	DP_AUX	DP B_AUXCH_DDC P	62
		DP_R8D	DP_AUX	DP B_AUXCH_DDC N	62
		TBTDP_R8D	TBTDP_RX	TBT B D2R1_AUXDDC P	62
		TBTDP_R8D	TBTDP_RX	TBT B D2R1_AUXDDC N	62

Thunderbolt IC Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
		DP_BSD	DP_TX	DP_TBTSRC ML C P<3..0>
		DP_BSD	DP_TX	DP_TBTSRC ML C N<3..0>
		DP_BSD	DP_AUX	DP_TBTSRC AUXCH C P
		DP_BSD	DP_AUX	DP_TBTSRC AUXCH C N
	TBT_SPT_CLK	TBT_SPT_45S	TBT_SPT	TBT_SPT_CLK
	TBT_SPT_M0S1	TBT_SPT_45S	TBT_SPT	TBT_SPT_M0S1
	TBT_SPT_M1S0	TBT_SPT_45S	TBT_SPT	TBT_SPT_MISO
	TBT_SPT_CS_L	TBT_SPT_45S	TBT_SPT	TBT_SPT_CS_L

MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MIP1_850	*	=S_OHM_DIFF	=S_OHM_DIFF	=S_OHM_DIFF	=S_OHM_DIFF	=S_OHM_DIFF	=S_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MIP1_20THER	*	=4X_DIELECTRIC	?
MIP1_2CLK	*	=6X_DIELECTRIC	?
MIP1CLK_20THER	*	=7X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_20CLK
CLK_MIPI	*	*	MIPICKL_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?	S2_DATA2SELF	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_DQS2OWNDATA	*	=2X_DIELECTRIC	?	S2_DQS2OWNDATA	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?	S2_CMD2CMD	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?	S2_CMD2CTRL	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?	S2_CTRL2CTRL	TOP, BOTTOM	=4X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?	S2_20THERMEM	TOP, BOTTOM	=6X_DIELECTRIC	?
S2MEM_2PWR	*	=2X_DIELECTRIC	?	S2MEM_2PWR	TOP, BOTTOM	=4X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?	S2MEM_2GND	TOP, BOTTOM	=4X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?	S2MEM_20THER	TOP, BOTTOM	=10X_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER
S2_MEM_DQS*	*	*	S2MEM_20THER
S2_MEM_CMD	*	*	S2MEM_20THER
S2_MEM_CTRL	*	*	S2MEM_20THER
S2_MEM_CLK	*	*	S2MEM_20THER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WINDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WINDATA

Memory to Power Spacing

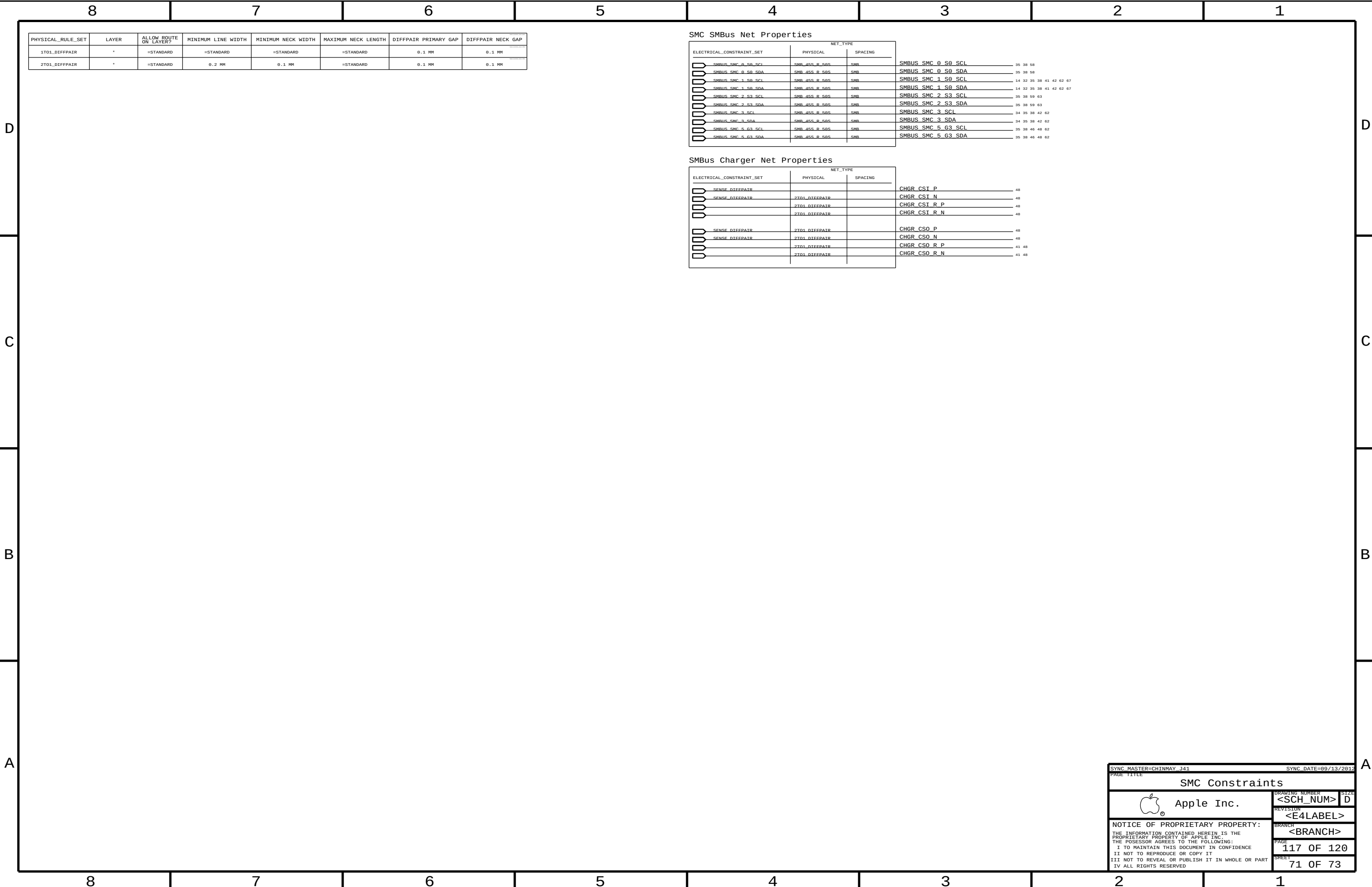
NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	S2_MEM_CLK_K	S2_MEM_BSD	S2_MEM_CLK_K	MEM_CAM_CLK_P	31 32
	S2_MEM_CLK_K	S2_MEM_BSD	S2_MEM_CLK_K	MEM_CAM_CLK_N	31 32
	S2_MEM_CMT1	S2_MEM_45S	S2_MEM_CTR1	MEM_CAM_CKE	31 32
	S2_MEM_CMT1	S2_MEM_45S	S2_MEM_CTR1	MEM_CAM_CS_L	31 32
	S2_MEM_CMT1	S2_MEM_45S	S2_MEM_CTR1	MEM_CAM_ODT	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTR1	MEM_CAM_CAS_L	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTR1	MEM_CAM_RAS_L	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	31 32
	S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	31 32
	S2_MEM_DQS0	S2_MEM_BSD	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	31 32
	S2_MEM_DQS0	S2_MEM_BSD	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	31 32
	S2_MEM_DQS1	S2_MEM_BSD	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	31 32
	S2_MEM_DQS1	S2_MEM_BSD	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	31 32
	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	31 32
	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	31 32
	S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14...0>	31 32
	S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7...0>	31 32
	S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15...8>	31 32
	MIPT_DATA_S2	MIPT_BSD	MIPT_DATA	MIPI_DATA_P	31 32
	MIPT_DATA_S2	MIPT_BSD	MIPT_DATA	MIPI_DATA_N	31 32
	MIPT_DATA_S2	MIPT_BSD	MIPT_DATA	MIPI_DATA_CONN_P	32 62
	MIPT_DATA_S2	MIPT_BSD	MIPT_DATA	MIPI_DATA_CONN_N	32 62
	MIPT_CLK_S2	MIPT_BSD	CLK_MIPT	MIPI_CLK_P	31 32
	MIPT_CLK_S2	MIPT_BSD	CLK_MIPT	MIPI_CLK_N	31 32
	MIPT_CLK_S2	MIPT_BSD	CLK_MIPT	MIPI_CLK_CONN_P	32 62
	MIPT_CLK_S2	MIPT_BSD	CLK_MIPT	MIPI_CLK_CONN_N	32 62
			S2_MEM_PWR	PP1V35_CAM	31 32
			S2_MEM_PWR	PP0V675_CAM_VREF	31 32
			S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	31 32
			S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	32



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D	Change List: <rdar://component/508389> J41 HW EE Schematic Proto 0 <rdar://component/512995> J41 HW EE Schematic Pre Proto 1 <rdar://component/508412> J41 HW EE Schematic Proto 1 <rdar://component/508413> J41 HW EE Schematic EVT <rdar://component/508414> J41 HW EE Schematic DVT Kismet: afp://kismet.apple.com/Kismet-Projects/J41-J43 Useful Wiki Links: Schematic Conventions - https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design MobileMac HW Radar: <rdar://component/497591> MobileMac HW Task <rdar://component/497587> MobileMac HW Schematic <rdar://component/497585> MobileMac HW New Bugs <rdar://component/497588> MobileMac HW Layout <rdar://component/497590> MobileMac HW Investigation <rdar://component/497589> MobileMac HW Architecture Other Info: Page Allocations - <rdar://problem/11791318> 2012 Schematic Page Allocations								D
C									C
B									B
A									A
	8	7	6	5	4	3	2	1	

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		BRANCH <BRANCH>	
		PAGE 120 OF 120	
		SHEET 73 OF 73	

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PAGE

120 OF 120

SHEET

73 OF 73